

When Statistical Trust Layers Go Silent: Feasibility and Padding Evasion in ML Intrusion Detection

Anonymous Submission

Abstract—Statistical trust layers can combine conformal evidence, grouped security hypotheses, and online false-discovery-rate control to turn ML detector scores into calibrated alerts. We show that this composition creates two system-level failure modes and characterize their security consequences.

First, we derive an ex-ante feasibility law for bounded conformal evidence under two families of arrival-time controllers. It yields a finite discovery horizon beyond which a rejection-free run cannot alert regardless of detector score, and an optimal calibration-to-horizon scaling; at $k = 1$, LOND/e-LOND require 20 calibration units per tested hypothesis even under the best cold-start allocation.

Second, we formalize adversarial robustness when an attacker can enlarge its own grouped hypothesis and prove a class-wide impossibility: no alert-capable symmetric e-merging family is robust to arbitrary zero-evidence padding. For the arithmetic mean, we derive the exact suppression cost.

Finally, we validate these effects on LSPR23 and AIT-LDSv2.0. Real-flow replay confirms padding suppression, while joint controller-state reruns show that suppressing early alerts can reduce campaign cost by roughly two orders of magnitude relative to independently pricing alerts.

These results show that statistical validity of individual components does not guarantee feasibility or adversarial robustness of the composed alerting system.

I. INTRODUCTION

A network intrusion detector produces scores; a security operations centre (SOC) must decide which activity to investigate. A useful alerting system therefore needs more than a classifier with good ranking performance. It must turn scores into evidence, decide which observations belong to the same investigation, and manage false alerts over a continuing stream. At the low base rates typical of intrusion detection, even a strong classifier can produce a queue dominated by false positives [1], [2].

Statistical methods offer a principled way to make these decisions. Conformal calibration converts detector scores into evidence under an exchangeability assumption [3], [4]. Grouping combines related flows into security hypotheses, such as communication between two hosts during a time interval. An online multiple-testing procedure then decides which hypotheses to reject while controlling a false-discovery criterion [5], [6], [7]. Together, these components form a *statistical trust layer* between the detector and the analyst (Fig. 1). Prior systems combine subsets of this pipeline [8], [9], [10], [11]. We assemble and evaluate the full composition to study the security consequences of its design choices.

The first consequence concerns whether an alert is possible at all. Finite calibration limits the strength of deterministic conformal evidence. Meanwhile, an uninterrupted online controller must allocate its error budget across an ongoing sequence of hypotheses. During a run without discoveries, the evidence required for rejection can eventually exceed the maximum the calibration permits. Beyond that point, even a detector that assigns the strongest possible score to an attack cannot produce an alert. We call this state *structural silence*. Its onset is a property of the calibration and controller, and can be calculated before deployment.

Grouping offers a natural response: testing episodes instead of individual flows reduces the number of decisions the budget must support. This choice also determines what an alert means and who can influence its evidence. Consider an attacker communicating with a victim inside a host-pair time bucket. Ordinary connections to the same victim enter the same hypothesis as the malicious activity. If those connections contribute little evidence, they dilute the group’s arithmetic mean. The attack flows retain their scores, but the alert disappears.

Changing the averaging rule cannot generally resolve this exposure. We show that any symmetric e-merging family that can attain an alert threshold is vulnerable to sufficiently many zero-evidence additions when membership is unbounded and attacker-controlled. The controller then creates a further interaction: suppressing an early alert removes the increase in testing levels that it would have provided. Later alerts can consequently become cheaper to suppress or disappear without further padding. Security depends on the composition of evidence, membership, and controller state.

Our contributions follow this chain.

A *deployment feasibility law* (Sections III and IV). We characterise finite discovery horizons for two families of arrival-time controllers and derive the best attainable cold-start calibration requirement. At rank depth $k = 1$ and error level 0.05, LOND/e-LOND require $20T - 1$ calibration units to keep every step of a horizon T feasible under the optimal allocation. Grouping reduces this requirement in direct proportion to the hypothesis count, at a measurable cost in alert resolution.

A *security impossibility across group sizes* (Section V). We formalise robustness to an attacker enlarging its own hypothesis and prove that no alert-capable symmetric e-merging family withstands arbitrary zero-evidence padding. For the arithmetic mean, the quantitative bound provides the exact suppression

cost. We also connect the feasibility ratio to a sufficient integer padding multiplier under stated conditions on the rest of the stream.

An empirical account of attack cost and capability (Sections VI and VII). We run real-flow replay on LSPR23 and AIT-LDSv2.0 to test the padding mechanism. We also rerun Joint controller to show that independently pricing alerts can overstate campaign cost by roughly two orders of magnitude. Restricting the attacker’s templates and controller knowledge reveals where practical suppression falls short of the algebraic bound. These results identify where mitigations must act and distinguish the tested capacity constraints from a complete admission policy.

II. SYSTEM AND THREAT MODEL

We study the pipeline from a scored flow to an issued alert. Its basic unit of investigation is an *episode*: flows sharing a pre-committed metadata key and time bucket. An episode is a true null when all its flows are benign; rejecting its null hypothesis issues an alert. Our main configuration uses source–destination host pairs in two-hour buckets. The following construction makes both the evidence available to each episode and the level at which the controller can reject it explicit.

A. From detector scores to bounded evidence

Flows $x_1, x_2, \dots$ arrive in timestamp order, and a detector assigns scores $s_i = f(x_i)$, with larger scores indicating more anomalous activity. A benign calibration set $\mathcal{C}$ is disjoint from training and precedes deployment. Let $K_i = 1 + \#\{c \in \mathcal{C} : s_c \geq s_i\}$ be the rank of the test score from the top of the calibration scores, resolving ties against the test flow. At rank depth $1 \leq k \leq |\mathcal{C}| + 1$, we use threshold conformal evidence

$$e_i = \frac{|\mathcal{C}| + 1}{k} \mathbf{1}\{K_i \leq k\}. \quad (1)$$

Under the null and the relevant exchangeability premise, $\mathbb{E}[e_i] \leq 1$, so e_i is an e-value. A flow receives positive evidence only if its score reaches the selected calibration tail. The construction therefore has two possible outputs, zero and the ceiling

$$M = \frac{|\mathcal{C}| + 1}{k}. \quad (2)$$

Once a score reaches that tail, increasing it further cannot increase its evidence. This finite resolution is inherited from deterministic rank-based conformal calibration [3], [12].

B. From flows to security hypotheses

At bucket close, the controller receives one hypothesis for each non-empty episode G_j . We combine its flow evidence using the arithmetic mean,

$$\text{Ev}(G_j) = \frac{1}{m_j} \sum_{i \in G_j} e_i, \quad m_j = |G_j|. \quad (3)$$

The mean preserves the ceiling M . It is also an e-merging rule: at any fixed arity, it combines marginally valid e-values into a valid e-value under arbitrary dependence. This property

accommodates flows that share a calibration set and can be strongly dependent within an episode.

Metadata-selected membership requires an additional premise. For the nominal FDR guarantee, each true-null group must have marginally valid evidence. We use the sufficient condition that every benign member of a true-null group remains e-valid conditional on the local metadata fixing that group’s key, membership, arity, order position, and horizon (Assumption 1). Linearity then gives group validity. Unconditional flow validity alone does not imply this condition, because metadata can correlate with detector features. Appendix A-A states the condition formally and gives a construction where it holds. The feasibility and suppression results concern realised evidence and rejection rules; they do not require this empirical validity premise.

C. Online decisions and the role of earlier alerts

Write α for the target error level and α_t for the testing level offered to the t -th episode. The controller rejects when

$$\text{Ev}(G_t) \geq 1/\alpha_t. \quad (4)$$

A larger testing level therefore makes rejection easier. Our principal controller is e-LOND, which controls FDR under arbitrary dependence given marginally valid group e-values [7]. FDR is the expected fraction of false discoveries among issued alerts, with that fraction defined as zero when no alert is issued. It controls false alerts without requiring any discoveries. If R_{t-1} is the number of earlier rejections, e-LOND’s level is

$$\alpha_t = \alpha \gamma_t (R_{t-1} + 1). \quad (5)$$

The non-negative weights γ_t have total mass at most one. Before the first alert, the level is $\alpha \gamma_t$; subsequent alerts increase the levels available to later hypotheses. This dependence on discovery history matters both for detection and for the cost of an attack.

We compare two allocations. The *horizon-free* allocation uses $\gamma_j = j^{-1.6}/\zeta(1.6)$ and supports uninterrupted operation without a specified terminal horizon. The *horizon-aware* allocation uses $\gamma_j = T^{-1} \mathbf{1}\{j \leq T\}$ for a known hypothesis count T . It gives every step the same cold-start weight and is the optimal cold-start benchmark derived in Section III-B. All experiments use $q = \alpha = 0.05$, $w_0 = 0.025$, and $k = 1$. We also analyse LOND, e-LORD, and LORD++; their definitions and additional procedures appear in Appendices B and C. For p-value procedures we use $p_i = \min(1, 1/e_i)$.

D. Attacker control and knowledge

The adversary controls traffic from an attacker-operated or compromised host. Before an episode’s bucket closes, it can open ordinary connections to the destination it is already attacking. These connections share the episode’s (SrcIP, DstIP, bucket) key and join its tested hypothesis. The defender implements its pipeline correctly; the attacker changes neither its code nor the original malicious flows. Its objective is to suppress alerts on its own activity.

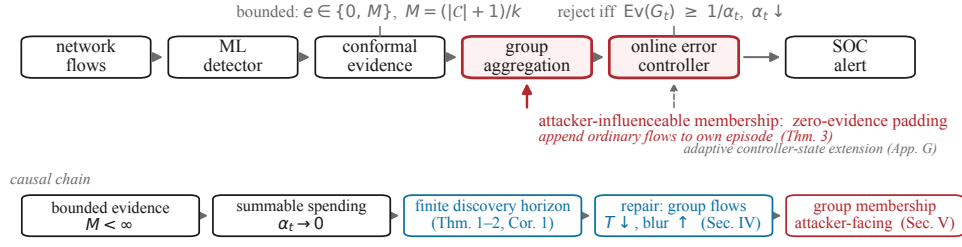


Fig. 1. The statistical trust layer and the argument of the paper. Bounded evidence and decaying testing levels create a finite discovery horizon (Section III). Grouping reduces the number of tests (Section IV), but also lets an attacker add ordinary traffic to its own hypothesis (Section V). Padding changes the aggregate while preserving the original malicious flows and their scores.

Our formal padding model permits unbounded additions carrying zero evidence. The experiments test whether real traffic supplies such additions and how capacity constraints restrict them. We distinguish an *oracle* that knows the realised evidence and live controller level from a *state-free* attacker that applies a fixed multiplier to its own flow count. Template availability is a separate capability: on AIT we compare a retrospective deployment-wide pool with traffic the victim had received before the attacker’s last flow in the target episode. Temporal availability does not establish that a compromised source could observe that traffic.

Groups closing in the same bucket are tested in a canonical order: a deterministic hash of each group’s metadata key, with collisions broken by the key. Appending flows to an existing group then leaves every other group’s position unchanged, preserving the local validity premise for untouched true nulls. This order does not prevent padding. We report first-flow ordering as an explicitly labelled sensitivity analysis; Appendix D-A examines ordering and key manipulation in detail.

E. Evaluation setting

LSPR23 [13] provides 16,353,511 flows from a 161.5-hour live-fire exercise. Of these, 1,644,599 are malicious: 10.06% of flows, or 0.48–0.81% of episodes. Each evaluation uses chronological training, benign calibration, and deployment blocks; calibration and deployment each span 15% of the dataset. The primary detector is HistGradientBoosting using protocol and 32 flow timing and volume features. Host-context experiments add six strictly causal features. AIT-LDSv2.0 [14] supplies an external setting with eight organisations where attacked hosts also receive benign traffic.

Positions 0.55 and 0.62 were fixed as the primary and secondary LSPR23 windows before validity diagnostics. They are overlapping views of one exercise, so the secondary window acts as a correlated robustness check. Position 0.85 violates marginal e-validity and is used only for explicitly labelled stress measurements. The remaining diagnostics do not establish the group-validity premise. Unless stated otherwise, results use seed 0, canonical order, and horizon-free spending. Appendices B, B-D and F give the preprocessing, split geometry, and validity diagnostics.

III. BOUNDED EVIDENCE CREATES A FINITE DISCOVERY HORIZON

Before asking how often the detector identifies attacks, we ask whether its evidence can ever be strong enough to trigger an alert. Combining the ceiling in (2) with the rejection rule in (4) gives the necessary condition

$$\alpha_t \geq 1/M. \quad (6)$$

We call a step *evidence-feasible* when this condition holds. It depends on the calibration size, rank depth, and offered level, independently of the detector’s ranking performance. Feasibility allows a rejection; the observed evidence still has to reach the threshold.

For e-LOND, the source of a finite horizon is already visible in (5). Before any rejection, $\alpha_t = \alpha \gamma_t$. As the spending weights decay, the required evidence $1/\alpha_t$ eventually exceeds M . A stronger score cannot overcome this ceiling, and a stream that cannot reject cannot obtain the increase in levels that an earlier rejection would provide.

A. When silence becomes permanent

On a rejection-free run, the last evidence-feasible step is the *discovery horizon* t^* . Beyond it, the run is absorbed: every admissible observation fails the rejection rule. We call this operational state *structural silence*. The following results establish finite horizons for two common forms of arrival-time controller. Their assumptions concern the level sequence and evidence ceiling, so they also apply when the evidence lacks a nominal validity guarantee.

Theorem 1 (Finite discovery horizon, multiplicative family). *Let the evidence be bounded by $M < \infty$ and the controller run without restart, with $\alpha_t = \alpha \gamma_t g(\mathcal{H}_{t-1})$, $\gamma \geq 0$, $\sum_t \gamma_t \leq 1$ and $g \leq c_g(R_{t-1} + 1)^d$ for the rejection count R and some $d \geq 0$ (which the step below uses; every procedure we cover has $d \in \{0, 1\}$). Since $R_{t-1} \leq t - 1$, a rejection is possible at t only if $\gamma_t t^d \geq 1/(\alpha c_g M)$. If $\gamma_t t^d \rightarrow 0$, only finitely many steps are feasible on a rejection-free run; if that set is non-empty it has a last element t^* and the run is absorbed from $t^* + 1$ onward, and if it is empty the run is absorbed from step 1. No monotonicity of γ is needed: a set of steps that is finite already has a maximum.*

Theorem 2 (Finite discovery horizon, lag-sum family). *Let the evidence be bounded by $M < \infty$ and $\alpha_t = w_0\gamma_t + \sum_j a_j\gamma_{t-\tau_j}$ over prior rejection times $\tau_j < t$, with $w_0 \geq 0$, $0 \leq a_j \leq \alpha$, $\gamma \geq 0$ and $\sum_t \gamma_t < \infty$. During a rejection-free run of length Δ every lag is distinct and at least Δ , so $\alpha_t \leq (w_0 + \alpha) \sum_{u \geq \Delta} \gamma_u$. The tail sum tends to zero, so $\Delta^* = \min\{\Delta : (w_0 + \alpha) \sum_{u \geq \Delta} \gamma_u < 1/M\}$ is finite and the run is absorbed once $\Delta \geq \Delta^*$, with no monotonicity assumption on γ .*

The proof mechanism is a bound on the largest level the controller could offer. For the multiplicative family, $R_{t-1} \leq t-1$ gives $\alpha_t \leq \alpha c_g \gamma_t t^d$, which eventually falls below $1/M$ under the stated decay condition, regardless of the preceding history. For the lag-sum family, a long gap without rejections leaves every reward at a large lag. The remaining tail of the spending sequence then bounds their total contribution and tends to zero. Once no rejection is possible, the gap can only grow. Appendix A gives the full proofs.

These scopes differ: the first bound is uniform over rejection histories, whereas the second requires a sufficiently long rejection-free gap. LOND, e-LOND, and the stated deterministic reparametrisation of e-LORD fall under Theorem 1; LORD++ falls under Theorem 2. The horizon-free sequence $\gamma_j \propto j^{-1.6}$ satisfies both sets of conditions. Coverage is always of a procedure together with its spending sequence (Appendix A-B).

Huo et al. [15] identify the conformal floor and its adverse effect on online testing. Our results characterise when this interaction makes rejection permanently impossible for the covered controllers. The next result translates that mechanism into a calibration requirement that can be checked before deployment.

B. The best attainable calibration requirement

A deployment with a known horizon can choose its allocation to keep the weakest cold-start step as strong as possible. Write c_0 for the cold-start coefficient, so $\alpha_t = c_0\gamma_t$ before the first rejection: $c_0 = \alpha$ for LOND/e-LOND and $c_0 = w_0$ for LORD++. Because the weights have total mass at most one, their minimum over T steps cannot exceed $1/T$. Uniform allocation attains this value. We call a deployment *cold-start feasible through T* if every step remains evidence-feasible before the first rejection.

Corollary 1 (Horizon-uniform feasibility). *Cold-start feasibility over a horizon T — every step $t \leq T$ evidence-feasible before the first rejection, which the ceiling decides and detection power does not — requires $|\mathcal{C}| \geq k/\alpha_T - 1$, for α_T the level offered at step T . Under the horizon-uniform allocation $\gamma_t = T^{-1}\mathbf{1}\{t \leq T\}$, where α_T is also the smallest level offered over the horizon, the condition is necessary and sufficient and reads*

$$|\mathcal{C}| \geq \frac{kT}{c_0} - 1.$$

The requirement is a scale-free count ratio k/c_0 : at $k = 1$ it is 20 calibration units per hypothesis for LOND/e-LOND and

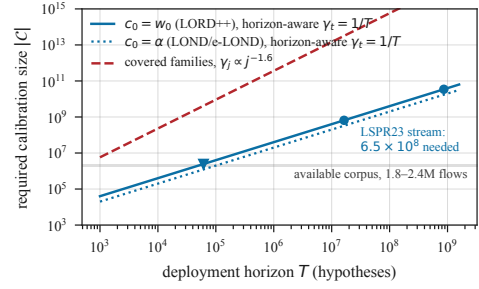


Fig. 2. Calibration needed to keep every cold-start step feasible through horizon T ($k = 1$, $w_0 = 0.025$). The optimal horizon-uniform requirement grows linearly; horizon-free $\gamma_j \propto j^{-1.6}$ requires growth as $T^{1.6}$. The grey band marks the 1.8–2.4M calibration flows available in the LSPR23 windows.

40 for LORD++, whatever the unit is — exactly $20T - 1$ and $40T - 1$ over a horizon T , the ratio being the rate rather than the count.

This is an attained sizing boundary. If calibration falls short, no pre-committed allocation can keep every cold-start step feasible. If it meets the requirement, horizon-uniform spending does so. The result concerns the possibility of detection, not a lower bound on recall.

For comparisons across configurations, define the allocation-optimal feasibility ratio

$$\rho_p = \frac{Mc_{0,p}}{T} = \frac{(|\mathcal{C}| + 1)c_{0,p}}{kT}. \quad (7)$$

Under horizon-uniform allocation, $\rho_p \geq 1$ is necessary and sufficient for cold-start feasibility through T . A different allocation has minimum-level ratio $\rho_p(\gamma) = Mc_{0,p} \min_{t \leq T} \gamma_t \leq \rho_p$. Throughout the paper, ρ denotes the horizon-uniform ratio.

At network scale, even the optimal requirement is substantial (Fig. 2). Testing all 16,353,511 LSPR23 flows in one uninterrupted run would require about 3.3×10^8 calibration flows for LOND/e-LOND and 6.5×10^8 for LORD++. The smaller requirement is already 20 times the entire exercise, compared with the 1.8–2.4 million benign calibration flows available before each evaluation window. Increasing calibration and deployment together does not remove this count-ratio constraint. Horizon-free spending is more demanding still; its full scaling appears in Appendix C.

C. Scope and alternatives

The theorems concern uninterrupted arrival-time decisions with the specified level forms. Restart, randomised smoothing, selective advancement of a spending index, and reconsideration of earlier hypotheses can change those premises. For example, ADDIS can stop advancing its effective index, while online e-BH revisits the observed history rather than fixing each decision at arrival. Appendix C-B compares these mechanisms and their measured costs; the separate ADDIS state attack remains an extension in Appendix G.

Grouping takes a different route. It preserves the evidence construction and controller, but reduces the number of hypotheses the calibration must support. We next examine

what this buys and what an alert loses as its unit becomes coarser.

IV. GROUPING RESTORES FEASIBILITY BY CHANGING THE SECURITY UNIT

The calibration requirement makes the alerting unit a consequential design choice. A SOC can investigate related flows together rather than spend a test on every connection. With flow calibration fixed, replacing individual-flow hypotheses by episodes reduces the horizon to T_{group} without changing the maximum flow evidence. The allocation-optimal requirement becomes

$$|\mathcal{C}|_{\min} + 1 = \frac{kT_{\text{group}}}{c_0}. \quad (8)$$

Thus grouping reduces the required calibration size in direct proportion to the hypothesis count.

A. Restoring feasibility and recovering detections

Two-hour host-pair grouping reduces the LSPR23 windows to 31,568–57,368 hypotheses. Horizon-uniform e-LOND is then cold-start feasible in all five windows, with $\rho = 2.13$ –3.34. Across five grouping families and seven bucket choices, each family becomes feasible once its buckets are sufficiently coarse (Appendix D).

This change creates the opportunity to alert, but the allocation still determines how widely that opportunity is available. Under horizon-free spending, only the first 748–902 hypotheses are feasible before the first rejection, or 1.6–2.6% of a window. Strong evidence must arrive within that prefix to start the controller’s discovery history. In the primary window, canonical e-LOND consequently detects three of 275 malicious episodes. With the same detector, groups, and order, horizon-aware spending detects 105, with no observed false discoveries (Table II).

The distinction matters when interpreting low recall. At flow granularity, the allocation-optimal calibration requirement is already too large. At episode granularity, grouping makes the optimal allocation feasible, while a decaying allocation can still leave most of the stream beyond its cold-start horizon. Detector quality influences which episodes supply sufficient evidence; it does not change ρ . Appendix C-A reports the detector and prevalence comparisons that separate these effects.

B. The resolution cost of an alert

Coarsening an episode also changes what the analyst learns from an alert. Ordinary episode recall cannot quantify this cost across bucket widths because the positive unit and its denominator both change. We therefore use a fixed reference: each malicious source–destination episode in a five-minute bucket is one *atom*. An alert’s *blur* is the number of distinct malicious atoms it represents. These atoms are an evaluation reference, not independently validated incidents.

Under first-flow order, which yields alerts at every evaluated width and is held fixed in this comparison, mean primary-window blur grows from 1.0 at five minutes to 7.2 at one hour, 16.1 at two hours, and 38.5 at one day (Fig. 3). Fixed atomic

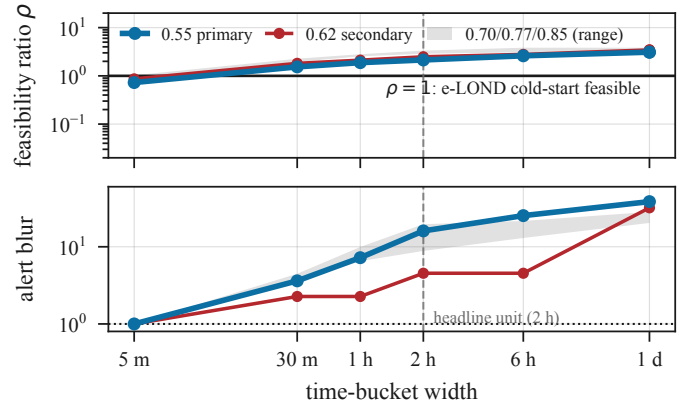


Fig. 3. Coarsening the source–destination bucket increases the feasibility ratio ρ through 1 and increases alert blur. Blur counts fixed five-minute malicious atoms per issued alert. Lines show the primary and secondary windows; shading covers the other three. Seed 0; the two-hour main configuration is marked.

coverage also increases: broader alerts cover more activity, but localise it less precisely.

There is also a security consequence. The grouping key decides which observations contribute to the same statistical decision. When an attacker can generate traffic with that key before the bucket closes, it can change the composition of its own hypothesis. The next section examines this control over membership.

V. PADDING EVASION THROUGH ATTACKER-INFLUENCEABLE MEMBERSHIP

A. How ordinary traffic suppresses an alert

Consider an episode whose malicious flows produce enough evidence for rejection. The attacker opens additional ordinary connections to the same destination before the bucket closes. Each added flow that misses the calibration tail contributes zero evidence. Under the arithmetic mean, these flows increase the denominator without increasing the evidence sum. Enough additions therefore move the episode below its alert threshold, even though every original malicious flow retains its score.

This mechanism raises two questions. Can another valid aggregation rule prevent dilution? And if padding is possible, how much traffic does the attacker need? We answer the first for the symmetric e-merging class and the second exactly for the arithmetic mean. Both results concern suppression at a specified controller level. They do not depend on whether the empirical null evidence is valid; Appendix A-A also constructs a stream where true-null validity holds and padding still erases an alert.

B. An impossibility across group sizes

An e-merging function combines valid input e-values into a valid output under arbitrary dependence. A symmetric rule gives the same output to every permutation of its inputs. Existing e-merging theory compares such rules at a fixed arity. Padding instead changes the arity: the same malicious activity is assessed by F_m before the attack and by F_{m+r} after r

additions. Robustness must therefore relate the rules across group sizes.

Definition 1 (θ -padding-robustness, attainment). *A family $\{F_N\}_{N \geq 1}$ of e-merging functions is θ -padding-robust, for $\theta > 1$, if for every $m \geq 1$, every finite $x \in [0, \infty)^m$ with $F_m(x) \geq \theta$ and every integer $r \geq 0$, $F_{m+r}(x, 0^r) \geq \theta$. It attains θ if some finite input has $F_m(x) \geq \theta$.*

Attainment requires that the rule can issue an alert at the threshold under discussion. Without it, a constant rule such as $F \equiv 1$ would be padding-invariant simply because it never alerts.

Theorem 3 (Symmetric padding impossibility). *No family of symmetric e-merging functions that attains θ is θ -padding-robust, for any $\theta > 1$. Quantitatively, for every symmetric e-merging family, every $m \geq 1$, every finite $x \in [0, \infty)^m$, every $\theta > 1$ and every integer $r \geq 0$,*

$$F_{m+r}(x, 0^r) < \theta \quad \text{whenever} \quad r > \frac{\sum_i x_i}{\theta} - m.$$

Proof idea. Vovk and Wang [16, Prop. 3.1] show that the arithmetic mean essentially dominates every symmetric e-merging function: $F(e) > 1$ implies $\text{mean}(e) \geq F(e)$. Splitting on whether $F(e) \leq 1$ gives $F_N(x) \leq \max\{1, \text{mean}(x)\}$ on finite inputs. After padding, the mean is $\sum_i x_i / (m + r)$, which falls below θ under the stated bound on r . The arithmetic mean attains the bound, so it is tight. Appendix A gives the full argument, including inputs containing zeros. $\square$

The security implication concerns membership as well as aggregation. Under attacker-controlled, unbounded membership, any alert issued by a symmetric e-merging family has a finite zero-padding budget that suppresses it. Replacing the mean with another member of this class cannot provide invariance to arbitrary additions. The inherited domination result supplies the mathematical bound; the cross-arity robustness definition identifies the adversarial property it rules out.

C. Exact cost for one alert

Let $S_t = \sum_{i \in G_t} e_i$ and $m_t = |G_t|$. After r zero-evidence additions, the mean is $S_t / (m_t + r)$. If the original episode is rejected at level α_t , suppression requires $S_t / (m_t + r) < 1/\alpha_t$. The smallest integer pad is therefore

$$r_t^* = \lfloor S_t \alpha_t \rfloor - m_t + 1. \quad (9)$$

Exact sizing requires the realised evidence and controller level. We consequently report r_t^* as an oracle per-alert cost. It is also a lower bound for replay with real traffic: a pad that contributes positive evidence can increase the required number of additions.

The level has a direct operational consequence. A larger α_t makes the original alert easier to obtain, but requires more padding to remove it. For the baseline true detections $\mathcal{D}$, the sum $\sum_{t \in \mathcal{D}} r_t^*$ prices every alert at its level on the unperturbed run. It does not account for what the controller does after an earlier alert is removed.

D. From individual alerts to the attacked trajectory

Equation (5) explains why campaign cost can be smaller than that sum. Suppressing an early rejection reduces R_{t-1} for later episodes, so their testing levels fall and their required evidence rises. Some later alerts need fewer pads; others cease to reject without being padded at all. We use J_{seq} for the total cost of a greedy sequential oracle that sizes each pad against the level on the attacked trajectory. It is one strategy's cost, hence an upper bound on the minimum joint cost.

An attacker without live controller knowledge can instead multiply each of its own episodes by a fixed integer c , appending $(c-1)m_t$ zero-evidence flows. At a given level this suppresses an alert whenever $c \geq 1 + r_t^*/m_t$. Under horizon-uniform e-LOND, keeping the entire trajectory at cold start yields a stronger result.

Corollary 2 (Horizon-uniform feasibility ratio as a dilution threshold). *Run e-LOND under the horizon-uniform allocation, so that $\alpha_t = (\alpha/T)(R_{t-1} + 1)$ with $R_0 = 0$, on arithmetic-mean group evidence whose flow e-values are bounded by M , and write $\rho = \rho_{e\text{-LOND}} = M\alpha/T$, the horizon-uniform ratio (7) for e-LOND. Suppose no hypothesis other than the attacker's own episodes carries evidence at or above the cold-start threshold T/α , and the attacker multiplies every one of its own episodes by an integer $c \geq 1$, appending $(c-1)m_t$ zero-evidence flows to an episode of arity m_t . If $c > \rho$, the attacked trajectory is rejection-free. If $c \leq \rho$ and some attacker episode attains M , it is not rejection-free. Hence the smallest integer multiplier that suffices for every stream satisfying the hypotheses is $c_{\text{int}} = \lfloor \rho \rfloor + 1$.*

The argument is inductive. With no earlier rejection, the threshold is T/α . Multiplication by $c > \rho$ brings every attacker episode below it, while the premise on other hypotheses prevents them from raising the controller level. Conversely, a ceiling-attaining episode still rejects if $c \leq \rho$. Thus the ratio that measures best-case feasibility also determines the universal integer dilution threshold under these assumptions.

For a particular stream, the smaller multiplier $\lfloor c_{\text{crit}} \rfloor + 1$ suffices, where c_{crit} is the largest attacker episode evidence multiplied by α/T and $c_{\text{crit}} \leq \rho$. Equality holds when an attacker episode attains the ceiling. Appendix A supplies the proof. The empirical question is now whether suitable pads are available and whether other alerts allow the controller to escape cold start.

VI. EXPERIMENTAL EVALUATION

We evaluate the attack in three stages. First, real-flow replay tests whether ordinary traffic provides the low-evidence additions assumed by the theory. Second, we compare allocations and rerun the controller to measure the difference between individual-alert and joint cost. Third, AIT tests how benign co-resident traffic, host context, and restricted attacker capabilities affect suppression. The principal comparisons hold the detector, grouping, and order fixed.

TABLE I

PADDING RESULTS UNDER HORIZON-FREE SPENDING AT SEED 0. ORDER IS CANONICAL UNLESS STATED. LSPR23 COSTS ARE EXACT ZERO-PAD ORACLE COSTS FROM (9); AIT COSTS MARKED “EMP.” ARE PER-ORGANISATION EMPIRICAL REPLAY MEDIANS AND ARE NO SMALLER THAN THE CLOSED FORM. SUPPRESSION USES PER-ALERT REAL-FLOW REPLAY; AIT USES THE RETROSPECTIVE DEPLOYMENT TEMPLATE POOL. FOR THE CANONICAL FLOW-ONLY AIT ROW, PER-DRAW SUCCESS IS 0.93–1.00. APPENDIX E GIVES ALL ARMS.

Setting	order	det. cost	suppr.
LSPR23 0.55 primary	canonical	323, 24, 33	3/3
LSPR23 0.62 secondary	canonical	11 med. 6	11/11
AIT flow-only, 8 orgs	canonical	79 med. 12–4,650 (emp.)	78/79
AIT host-cond., 8 orgs	first-flow	63 med. 43–8,471.5 (emp.)	57/63

A. Real-flow replay tests the padding model

The closed form predicts a traffic count, but its operational relevance depends on how the detector scores the added traffic. For LSPR23, we choose the most frequent protocol–port pair in the training prefix and use real flows from that service as padding templates. This selection precedes calibration and deployment and uses only network-observable service frequency, with no attack labels or detector outputs.

At their original controller levels, the three canonical primary-window alerts require 23, 24, and 33 zero-evidence additions. Replaying real flows suppresses all three at exactly those counts. All eleven secondary-window alerts also suppress at their predicted costs, with median six (Table I and Fig. 4). Under horizon-aware spending, the same per-alert replay suppresses 105/105 and 107/107 true detections in the primary and secondary windows on every draw.

None of the 845,691 available primary-window pool flows reaches the conformal tail. For this finite pool, replay therefore agrees exactly with the zero-evidence model. A one-sided 95% upper bound on a common firing probability is 3.5×10^{-6} ; extending that interpretation to future traffic requires a common-rate sampling assumption. Appendix E-C gives the sampled-pool diagnostics, alternative pools, and residual-risk analysis.

This experiment scores real traffic and suppresses each alert separately at its original level. It establishes the pad model used in the LSPR23 joint rerun below. LSPR23’s attacked host pairs contain no benign traffic, so these templates come from a common service across the window. The AIT experiments address the resulting limitation using actual benign traffic to attacked victims.

B. Allocation raises individual cost; feedback lowers joint cost

A horizon-aware allocation gives the detector more opportunities to alert. Does it also make padding substantially harder? Table II separates the effect on each alert from the effect on the whole trajectory. In the primary window, changing only the allocation increases true detections from three to 105 and the median per-alert cost from 24 to 2,946 flows. No false discovery occurs in either primary-window run. The secondary window shows the same increase in detections and cost, with one false discovery under horizon-aware spending (realised FDP 0.009).

The individually priced pads sum to 627,495 flows in the primary horizon-aware run. Applying them together and rerunning e-LOND suppresses every true detection, but that sum overstates what the sequential oracle needs. Once earlier alerts disappear, later episodes receive lower testing levels. The sequential oracle pays 5,650 flows in total; the secondary-window total is 9,992, compared with a per-alert sum of 977,567. These are roughly two orders of magnitude smaller than the independent sums.

The change is visible at the episode level. Among episodes the sequential oracle actually pads, median additions are 48.5 and 33 flows in the primary and secondary windows. A further 13 and 4 baseline alerts need no pad because they fired only after earlier rejections had raised their levels. The secondary window’s false discovery also disappears. Suppression thus affects both the target episode’s evidence and the testing levels offered to later episodes.

Better allocation still raises the cost of this joint strategy: the corresponding horizon-free totals are 30 and 859. It does not remove the finite-padding vulnerability. These joint results use zero-evidence additions, validated for the replay pool, and one greedy oracle strategy; they are not claims of minimum campaign cost. Appendix E-G reports the complete reruns and checks that padding preserves the hypothesis keys and order.

C. Suppressing without live controller state

The primary and secondary windows have $\rho = 2.13$ and 2.49. Both satisfy the conditions of Corollary 2, and both contain a ceiling-attaining attacker episode. Multiplying every attacker-owned episode by $c = 3$ therefore suppresses the horizon-aware trajectory in each window, whereas $c = 2$ does not. Although the unperturbed secondary run contains a false discovery, that episode reaches its rejection threshold only after earlier alerts raise the level. It remains below the cold-start threshold, as the corollary requires.

Executing this fixed multiplier needs only the attacker’s episode size. On LSPR23, attacked host pairs contain no benign traffic, so the attacker knows that size from its own flows. No live evidence sum or testing level is needed. This result depends on multiplying every own episode, obtaining zero-evidence pads, and preventing other hypotheses from initiating a rejection history. Appendix E-B gives the horizon-free multiplier and flat-pad sensitivity analyses. The next experiment tests where these conditions break.

D. Transfer to benign-inclusive victims and host-aware detection

AIT-LDSv2.0 provides eight organisations in which attacked hosts also receive ordinary traffic. For each held-out organisation, we train on the other seven, calibrate on in-organisation benign flows preceding the first attack, and run the same two-hour host-pair/e-LOND pipeline. Per-alert replay uses real benign-to-victim templates over 200 draws.

With the flow-only detector and canonical order, 78 of 79 true detections are suppressible using the full deployment template pool (Table I). This count records episodes with a successful

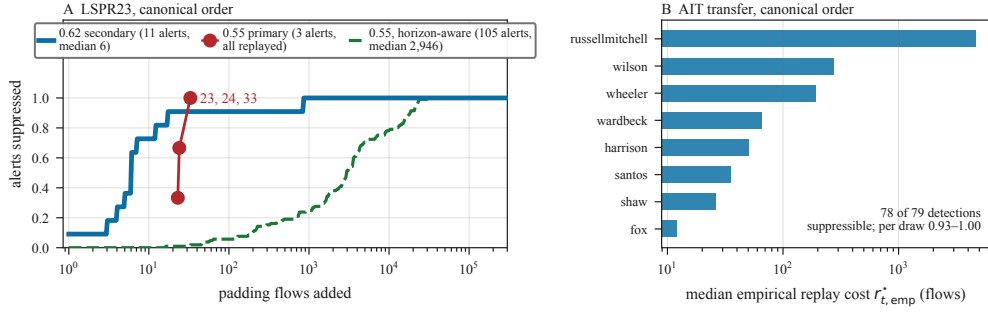


Fig. 4. Per-alert suppression cost under canonical order, seed 0. **A:** LSPR23. Points mark the three primary-window horizon-free alerts suppressed by real-flow replay at 23/24/33 added flows; the solid curve shows the secondary window and the dashed curve the primary window under horizon-aware allocation. **B:** AIT empirical replay medians by organisation. Replay cost is at least the zero-pad bound and equals it in seven of eight organisations. Suppression success is reported separately in Table I; Fig. 5 gives sensitivity results.

TABLE II

DETECTION AND PADDING COST WITH DETECTOR, GROUPING, CANONICAL ORDER, AND SEED 0 FIXED. r_t^* IS THE ORACLE COST OF ONE ALERT; $\mathcal{D}$ IS THE SET OF TRUE DETECTIONS; AND J_{seq} IS THE COST OF A GREEDY SEQUENTIAL ORACLE ON THE ATTACKED TRAJECTORY (SECTION VI-B AND TABLE XXXVII). THE PER-ALERT SUM UPPER-BOUNDS THE JOINT COST; FDP IS REALISED.

Window	allocation	true det.	FDP	median r_t^*	$\sum_{t \in \mathcal{D}} r_t^*$	J_{seq}
0.55 primary	horizon-free	3	0.000	24	80	30
	horizon-aware	105	0.000	2,946	627,495	5,650
0.62 secondary	horizon-free	11	0.000	6	926	859
	horizon-aware	107	0.009	3,908	977,567	9,992

replay; per-draw success is 0.93–1.00. Per-organisation median empirical costs range from 12 to 4,650 flows. A template can contribute positive evidence, in which case empirical cost exceeds the zero-pad bound. The baseline runs issue 83 rejections, including four false discoveries, for pooled realised FDP 0.048.

Host context could make repeated connections increasingly anomalous. We test this by recomputing six causal host-context features as pads accumulate. On LSPR23, both primary-window host-conditioned targets remain suppressible under first-flow order. On AIT, the host-conditioned arm uses first-flow order across all eight organisations and suppresses 57 of 63 true detections. The other six have no successful draw within twice the zero-pad cost; that is a limit of the evaluated replay budget, not proof that padding is impossible. Appendix E-D gives the accumulation model, scored ranges, and order comparisons.

E. Template availability and knowledge limit the joint attack

The deployment-wide template pool is retrospective. To examine a more constrained attacker, we rerun the AIT controller using only traffic received by the victim before the attacker’s last flow in each target episode. An episode with fewer than 20 eligible prior flows is not padded. The joint runs propagate both sampled pad evidence and the changed controller state.

With the full deployment pool, the sequential oracle leaves one and zero baseline attacker-owned alerts under horizon-free and horizon-aware spending, respectively (Table III). Restricting the same oracle to victim-prior traffic leaves 24 and 27. Across the 95 attacker-owned episodes, 30 have fewer

TABLE III

AIT FLOW-ONLY JOINT-ATTACK BOUNDARY UNDER CANONICAL ORDER, SEED 0: ATTACKER-OWNED ALERTS REMAINING (LOWER IS MORE EFFECTIVE). BASELINES ARE EXACT SUMS ACROSS EIGHT ORGANISATIONS; ATTACKED COUNTS SUM THE PER-ORGANISATION MEDIAN OVER 100 SAMPLED TRAJECTORIES AND ARE NEITHER ONE COMBINED TRAJECTORY NOR THE MEDIAN OF AN AGGREGATE TOTAL; TABLE XXXIV GIVES THE PER-TRAJECTORY RELIABILITY. THE DEPLOYMENT POOL IS RETROSPECTIVE; THE PRIOR POOL CONTAINS ONLY VICTIM TRAFFIC OBSERVED BEFORE THE ATTACKER’S LAST FLOW IN THE TARGET EPISODE.

Attacker arm	horizon-free	horizon-aware
Baseline	79	80
Sequential oracle, deployment pool	1	0
Sequential oracle, victim-prior pool	24	27
State-free multiplier, victim-prior pool	65	60

than 20 prior templates and 18 have none. Missing cover traffic prevents the attacker from suppressing some early alerts, allowing those alerts to raise later testing levels.

Removing live controller knowledge further reduces suppression. Applying the fixed multiplier $c = \lfloor \rho \rfloor + 1$ with the prior pool leaves 65 and 60 alerts. Its use under horizon-free spending is a heuristic transfer, since Corollary 2 is proved for horizon-uniform spending. These failures have identifiable causes: some episodes cannot be padded, some templates carry positive evidence, and some non-attacker episodes can reject at cold start. Sizing from the full group arity instead of the attacker’s own flows leaves the same residual counts, so unknown co-resident arity is not the binding constraint in this arm. Table XXXV tests the premises directly.

The table sums per-organisation medians over 100 sampled

trajectories; its totals are not one combined campaign. Table XXXIV reports per-trajectory reliability. Even the prior-pool arm assumes access to traffic that a compromised source may not observe. The results therefore locate capability boundaries: they demonstrate suppression with available templates, and quantify how restricting those templates and controller knowledge limits the attack.

VII. DESIGN IMPLICATIONS AND LIMITATIONS

A. Where a mitigation must act

The results make calibration, allocation, and membership joint design decisions. Computing ρ before deployment establishes whether a covered controller can keep every cold-start step feasible under its best allocation. Choosing a coarser hypothesis can improve this ratio, but also changes the activity represented by each alert and the traffic an attacker can add to it. A deployment therefore needs both a feasibility calculation and an explicit model of who controls episode membership.

The padding theorem narrows the available remedies. Under unbounded attacker-controlled membership, substituting another alert-capable symmetric e-merger cannot provide arbitrary padding robustness. A mitigation must change membership, introduce asymmetry, or change the calibrated unit. Our evaluated alternatives illustrate the resulting tradeoffs (Table XXI). Pre-committed asymmetric weights can ignore appended flows, but leading flows defeat the tested schemes that retain the mean’s recall. Calibrating a group maximum protects against dilution, but replaces calibration flows with far fewer calibration groups; the resulting pipeline produces no alerts at the primary or secondary window. Appendices E-E and F-B give the full comparisons.

Changing the controller requires similar care. A restart with pre-committed epoch budgets satisfying $\sum_i q_i \leq q$ can retain deployment-wide FDR control under the validity premise. A fresh budget q in every epoch instead supplies a per-epoch guarantee. Appendix C-B measures the power gained under each choice. These alternatives should be evaluated with their evidence scale, decision timing, and error guarantee made explicit.

B. What the capacity experiment establishes

A membership limit changes the padding model by making some required additions infeasible. We test this boundary with a sequential attacker that pads an episode only if its final size fits a cap n ; otherwise it allows the original alert to fire. Those surviving alerts can raise the cost of later pads, so the effect depends on the whole attacked trajectory.

At $n = 100$, 96 of the 105 primary-window horizon-aware alerts remain, as do 79 of 107 in the secondary window. Some required pads exceed the cap even at cold start; the resulting early alerts raise later testing levels and make further pads too large. At $n = 1,000$, the sequential attacker suppresses both windows. Under horizon-free spending, all three primary-window alerts remain suppressible at $n \geq 100$ (Appendix E-F).

These results describe a capacity-constrained attacker. We do not truncate or rescore the complete clean and attacked

streams under an admission policy. A cap of 100 would alter 3.4% and 2.9% of benign episodes as currently formed, but those shares do not measure the effect on clean detection, false discoveries, or service. An operational membership defence therefore still requires end-to-end evaluation.

C. Validity and empirical scope

The nominal FDR interpretation of e-LOND requires valid true-null group evidence. Our sufficient metadata-conditional premise is stated in Assumption 1; the diagnostics do not establish it. The horizon and suppression calculations remain properties of the rejection rule and observed evidence regardless of that premise. Table XXXVIII records which assumptions each result uses. This distinction allows us to study structural failure without treating low observed FDP as evidence that the formal guarantee holds.

LSPR23 supplies one exercise with overlapping evaluation windows. The secondary window is a correlated robustness check, and the known-invalid stress window supports only mechanism and cost measurements. AIT adds a benign-inclusive setting across eight organisations, but does not establish broad cross-domain generality. LSPR23 also lacks reliable flow-to-campaign identifiers, so our episodes and five-minute atoms describe operational and evaluation units rather than verified incident boundaries.

Attack cost is conditional on capability. Exact per-alert and sequential sizing use oracle information. The LSPR23 joint reruns cover two windows and one seed with zero-evidence pads; a positive pad firing rate can raise their cost. AIT joint runs sample real pads, but the deployment pool is retrospective and the victim-prior pool can include traffic a compromised source cannot observe. The host-conditioned experiments cover six causal features and an already-active attacker–victim pair; they do not characterise all host-aware detectors or padding from many new peers. The detailed replay ranges and censored costs appear in Appendix E-D.

Finally, the horizon results apply to the specified controller–sequence pairs and evidence ceiling. Smoothing, restart, deferred decisions, and different evidence constructions can change those premises. Appendix G separately examines an attack on ADDIS’s evidence-conditioned index; that extension is not required for the main padding argument.

VIII. RELATED WORK

Online testing and finite evidence. Online FDR procedures allocate testing levels under different assumptions about evidence and dependence. LOND, LORD, SAFFRON, ADDIS, e-LOND, online e-BH, and e-GAI develop this framework from alpha-investing and offline FDR control [17], [18], [5], [19], [6], [20], [21], [7], [22], [23]. E-values support testing under optional continuation and arbitrary dependence [24], [16], [25]. Alpha-death and loss of power at small levels are established concerns. Huo et al. [15] study how finite conformal resolution affects online testing, and Hennhofer and Preisach [12] examine resolution collapse and smoothing in low-data batch tests. Our horizon results identify when bounded evidence makes rejection

impossible for the covered controller families and yield an attained deployment sizing law. Krönert et al. [26] derive a related calibration equality for a windowed procedure; our law concerns one uninterrupted controller. Appendix C-B also evaluates recent closure and donation constructions [27], [28].

Statistical layers for intrusion detection. Chronological evaluation follows the concerns identified by TESSERACT [29]; conformal drift systems such as Transcend(ent) [30], [31] likewise separate detector scores from statistical evidence. The closest alerting systems implement subsets of our composition. Rebjock et al. [8] apply online FDR to anomaly statistics, and C-PP-COAD [9] uses conformal p-values and active statistics. Online multi-layer FDR [10] partitions hypotheses rather than merging observations. Gao et al. [11] combine conformal novelty detection with batch FDR procedures. We study an evaluated design assembled from these established kinds of components, rather than claiming an attack on an existing deployment. Our data-quality audit also addresses the label concerns documented in intrusion datasets [32], [33].

Aggregation and adversarial testing. E-merging theory characterises symmetric and asymmetric rules at fixed arity [16], [34]. Our adversary changes the arity by adding observations to its own hypothesis. This differs from manipulating scores, perturbing p-values, or corrupting reports [35], [36], [37], [38]. Our cross-arity robustness formulation turns essential domination by the mean into a quantitative impossibility for hypotheses whose membership an attacker can enlarge.

Batching can improve online-testing power [39], although the dependence conditions of standard batch procedures need not hold for co-located flows sharing calibration. Decision deadlines offer a way to defer decisions under arbitrary dependence [40], [27]; Appendix C-B compares them with immediate and history-wide decisions. Feedback-aware procedures incorporate revealed labels with formal control [41], [42]. The analyst-delay baselines in Appendix H instead measure simpler threshold controllers and are not claims about those procedures.

IX. CONCLUSION

A statistical trust layer must be designed around the evidence and security unit it actually receives. For the controller families studied here, bounded conformal evidence creates a finite discovery horizon, and keeping a deployment cold-start feasible requires calibration to grow at least linearly with its hypothesis count. Grouping reduces that count, but also determines how an attacker can influence the evidence within a hypothesis. With unbounded attacker-controlled membership, every alert-capable symmetric e-merging family is vulnerable to sufficient zero-evidence padding.

The experiments show why this interaction matters beyond the aggregation formula. Real traffic can supply the padding, and suppressing early alerts can keep later controller levels near cold start, substantially reducing joint cost. AIT also shows that template availability and attacker knowledge constrain realised suppression. Feasibility, alert resolution, membership, and controller feedback therefore need to be evaluated together when turning detector scores into operational alerts.

The artifact is published at

https://anonymous.4open.science/r/SaTML_2027-EFD9/

and contains the complete set of code and results needed to reproduce every number and figure in this paper: a single executable notebook and optionally a runner script that carries the paper’s derivations and measurements, the library of measurement scripts it imports, the cached results of a full run, and environment details. Cached-result mode executes in about half a minute, renders every table and figure directly from the cache and requires no raw dataset. Full mode recomputes those from the original flow data; we executed both the modes end to end and verified that it reproduces the published results.

Regarding third-party data, LSPR23 is publicly available under CC-BY-4.0 (Zenodo record 8042347), and the artifact contains the exact preprocessing commands that turn the raw CSV into the pipeline’s inputs. The transfer test additionally uses AIT-LDSv2.0 [14] (Zenodo record 5789064, CC BY-NC-SA 4.0): the artifact contains the per-organisation `tstat` netflow extraction and the exact malicious/benign label mapping (Appendix B) that turn its released packet captures into the eight-organisation netflow inputs, following the dataset’s own recipe. Full-mode reproduction therefore requires downloading both records. The anonymised repository is kept accessible and unedited for the duration of review and, if accepted, will be deposited on Zenodo by the venue’s final-artifact deadline.

LLM USAGE CONSIDERATIONS

Hosted LLMs were accessed through Claude Code for editorial refinement, code generation, evaluation support, and read-only code audits. The LLMs were used for editorial purposes in this manuscript and the authors inspected all outputs to ensure precision and originality. They are not part of the proposed method, the released pipeline’s operation, or the dataset labeling. The released artifact contains the exact code used to produce the reported results. The authors take responsibility for all content and any remaining errors.

Necessity, model choice and footprint. We have used `claude-opus-5` and `claude-sonnet-5` via Claude code. The use of LLM was motivated by the volume of automation, evaluation and consistency-checking code the project required within its timeframe. We limited query volume by working in long sessions that batch related requests, by reusing generated code and the cached result objects rather than regenerating them, by running every consistency checks locally as a script rather than by querying a model.

Provider-side energy measurements for the assistants were not available to us, so we do not estimate their emissions. The reported experiments themselves are CPU-only scikit-learn computations on a single workstation, a MacBook Pro with an Apple M5 Pro system-on-chip (18 CPU cores, 48 GB unified memory, no discrete GPU; the pipeline uses no GPU): a full recomputation from the flow data takes about two to three hours and the cached-result reproduction about half a minute.

ETHICAL CONSIDERATIONS

This paper describes working attacks against a statistical trust layer for intrusion detection, so we state our reasoning against the Menlo Report framing [43].

Respect for persons and justice. All measurements are made on public research datasets released for this purpose, captured during authorised cyber-defence exercises on purpose-built ranges rather than on production networks carrying third-party traffic. We rely on the publishers’ own ethical review and release terms and did not conduct a separate consent audit, so we make no claim about individual consent beyond what those releases state. No attack was executed against any live network, and no personal data is used, disclosed or re-identified.

Beneficence and responsible disclosure. This work does not target a deployed product, and the class of system it studies is, to our knowledge, not yet widely deployed. It introduces no new attack on the underlying detector: it identifies a security consequence that emerges when public statistical components are composed — bounded evidence, attacker-influenceable grouping and online error spending together produce an alert layer that can become infeasible or suppressible. We report both the inexpensive and the expensive attack regimes and evaluate representative mitigations, so that designers can assess the exposure before deploying such a composition.

Law and public interest. We identify a small set of host pairs whose labels we believe to be wrong and deliberately do not publish corrected labels. We report the direction of the error instead.

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TABLE IV

THE CONTRIBUTION BOUNDARY: INHERITED MATHEMATICS, OUR SECURITY FORMALISATION, AND THE RESULTING SYSTEM-LEVEL CONSEQUENCES.

Inherited result	Our formalisation $\rightarrow$ consequence
Finite conformal resolution [3], [12], [15]; alpha-death and low power at small levels [17], [6]	<i>Formalise</i> : cold-start feasibility for specified controller–sequence pairs. <i>Derive</i> : an absorbing horizon for two families and an attained ex-ante sizing law (Theorems 1 and 2 and Corollary 1).
Fixed-arity domination of symmetric e-merging rules [16]	<i>Formalise</i> : cross-arity robustness when an attacker enlarges its own hypothesis (Definition 1). <i>Derive</i> : a class-wide impossibility and the mean’s exact cost bound (Theorem 3).
Batching can recover online-testing power [39]; zero inputs dilute an arithmetic mean	<i>Formalise</i> : grouping, attacker-controlled membership, and controller state jointly. <i>Derive</i> : one ratio for best-case feasibility and universal integer dilution under Corollary 2, plus the joint-cost feedback in Section VI-B.

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APPENDIX A

PROOFS AND EXACT SCOPE

Question answered. What exactly do the body’s core formal results assume, and what does the FDR reading of an empirical e-LOND run assume on top of them? *Takeaway.* The feasibility theorems need only a finite evidence ceiling and the stated level form; the padding impossibility needs only the symmetric e-merging class; the group-validity premise below is needed by neither, and only by the nominal guarantee.

The body states Theorems 1 to 3 and Corollaries 1 and 2. This appendix adds the group-validity premise, the exact scope of each body result, and five further results the body cites; all proofs follow, verified numerically in the released artifact.

A. The group-validity premise

Using the mean at variable, metadata-selected arity needs a group-level premise, and different controllers need different things of it: e-LOND and online e-BH control FDR under

arbitrary dependence given only *marginally* valid group e-values, whereas the p-value procedures need conditional super-uniformity, independence or monotonicity. The guarantee-bearing claims therefore use e-LOND.

The conditioning is stated *per hypothesis*. For group j let $\mathcal{M}_j$ be the σ -field generated by the pre-committed metadata that determines that hypothesis and nothing else: the grouping key, the membership indicators and arity m_j , the group’s position in the pre-committed within-bucket order (hence the spending weight γ_j it is offered), and the horizon T . It carries no information about any other group’s composition. Pre-committed means chosen before the scores and not score-adaptive.

Assumption 1 (Group-local metadata-conditional validity). *For every benign flow i of a true-null group G_j , $\mathbb{E}[e_i \mid \mathcal{M}_j] \leq 1$; for the two-point e-value, equivalently $\Pr(K_i \leq k \mid \mathcal{M}_j) \leq k/(|\mathcal{C}| + 1)$.*

Lemma 1 (Group-level marginal e-validity). *Under Assumption 1, every true-null group with $m_j > 0$ has a marginally valid group e-value, $\mathbb{E}[\text{Ev}(G_j)] \leq 1$, for random membership and arity and under arbitrary within- and between-group dependence.*

Conditioning on $\mathcal{M}_j$ fixes membership and m_j , so linearity gives $\mathbb{E}[\text{Ev}(G_j) \mid \mathcal{M}_j] \leq 1$ and the tower property removes the conditioning (proof below). The premise is stronger than the unconditional validity split conformal supplies, it is a genuine modelling assumption (endpoint and service metadata can correlate with the timing and volume features that drive the score), and it is *not* established by the dataset. Table XXXVIII separates every claim by what it needs.

A construction in which the premise holds and the attack still succeeds. The coexistence the paper claims—valid true-null evidence and a suppressible alert in the same run—can be exhibited without data. Draw $|\mathcal{C}| = 99$ calibration scores and the benign test scores independently from $\text{Uniform}(0, 1)$, take $k = 1$, so $M = 100$, and fix the groups and the within-bucket order in advance, independently of every score. Run e-LOND under the horizon-uniform allocation with $T = 4$ hypotheses and $\alpha = 0.05$: the cold-start level is $\alpha/T = 0.0125$ and the threshold is 80. Place a singleton attack flow with score 2 first. It exceeds every calibration score, so $K = 1$, $e = 100 \geq 80$, and it is rejected. Now let the attacker append one flow of its own with score -1 to that episode: the flow ranks below every calibration score, $K = 100 > k$, its e-value is 0, and the group mean becomes $50 < 80$, so the alert is suppressed; nothing else in the stream has changed. Assumption 1 holds by construction before and after the pad. Every true-null group is untouched by the attack, its membership, arity and position were fixed before the scores were drawn, so conditioning on $\mathcal{M}_j$ conditions on nothing that depends on the scores, and each benign score is exchangeable with the calibration scores, giving $\Pr(K_i \leq 1 \mid \mathcal{M}_j) = 1/(|\mathcal{C}| + 1)$ and $\mathbb{E}[e_i \mid \mathcal{M}_j] = 1$ exactly. By Lemma 1 every true-null group e-value is marginally valid and the nominal FDR guarantee is intact, while the one alert

the guarantee we meant to deliver has been removed by a single flow. The validity of the nulls and the suppressibility of the alert are properties of different objects, and neither constrains the other; Section V measures the same separation on real streams.

B. Exact scope of the finite-horizon theorems

Theorem 1 covers LOND, e-LOND and e-LORD, the last through the equivalence $\text{e-LORD} = \text{e-LOND}$ under $\gamma_t = \omega_t \prod_{j < t} (1 - \omega_j)$ [23]. That equivalence is a hypothesis and not a remark, and it needs two things of the weights, not one. They must be *deterministic functions of the index*, fixed before deployment — predictability in the filtration, which is all online FDR normally asks, would make γ a history-dependent random object, and Theorem 1’s condition is a statement about a sequence. And they must satisfy $\omega_j \in [0, 1]$: pre-commitment alone gives fixedness only, and a pre-committed $\omega_1 = 2$ yields $\gamma_1 = 2$ and $\gamma_2 < 0$, breaking both $\gamma \geq 0$ and the budget. Under $\omega_j \in [0, 1]$ the partial sums telescope, $\sum_{t \leq n} \gamma_t = 1 - \prod_{j \leq n} (1 - \omega_j) \leq 1$, which gives Theorem 1 the non-negativity and the budget. It does not give the decay condition, and the sparse sequence of the next paragraph is itself induced by a deterministic $\omega \in [0, 1]$, so $\gamma_t t^d \rightarrow 0$ has to be checked for the induced γ , as it is for the $\gamma_j \propto j^{-1.6}$ pair we run.

Theorem 2 covers LORD++ ($a_1 = \alpha - w_0$, $a_j = \alpha$ for $j \geq 2$). What is covered is a procedure *together with* its spending sequence, not a named procedure in the abstract: summability alone is not enough for Theorem 1 at $d = 1$, since $\gamma_{2^n} = n^{-2}/\zeta(2)$ for $n \geq 1$, with $\gamma_t = 0$ elsewhere, sums to exactly one yet has $2^n \gamma_{2^n} \rightarrow \infty$. That example refutes the *hypothesis*, not the *conclusion*: the same sequence has $\gamma_t \rightarrow 0$ and so is still absorbed at $d = 0$. The decay condition therefore has to be checked per pair, and it is asymptotic: for $\gamma_j \propto j^{-1.6}$, $t\gamma_t \rightarrow 0$ so $d = 1$ holds, whereas a level growing quadratically in the rejection count would escape the argument. A horizon-uniform γ is supported on $t \leq T$, so an asymptotic decay condition is met trivially and decides nothing; it needs none, because before any rejection the level is the constant c_0/T across the horizon, feasibility there is all-or-nothing, and Corollary 1 decides it exactly.

The two families are not equally strong, and the difference is worth stating. Theorem 1’s bound $\alpha_t \leq \alpha c_g \gamma_t t^d$ uses only $R_{t-1} \leq t - 1$, so it holds whatever the history: the feasible set is a fixed set of indices and absorption does not depend on the run being rejection-free. Theorem 2’s bound is driven by the length Δ of the *current* rejection-free gap, so its absorption is conditional on such a gap occurring. A lag-sum procedure that keeps rejecting keeps its level up — for LORD++ under $\gamma_j \propto j^{-1.6}$ with a rejection at every step, α_t climbs from its cold-start value towards 0.039, and every step of that run is feasible once $|\mathcal{C}| \geq 91$, the binding step being the cold start itself. This is why the paper’s claims are cold-start claims: the horizon is the horizon of a rejection-free prefix, which is the state a deployment starts in and, on these streams, mostly stays in. The theorems are scoped deliberately: they do not

say every online FDR procedure dies, and Appendix C locates the procedures that escape.

C. Attainment and the symmetric class

Every rejection threshold in our setting is $\theta = 1/\alpha_t > 1$, and attainment is the hypothesis the deployment supplies whenever an alert is possible at all: if a symmetric family ever issues an alert then some input it receives has $F \geq \theta$, so it attains θ and Theorem 3 applies to it. *Not* being θ -padding-robust and attaining θ are therefore the same condition, since a family reaching θ nowhere — $F \equiv 1$, which is padding-invariant but can never fire, because rejection needs evidence $\geq 1/\alpha_t > 1$ — satisfies Definition 1 vacuously. For the mean rule the body evaluates the witness is explicit: the all-firing group $x = (M, \dots, M)$ attains θ at every evidence-feasible step (6). The implication runs only that way. Attainment is strictly the weaker condition, since it quantifies over all of $[0, \infty)^m$ — a scaled rule such as $\frac{1}{2}$ mean is a valid symmetric e-merging family that attains θ somewhere yet, on two-point evidence with $\theta > M/2$, can never fire — so what attainment excludes is only the rules that reach θ nowhere, which is all the theorem needs. Ordinary domination by the mean is false ($\kappa + (1 - \kappa)$ mean, for $\kappa \in (0, 1)$, is a counterexample), which is why the proof splits cases.

D. Further results used in the body

Proposition 1 (Donation is bounded before the first rejection). *For donation e-LOND with parameter $\delta \in (0, 1)$ and a spending sequence with $\gamma \geq 0$ and $\sum_t \gamma_t \leq 1$ — the normalisation the bound relies on — on a rejection-free prefix $\alpha_t \leq \delta \gamma_t / (1 - \delta)$. The restriction $\delta < 1$ is necessary, not cosmetic: at $\delta = 1$ the donated wealth can drive the denominator to zero and hold α_t away from zero along a rejection-free run, so neither the bound nor absorption survives (Appendix A). Hence donation e-LOND satisfies Theorem 1’s hypotheses along a rejection-free prefix, with $c_g = \delta / [\alpha(1 - \delta)]$ — which is $1/(1 - \delta)$ exactly when $\delta = \alpha$, the setting we run, since Theorem 1 writes the level as $\alpha \gamma_t g$ — and $d = 0$, because that bound holds uniformly in t — g itself is not constant along the prefix, since the donated wealth grows, but $d = 0$ asks for a constant bound, not a constant g . The instantiation matters: $d = 0$ asks only for $\gamma_t \rightarrow 0$, which summability supplies, whereas $d = 1$ would ask for $t\gamma_t \rightarrow 0$, which it does not (take $\gamma_{2^j} = 2^{-j}$ and $\gamma_t = 0$ elsewhere). That is all Theorem 1 needs, since its conclusion is itself a statement about rejection-free runs — so the horizon stays finite and the run is absorbed. It is not a global member of the family: past $1/\delta$ rejections no finite $c_g(R + 1)^d$ bounds the level. The bound is a cold-start statement only — past $1/\delta$ rejections the denominator can vanish — and it is sharp where it applies. On our streams the realised donation wealth stays at most 0.12 of a budget of 1.*

Proposition 2 (Closure is bounded by its zero-evidence subset). *For closed e-LOND whose subset aggregator is zero on all-zero input — $E_S = 0$ whenever $E_i = 0$ for every $i \in S$, including $S = \emptyset$ — let $Z_t = \#\{i < t : E_i = 0\}$. Then $\alpha_t \leq \delta \gamma_{Z_t+1} (R_{t-1} + 1)$. The hypothesis is not implied by*

e-merging validity: $\kappa + (1 - \kappa)$ mean is symmetric and valid yet returns $\kappa > 0$ on all-zero input, and the zero-evidence subset then certifies nothing.

Corollary 3 (When the closure is absorbing). *Let $H = \min\{j : \gamma_j < 1/(\delta M)\}$ be the first index at which the spending weight dips below the floor — Theorem 1’s index horizon when γ is decreasing, as ours is — and $P_T = \#\{i \leq T : E_i > 0\}$ the number of hypotheses carrying any evidence. (i) Along a rejection-free run, closed e-LOND is absorbed by step $H + P_T$ whenever $H + P_T \leq T$: an additive offset, holding pathwise with no assumption beyond counting. (ii) Asymptotically, if $\liminf_t Z_t/t > 0$ then for every $\rho' < \liminf_t Z_t/t$ there is an onset t_1 beyond which the closure’s effective spending index is at least $\rho't$, so absorption occurs by $\max\{t_1, \lceil (H - 1)/\rho' \rceil\}$ — the index is rescaled by ρ' and the absorption time therefore stretched by $1/\rho'$, not shortened. A positive $\liminf$ permits an arbitrarily long initial transient, which is why t_1 cannot be dropped. On our streams $P_T \in \{0, \dots, 152\}$, at most 0.48% of T , so form (i) applies and the offset is at most 152 hypotheses.*

Theorem 4 (Reach). *Fix one hypothesis, offered level α_t , whose episode has flow positions i . For a pre-committed weight sequence with $\sum_i w_i \leq 1$, the number P of positions at which a lone attack flow can fire the rule $F(e) = \sum_i w_i e_i$ is at most $\alpha_t M$, the quantity that governs Corollary 1.*

Theorem 5 (Front-load cost). *At that same hypothesis, detection is impossible once $L \geq L^*(w) := \min\{L : \sum_{i>L} w_i < \beta\}$, with $\beta = 1/(\alpha_t M)$; L^* is finite for every summable w . The threshold is sufficient for impossibility, and exact only when every non-prepended position carries firing evidence: with fewer firing flows detection can already be impossible at smaller L , so L^* is an upper bound on what the attacker must prepend.*

E. Group-level marginal e-validity (Lemma 1)

Proof. For a tested group G_j , recall $\mathcal{M}_j$ is the σ -field generated by the pre-committed metadata that determines *this* hypothesis: the grouping key, the membership indicators $1\{i \in G_j\}$, the arity m_j , the group’s position in the pre-committed stream order (and hence the spending weight γ_j it is offered), and the horizon T . It carries no information about any other group’s composition or arity, and — as the scope note below records — none about the realised calibration scores. Fix a true-null group G_j (all constituent flows benign) with $m_j > 0$. By linearity of conditional expectation and $\mathcal{M}_j$ -measurability of the membership indicators and of m_j ,

$$\begin{aligned} \mathbb{E}[\text{Ev}(G_j) \mid \mathcal{M}_j] &= \mathbb{E}\left[\frac{1}{m_j} \sum_{i \in G_j} e_i \mid \mathcal{M}_j\right] \\ &= \frac{1}{m_j} \sum_{i \in G_j} \mathbb{E}[e_i \mid \mathcal{M}_j] \leq 1, \end{aligned}$$

the inequality by Assumption 1. Taking expectations and applying the tower property gives $\mathbb{E}[\text{Ev}(G_j)] = \mathbb{E}[\mathbb{E}[\text{Ev}(G_j) \mid \mathcal{M}_j]] \leq 1$. No independence or exchangeability *among* the $\{e_i\}$ is used, so within-group dependence is unrestricted; and because e-LOND and online e-BH control FDR under arbitrary dependence given marginally valid per-hypothesis e-values,

arbitrary between-group dependence is permitted as well. An empty group ($m_j = 0$) tests no hypothesis; assigning it the trivial $\text{Ev} = 1$ is harmless. For the two-point conformal e-value $e_i = M 1\{K_i \leq k\}$ the premise $\mathbb{E}[e_i \mid \mathcal{M}_j] \leq 1$ is exactly $\Pr(K_i \leq k \mid \mathcal{M}_j) \leq k/(|C| + 1)$, since $M = (|C| + 1)/k$. $\square$

Why the conditioning is per hypothesis. A single global metadata σ -field $\mathcal{M}$ will not support the attack analysis of Section V. $\mathbb{E}[e_i \mid \mathcal{M}] \leq 1$ does *not* imply $\mathbb{E}[e_i \mid \mathcal{M}'] \leq 1$ for a richer $\mathcal{M}'$ — take $e = 21_A$ with $\Pr(A) = \frac{1}{2}$: against the trivial σ -field the conditional expectation is 1, against $\sigma(A)$ it is 2 on A — so an adversary who enriches the stream’s metadata anywhere would, on the global reading, void the premise even for groups it never touched. With $\mathcal{M}_j$ the statement is local and survives: padding adds flows carrying the attacked group’s own key, so it changes no other group’s key, membership or arity, and adds no hypothesis, leaving T fixed. Provided the within-bucket order is a function of group metadata alone (the deterministic hashed-key canonical order of Appendix D-A, with collisions broken on the key itself), every true-null group’s position and γ_j are likewise fixed, so $\mathcal{M}_j$ is literally the same σ -field before and after the attack. Under an arrival-ordered tie-break this fails: a pad placed before the attacked group’s first flow moves that group within its bucket and can shift a true-null group’s index, hence γ_j , hence $\mathcal{M}_j$. Creating a *new* host pair is likewise outside the padding claim — it adds a hypothesis and changes T , which the horizon-uniform $\gamma_j = 1/T$ reads.

Scope. The lemma certifies the *e-value* controllers (e-LOND, online e-BH) only. The p-value procedures carry their own conditions — conditional super-uniformity for mFDR, and independence with monotonicity for FDR — which this argument does not supply, and which Section III-C shows ADDIS violates on this stream. The premise $\mathbb{E}[e_i \mid \mathcal{M}_j] \leq 1$ is marginal over the calibration draw and test score; it is *not* validity conditional on the realised calibration scores, which are deliberately excluded from $\mathcal{M}_j$ (the Bates adjustment supplies that stronger, calibration-conditional guarantee at a power cost and is not required here). $\mathcal{M}_j$ is also *smaller* than a global metadata σ -field, so Assumption 1 is the weaker of the two conditional premises, while remaining stronger than unconditional $\mathbb{E}[e_i] \leq 1$.

F. Feasibility (Theorems 1 and 2, Corollary 1)

Theorem 1. With $\alpha_t = \alpha \gamma_t g(\mathcal{H}_{t-1})$, $g \leq c_g(R_{t-1} + 1)^d$, and $R_{t-1} \leq t - 1$, we have $\alpha_t \leq \alpha c_g \gamma_t t^d$. A rejection at t needs $\alpha_t \geq 1/M$, hence $\gamma_t t^d \geq 1/(\alpha c_g M)$. If $\gamma_t t^d \rightarrow 0$ then only finitely many t satisfy that inequality, since a sequence tending to zero exceeds a fixed positive constant finitely often. A finite set of indices has a maximum, so there is a last feasible index t^* (or none at all), and every $t > t^*$ fails the necessary condition: the run is absorbed from $t^* + 1$. No monotonicity of γ is needed. Requiring permanence from the *first* infeasible step would be a stronger property — false for non-monotone γ , since feasibility can fail once and return — and it is not what Section III defines. $\square$

Theorem 2. Let $\alpha_t = w_0\gamma_t + \sum_j a_j\gamma_{t-\tau_j}$ with $w_0 \geq 0$ and $0 \leq a_j \leq \alpha$ over rejection times $\tau_j < t$. During a rejection-free run of length Δ the last rejection is at least Δ before t , so every lag $t - \tau_j \geq \Delta$ and the lags are distinct. With $\Gamma(\Delta) = \sum_{u \geq \Delta} \gamma_u$ we have $\gamma_{t-\tau_j} \leq \Gamma(\Delta)$ termwise and $\gamma_t \leq \Gamma(t) \leq \Gamma(\Delta)$, so, using $a_j \leq \alpha$ and the distinctness of the lags, $\alpha_t \leq w_0\Gamma(\Delta) + \alpha\Gamma(\Delta) = (w_0 + \alpha)\Gamma(\Delta)$; the first term is where $w_0 \geq 0$ is used, since $\gamma_t \leq \Gamma(\Delta)$ may only be multiplied through by a non-negative w_0 . Since Γ is non-increasing and $\sum \gamma < \infty$ forces $\Gamma(\Delta) \rightarrow 0$, the quantity $\Delta^* = \min\{\Delta : (w_0 + \alpha)\Gamma(\Delta) < 1/M\}$ is finite; Δ only grows during a rejection-free run, so the state is absorbing. No monotonicity of γ is used. $\square$

Corollary 1 and max-min optimality of $\gamma_t = 1/T$. Before any rejection, family I has $\alpha_t = \alpha\gamma_t$ and family II has $\alpha_t = w_0\gamma_t$; write the cold-start coefficient $c_0 \in \{w_0, \alpha\}$, so $\alpha_T = c_0\gamma_T$. Feasibility at T requires $\alpha_T \geq 1/M = k/(|\mathcal{C}|+1)$, i.e. $|\mathcal{C}| \geq k/\alpha_T - 1 = k/(c_0\gamma_T) - 1$. Under $\gamma_T = 1/T$ this is $|\mathcal{C}| \geq kT/c_0 - 1$; $kT/w_0 - 1$ for a level- w_0 procedure such as LORD++, and $kT/\alpha - 1$ for LOND/e-LOND. For optimality, a cold-start rejection is possible at every $t \leq T$ iff $\min_{t \leq T} \gamma_t \geq k/((|\mathcal{C}|+1)c_0)$; among sequences with $\sum_{t \leq T} \gamma_t \leq 1$ the value $\min_{t \leq T} \gamma_t$ is maximised, at $1/T$, by the uniform allocation: $\min_{t \leq T} \gamma_t \leq T^{-1} \sum_{t \leq T} \gamma_t \leq 1/T$, since a minimum is at most an average, and $\gamma_t = 1/T$ attains it. Constraining the mass over all t rather than over $t \leq T$ changes nothing, since $\sum_t \gamma_t \leq 1$ implies $\sum_{t \leq T} \gamma_t \leq 1$, and the attaining sequence satisfies the global constraint too. Hence $\gamma_t = 1/T$ maximises the feasible horizon. That same criterion supplies the corollary's *sufficiency*, which is otherwise easy to miss: under $\gamma_t = 1/T$ it reads $1/T \geq k/((|\mathcal{C}|+1)c_0)$, i.e. $|\mathcal{C}| \geq kT/c_0 - 1$ once more, so this one inequality is not merely necessary at step T but sufficient for feasibility at every $t \leq T$. The two directions coincide here only because a cold-start uniform level is constant over the horizon; for a general γ the necessary condition at T is strictly weaker than feasibility throughout, which is why the corollary states the equivalence for $\gamma_t = 1/T$ alone. $\square$

G. The UAI 2026 procedures (Propositions 1 and 2)

Proposition 1. Donation e-LOND sets $\alpha_t = \delta\gamma_t(|R_{t-1}| + 1)/(1 - (\delta(|R_{t-1}| + 1)\bar{W}_t \wedge 1))$ with $\bar{W}_t = \sum_{i \in R_{t-1}} \gamma_i((E_i - \frac{1}{\delta\gamma_i(|R_{t-1}|+1)}) \wedge 1) + \sum_{i \notin R_{t-1}} \gamma_i(E_i \wedge 1)$, with both sums running over $i < t$. Every summand is γ_i times a quantity capped at 1, so $\bar{W}_t \leq \sum_{i < t} \gamma_i \leq 1$ *always*. On a rejection-free prefix $R_{t-1} = \emptyset$, so for $\delta < 1$ the denominator is at least $1 - \delta > 0$ and $\alpha_t \leq \delta\gamma_t/(1 - \delta)$, which is Theorem 1 with $c_g = \delta/[\alpha(1 - \delta)]$ and $d = 0$: Theorem 1 writes the level as $\alpha\gamma_t g(\mathcal{H}_{t-1})$, so the constant must absorb the ratio δ/α , and it reduces to $1/(1 - \delta)$ exactly in the $\delta = \alpha$ setting we run. $d = 0$ is the right instantiation, not merely a valid one: on a rejection-free prefix $R_{t-1} = \emptyset$, so the displayed bound $g \leq c_g$ holds uniformly in t — g itself rises with $\bar{W}_t$ and is not constant — and Theorem 1's conclusion then needs only $\gamma_t \rightarrow 0$, which $\sum_t \gamma_t \leq 1$ gives. Taking $d = 1$ instead

would demand $t\gamma_t \rightarrow 0$, which summability does *not* imply ($\gamma_{2^j} = 2^{-j}$, zero elsewhere, sums to one with $t\gamma_t = 1$ infinitely often).

Why $\delta < 1$ is required. At $\delta = 1$ the denominator is $1 - \bar{W}_t$, which the donated wealth drives to zero, and the conclusion fails outright. Take $\delta = 1$, $\gamma_t = 2^{-t}$ and $E_t = 1$ for every t . Then $\bar{W}_t = \sum_{i < t} \gamma_i(E_i \wedge 1) = \sum_{i < t} \gamma_i = 1 - 2^{-(t-1)}$ and $\alpha_t = \gamma_t/(1 - \bar{W}_t) = 2^{-t}/2^{-(t-1)} = \frac{1}{2}$ for *every* t . The run is rejection-free because rejection needs $E_t \geq 1/\alpha_t = 2$ and $E_t = 1$ — this is why the evidence is pinned at exactly 1 rather than merely bounded below by it — yet the offered level never decays, so with any $M \geq 2$ every step stays evidence-feasible and the run is never absorbed. The proposition therefore quantifies over $\delta \in (0, 1)$; our runs use $\delta = \alpha = 0.05$, where $1/(1 - \delta) = 1.05$.

The restriction to the rejection-free prefix is not cosmetic. With $R = |R_{t-1}|$ the denominator is $1 - (\delta(R + 1)\bar{W}_t \wedge 1)$, which vanishes once $\delta(R + 1)\bar{W}_t \geq 1$; since $\bar{W}_t$ can approach 1, that is reachable at $R + 1 \geq 1/\delta$. The bound is therefore a cold-start statement, which is exactly the state Corollary 1 concerns. It is also sharp: taking $E_i \geq 1$ for every i makes $\bar{W}_t = \sum \gamma_i \rightarrow 1$ and the bound tight. $\square$

Proposition 2. Closed e-LOND sets $\alpha_t = \min_{S \subseteq [t-1]: D_t(S) > 0} \delta\gamma_{|S|+1}(|R_{t-1}| + 1)/D_t(S)$ with $D_t(S) = 1 + |S \cap R_{t-1}| - \delta E_S(|R_{t-1}| + 1)$, where E_S aggregates the evidence on S . Because α_t is a *minimum* over admissible subsets, any single admissible S furnishes an upper bound.

Take $S = \{i < t : E_i = 0\}$, of size Z_t . A hypothesis carrying zero evidence is never rejected, so $S \cap R_{t-1} = \emptyset$; and $E_S = 0$ because the aggregator is zero on all-zero input, including the empty subset that arises when $Z_t = 0$. Hence $D_t(S) = 1 > 0$, S is admissible, and $\alpha_t \leq \delta\gamma_{Z_t+1}(|R_{t-1}| + 1)$.

This is the decay of Theorem 1 with the time index rescaled from t to $Z_t + 1$. The bound itself is unconditional and pathwise; the *conclusion* that the state is absorbing is not, and Corollary 3 supplies its hypothesis. $\square$

Corollary 3. By Proposition 2, $\alpha_t \leq \delta\gamma_{Z_t+1}(|R_{t-1}| + 1)$ along any run. On a rejection-free run $|R_{t-1}| = 0$, so $\alpha_t \leq \delta\gamma_{Z_t+1}$. If $\gamma_{Z_t+1} \rightarrow 0$ then $\alpha_t \rightarrow 0$, and since the evidence is bounded above by M (Section III) the offered level falls below $1/M$ after finitely many steps and no rejection is possible thereafter: the state is absorbing, which is Theorem 1's conclusion.

Form (i), exact and finite-horizon. Along the rejection-free run $|R_{t-1}| = 0$, so $\alpha_t \leq \delta\gamma_{Z_t+1}$. Our $\gamma_j \propto j^{-1.6}$ is decreasing, so no rejection is possible once $\gamma_{Z_t+1} < 1/(\delta M)$, i.e. once $Z_t \geq H - 1$ for $H = \min\{j : \gamma_j < 1/(\delta M)\}$, which is exactly Theorem 1's index horizon. Write $P_T = \#\{i \leq T : E_i > 0\}$. Of the $t - 1$ hypotheses before step t , at most P_T carry positive evidence, so $Z_t \geq t - 1 - P_T$ for every $t \leq T$ — a counting statement, with no assumption on the stream. Hence $Z_t \geq H - 1$ as soon as $t \geq H + P_T$, and

$$t_{\text{closed}} := \inf\{t : Z_t \geq H - 1\} \leq H + P_T.$$

The closure therefore reaches the absorbing state within an *additive* P_T steps of the horizon Theorem 1 gives, whenever $H + P_T \leq T$. Nothing is asymptotic and no density is assumed. For a non-monotone γ the same conclusion follows without the decreasing step, because α_t is a *minimum* over admissible subsets: any $H - 1$ zero-evidence indices furnish an admissible S and hence the same bound.

Form (ii), asymptotic. Suppose $\liminf_t Z_t/t = \rho > 0$ and fix any $\rho' < \rho$. By the definition of the $\liminf$ there is a t_1 with $Z_t \geq \rho't$ for all $t \geq t_1$; so $Z_t \rightarrow \infty$ and, $\gamma_j \propto j^{-1.6}$ being decreasing to 0, $\gamma_{Z_t+1} \leq \gamma_{\lceil \rho't \rceil} \rightarrow 0$. Substituting into Theorem 1's discovery-horizon computation replaces the index t by $\rho't$: the effective spending index is rescaled by ρ' , so absorption is reached once both $t \geq t_1$ and $\rho't \geq H - 1$, i.e. by $\max\{t_1, \lceil (H - 1)/\rho' \rceil\}$. The onset t_1 cannot be dropped: a positive $\liminf$ constrains only the eventual density and permits an arbitrarily long initial transient. The *time* to absorption is therefore stretched by $1/\rho'$, not shortened — a sparser zero-evidence stream buys the closure more steps, not fewer. This holds for every $\rho' < \rho$ and *not*, from this argument, at ρ itself: the $\liminf$ gives eventual domination below its value, not at it.

On our streams form (i) is the operative one and needs no density at all. The number of hypotheses carrying *any* evidence is $P_T \in \{0, \dots, 152\}$ over the ten window-seed cells, at most $\hat{p} = 4.82 \times 10^{-3}$ of T , attained at position 0.85, and $P_T = 0$ at 0.70 seed 1; so the horizon offset is at most 152 hypotheses and the cap is attained (ratio 1.000000 over every configuration measured). The argument is specific to *sparse* evidence: a stream in which most hypotheses carry positive evidence has P_T close to T , Z_t small, and this bound says little — which is the regime the closure was designed for, and not a security stream. $\square$

H. Symmetric padding impossibility (Theorem 3)

We give a full proof (Route A) and an independent lemma (Route B) that needs no domination result. Route B's scope is narrow and worth stating exactly: it shows that a symmetric family maps any permutation of a mean-one two-point vector to at most 1. That yields the impossibility only for families whose *unpadded* value on that vector already reaches θ — the arithmetic mean among them — and not for the class. A symmetric valid family such as $F_N(e) = \text{mean}(e) \mathbf{1}\{\text{not all } e_i \text{ equal}\}$ attains θ elsewhere yet returns 0 on Route B's unpadded vector, so Route B produces no violation for it and Route A must. The class-wide statement and the quantitative bound are Route A's alone.

Route A (essential domination). Vovk and Wang's Proposition 3.1 [16] states that the arithmetic mean M_K *essentially* dominates every symmetric e-merging function: $F(e) > 1 \Rightarrow M_K(e) \geq F(e)$. Ordinary domination is false ($\kappa + (1 - \kappa)M_K$, for any $\kappa \in (0, 1)$, is a symmetric valid counterexample), so we use it through a case split: for finite x , either $F_N(x) \leq 1$, or $F_N(x) > 1$ and then $M_K(x) \geq F_N(x)$; in both cases $F_N(x) \leq \max(1, \text{mean}(x))$.

We need that inequality at $(x, 0^r)$, a boundary point of $[0, \infty)^N$, so we prove it directly rather than rely on the domain

of the cited proposition. Let $y \in [0, \infty)^N$, $\bar{y} = \text{mean}(y)$, and let Π be a uniformly random permutation of y ; each coordinate of Π has mean $\bar{y}$, and $F_N(\Pi) \equiv F_N(y)$ by symmetry. If $\bar{y} \leq 1$ then Π is an admissible input, $F_N(\Pi)$ is a constant e-variable, so $F_N(y) \leq 1$. If $\bar{y} > 1$, take $E = \Pi$ with probability $1/\bar{y}$ and $E = 0$ otherwise; every coordinate of E then has mean exactly 1, so validity gives $F_N(y)/\bar{y} + (1 - 1/\bar{y})F_N(0) \leq 1$, and $F_N(0) \geq 0$ yields $F_N(y) \leq \bar{y}$. Either way $F_N(y) \leq \max(1, \bar{y})$ on all of $[0, \infty)^N$, zeros included, which is what the padding step uses. This also removes any need for monotonicity or measurability assumptions on F_N . Padding $x \in [0, \infty)^m$ with r zeros gives $\text{mean}(x, 0^r) = (\sum_i x_i)/(m + r)$, so

$$F_{m+r}(x, 0^r) \leq \max\left(1, \frac{\sum_i x_i}{m+r}\right) < \theta \text{ once } r > \frac{\sum_i x_i}{\theta} - m.$$

Attainment supplies an x with $F_m(x) \geq \theta$ to which this applies, so no attaining symmetric family is θ -padding-robust for any $\theta > 1$. $\square$

Route B (a negatively-dependent special case). Fix $\theta > 1$ and m . Draw a uniformly random size- m subset $S \subseteq [N]$ and set $e_i = (N/m) \mathbf{1}\{i \in S\}$. Then $\mathbb{E}[e_i] = (N/m) \Pr(i \in S) = 1$, so each e_i is an e-variable, and the vector is a permutation of $((N/m) \mathbf{1}_m, 0^{N-m})$. A symmetric F_N maps every realisation to the same constant, and a constant e-variable is ≤ 1 . Hence $F_N((N/m) \mathbf{1}_m, 0^{N-m}) \leq 1 < \theta$ for all $N \geq m$: appending $N - m$ zeros to the m firing positions drives any symmetric rule below θ . $\square$

Tightness and attainment. The bound is tight: the mean itself has $F_{m+r}(x, 0^r) = (\sum x)/(m + r) \geq \theta$ exactly while $r \leq (\sum x)/\theta - m$. Attainment (Section V) is necessary, not cosmetic: the constant rule $F \equiv 1$ is symmetric, valid and padding-invariant but never reaches any $\theta > 1$, satisfying θ -padding-robustness vacuously; since rejection needs evidence $\geq 1/\alpha_t > 1$, restricting to attaining families excludes exactly the rules that cannot fire.

I. Feasibility ratio as a dilution threshold (Corollary 2)

Corollary 2. Under $\gamma_t = 1/T$ the level is $\alpha_t = (\alpha/T)(R_{t-1} + 1)$, so on a rejection-free prefix $\alpha_t = \alpha/T$ and (4) rejects G_t exactly when $\text{Ev}(G_t) \geq T/\alpha$; the rule is inclusive. *Sufficiency.* Let $c > \rho$ be an integer; we show by induction on t that no rejection occurs. Suppose none has occurred before t , so $R_{t-1} = 0$ and the level is α/T . If G_t is not an attacker episode, $\text{Ev}(G_t) < T/\alpha$ by hypothesis. If it is, with unpadded sum S_t and arity m_t , each of its flow e-values is at most M by (2), so $S_t \leq Mm_t$; its padded arity is the integer cm_t and the padded mean is $S_t/(cm_t) \leq M/c < M/\rho = T/\alpha$, using $c > \rho$ and $\rho = M\alpha/T$. In either case (4) fails at t and the induction closes. *Necessity.* Let $c \leq \rho$ be an integer and let some attacker episode G_u attain the ceiling, $S_u = Mm_u$. If the attacked trajectory were rejection-free, G_u would be tested at α/T with padded mean $Mm_u/(cm_u) = M/c \geq M/\rho = T/\alpha$ and rejected by the inclusive rule, a contradiction. Hence no integer $c \leq \rho$ suffices for every stream satisfying the hypotheses, whereas the integer $\lfloor \rho \rfloor + 1 > \rho$ does. Two remarks. The hypothesis on the

other hypotheses cannot be dropped: a benign group rejected at the cold-start level raises R and with it every later level, and the attacker's pads are then priced above α/T . And the threshold is uniform over streams: for one given stream with at least one attacker episode, the same two arguments, with M replaced by $\max_u \text{Ev}(G_u)$ over the attacker's episodes, show that the smallest sufficient integer is $\lfloor c_{\text{crit}} \rfloor + 1$ with $c_{\text{crit}} = \max_u \text{Ev}(G_u) \alpha/T \leq \rho$, and $c_{\text{crit}} = \rho$ exactly when some attacker episode attains M . A real-valued c would make $(c-1)m_t$ non-integral; the statement is therefore made for integers, which is what an attacker can append. $\square$

J. Reach and front-load (Theorems 4 and 5)

Let $w = (w_1, w_2, \dots)$ be precommitted with $w_i \geq 0$ and $\sum_i w_i \leq 1$, let $F(e) = \sum_i w_i e_i$ with each $e_i \in \{0, M\}$, and write $\beta = 1/(\alpha_t M)$ for the weight mass a detection needs: a lone flow firing at position i gives $F = w_i M$, which clears $1/\alpha_t$ iff $w_i \geq \beta$.

Theorem 4. The positions at which a lone attack flow fires are $\{i : w_i \geq \beta\}$. Since $\sum_i w_i \leq 1$, at most $1/\beta$ indices can carry mass $\geq \beta$, so $P = |\{i : w_i \geq \beta\}| \leq 1/\beta = \alpha_t M$, the quantity that governs Corollary 1. $\square$

Theorem 5. Prepending L zero-evidence flows moves every attack flow to a position $> L$. Every $e_i \leq M$, so $F(e) \leq M \sum_{i>L} w_i$, and detection requires $\sum_{i>L} w_i \geq \beta$; it is impossible once $L \geq L^*(w) = \min\{L : \sum_{i>L} w_i < \beta\}$. Summability gives $\sum_{i>L} w_i \rightarrow 0$, so the set is non-empty and L^* is finite for every summable w .

Sufficient, and exact only under a full-firing episode. The displayed bound is attained only when every position $> L$ fires, so L^* is the exact minimum in that case and an upper bound otherwise. The gap is real: for $w = (0.4, 0.3, 0.3, 0, \dots)$ and $\beta = 0.5$ the tail first falls below β at $L = 2$, yet an episode carrying a lone firing flow is already undetectable at $L = 0$, since $\max_i w_i = 0.4 < \beta$ and Theorem 4 gives $P = 0$. We therefore quote L^* as an attacker's upper bound throughout. $\square$

Consequently a precommitted scheme cannot simultaneously be padding-invariant (weights fixed by absolute position), have usable power (some $w_i \geq \beta$), and make ordering robustness unbounded ($L^* = \infty$). Concentrating weight early gives $L^* = 1$ but $P = 1$; spreading it over more than $1/\beta$ positions drives every $w_i < \beta$, so $P = 0$.

K. ADDIS state budget (Theorem 6) and the coincidence

Theorem 6. ADDIS offers $\alpha_t = \min\{\lambda, (\tau - \lambda)[W\gamma_{S^t - C_{0+}} + \dots]\}$. The theorem assumes $\lambda \geq k/(|\mathcal{C}| + 1)$, i.e. the cap sits at or above the conformal floor, so that the driving term and not the cap is what the attack has to push below $1/M$; if the cap were already below the floor no target could be rejected at $D = 0$ and the budget would be zero rather than the displayed value. With $W = w_0$ before the first rejection and $W = \alpha R$ after R of them, and 0-indexed $\gamma_j = (j + 1)^{-1.6}/\zeta(1.6)$. Precursors injected at index 0 hold the spending index at $S^t - C_{0+} = D$, so the driving level is $(\tau - \lambda)W\gamma_D$. The collapse of ADDIS's

reward sum to the single term $W\gamma_D$ is exact in the pre-rejection state, where $W = w_0$ and there is one active term. After $R \geq 1$ rejections each reward term carries its own selected-non-candidate lag $\ell_j \geq 0$, sitting at index $D - \ell_j$; the coefficients sum to exactly $w_0 + (\alpha - w_0) + \alpha(R - 1) = \alpha R$, but γ is decreasing, so $\sum_j a_j \gamma_{D - \ell_j} \geq \alpha R \gamma_D$. The drive is therefore larger than the collapsed expression, feasibility survives further, and the displayed value at $W = \alpha R$ is a lower bound on B^* — equality only when every $\ell_j = 0$. Numerically, at $\lambda M = (\tau - \lambda)M = 4.5 \times 10^5$, $R = 1$ and a single lag of 100, the expression returns 313 while the true budget is 373. The attack is mounted in the pre-rejection state, since the precursors precede every target. A rejection is possible only while $(\tau - \lambda)W\gamma_D \geq 1/M$, i.e. $(D + 1)^{-1.6}/\zeta(1.6) \geq k/((\tau - \lambda)W(|\mathcal{C}| + 1))$, i.e. $D + 1 \leq [(\tau - \lambda)W(|\mathcal{C}| + 1)/(k\zeta(1.6))]^{1/1.6}$. The smallest D violating this is B^* . Each precursor is a group carrying exactly one firing flow, so its evidence is M/m and its p-value m/M ; landing in $(\lambda, \tau]$ needs $m \in (\lambda M, \tau M]$, whose smallest integer is $m^* = \lfloor \lambda(|\mathcal{C}| + 1)/k \rfloor + 1$ (the interval is non-empty because $(\tau - \lambda)M \gg 1$). The total is $B^* m^*$, scaling as $\lambda |\mathcal{C}| [(\tau - \lambda)w_0 |\mathcal{C}|]^{0.625}/k^{1.625}$. $\square$

Coincidence. When ADDIS's cap binds, $\alpha_t = \lambda$, and the minimal suppressing pad raises the padded p-value just past λ . Because the p-value moves in steps of one conformal floor $1/(|\mathcal{C}| + 1) \ll \tau - \lambda$, it lands in $(\lambda, \tau]$ and advances the index: cap-saturation implies landing. The converse is not forced, but on this stream every unsaturated detection has a level four orders of magnitude below λ , so none can land, and the observed equality (all 101 saturated detections land) is the generic case rather than a coincidence of this window.

L. Restart FDR composition

Proof. Restart runs n controllers on disjoint epochs, each controlling its own FDR at q . The deployment-wide proportion $\text{FDP} = \sum_i V_i / (\sum_i R_i \vee 1)$ is a weighted average of the per-epoch FDP_i with data-dependent weights $R_i / \sum_j R_j$, so per-epoch control does not bound $\mathbb{E}[\text{FDP}]$. Indeed, suppose each epoch makes one false rejection with probability q and no rejection otherwise. Each epoch then has $\text{FDR}_i = \mathbb{E}[V_i / (R_i \vee 1)] = q$, yet whenever any epoch rejects every pooled rejection is false, so the deployment-wide FDP is 1 on that event and 0 otherwise, and with independent epochs $\text{FDR} = \Pr(\text{some epoch rejects}) = 1 - (1 - q)^n$. (The neighbouring construction — each epoch always rejecting once, false with probability q — instead gives pooled $\text{FDR} = q$, so non-composition requires the all-or-nothing epoch above and does not follow from per-epoch control alone.) In contrast, mFDR pools: if $\mathbb{E}[V_i] \leq q \mathbb{E}[R_i]$ for every i , the median inequality gives $\sum_i \mathbb{E}[V_i] / \sum_i \mathbb{E}[R_i] \leq q$.

This non-composition is specific to *budget reset* — every epoch spends its own q , total nq . A *preallocated* restart — pre-committed epoch budgets $q_1, \dots, q_n$ with $\sum_i q_i \leq q$, fixed before deployment — instead *preserves* the deployment-wide guarantee. Since $\sum_j R_j \geq R_i$, the pooled proportion is

TABLE V

TERMS HELD DISTINCT THROUGHOUT THE PAPER. DEFINITIONS ONLY, SO **ORDER-INVARIANT**: THE TABLE REPORTS NO MEASUREMENT, AND THE QUANTITIES DEFINED HERE THAT ARE MEASURED ON A SEQUENCED STREAM ARE REPORTED ELSEWHERE. VALIDITY IS AN ASSUMPTION; FEASIBILITY, POWER AND RESOLUTION ARE SEPARATE CONSEQUENCES.

Term	Meaning
Statistical validity	the assumptions the stated FDR/e-value guarantee needs (Assumption 1 and Lemma 1)
Feasibility	whether <i>any</i> admissible evidence can cross the controller threshold
Detection power	whether malicious episodes actually generate sufficient evidence
Episode recall	recall under the <i>current</i> grouping definition (moving denominator)
Atomic coverage	coverage of a <i>fixed</i> fine-grained attack-unit denominator
Alert blur	number of fixed atomic <i>reference</i> units one issued alert represents
Per-alert padding cost r_t^*	zero-evidence flows that suppress one alert at the level it received on the unperturbed run, (9); an oracle lower bound (Surface A). On AIT, where a pad flow can fire, the empirical replay cost $r_{t,\text{emp}}^* \geq r_t^*$ is reported and labelled
Per-alert sum $\sum_{t \in \mathcal{D}} r_t^*$	the per-alert costs of the baseline true detections $\mathcal{D}$ added up on the unperturbed trajectory; an upper bound on the joint cost
Joint padding cost J_{seq}	total cost of the greedy sequential oracle when e-LOND is re-run over the padded stream, so that R_{t-1} responds to the pads (the <i>joint controller-state rerun</i> , Surface A)
State-free multiplier c	the attacker multiplies its own episode size by c and reads no controller state (Corollary 2)
Controller-state attack cost B^*	precursor hypotheses that advance ADDIS's evidence-conditioned spending index (Surface B, Appendix G); a different mechanism from the e-LOND rerun above: padding changes e-LOND's rejection state indirectly by preventing discoveries, whereas the ADDIS attack directly advances its evidence-conditioned spending index

dominated termwise,

$$\text{FDP} = \frac{\sum_i V_i}{\sum_j R_j \vee 1} \leq \sum_i \frac{V_i}{R_i \vee 1} = \sum_i \text{FDP}_i$$

(both sides 0 when no epoch rejects), so by linearity of expectation $\text{FDR} \leq \sum_i \text{FDR}_i \leq \sum_i q_i \leq q$ — no independence across epochs is required (independence enters only the $1 - (1 - q)^n$ counterexample above). Each e-LOND epoch delivers $\text{FDR}_i \leq q_i$ under the same group e-validity premise (Assumption 1); the shared calibration set only induces between-epoch dependence, which this bound tolerates, provided the epoch boundaries and the q_i are not data-dependent. Restart therefore does not simply void the deployment-wide guarantee: it forces a *choice* — budget reset recovers more power but controls only a per-epoch FDR, whereas preallocation keeps deployment-wide $\text{FDR} \leq q$ at a power cost (Table XII). $\square$

APPENDIX B PROCEDURES, DATA, AND REPRODUCIBILITY

A. Terms

B. Procedure definitions

Each procedure offers a level α_t and rejects H_t when the evidence exceeds $1/\alpha_t$ (p-value procedures reject when $p_t \leq \alpha_t$). We write R_{t-1} for the rejection count, τ_j for the j -th rejection time, and, for the candidate-based procedures, $C_{j+}(t)$ for the number of candidates after τ_j . Transcriptions were unit-tested against literal $O(T^2)$ implementations of the published formulas and against four identities the sources state (offline e-BH as the $t = K$ case of online e-BH; $R(\text{e-LOND}) \subseteq R(\text{online e-BH})$; e-LORD = e-LOND under $\gamma_t = \omega_t \prod_{j < t} (1 - \omega_j)$; e-SAFFRON($\lambda=0$) = e-LORD).

LOND, e-LOND: $\alpha_t = \alpha \gamma_t (R_{t-1} + 1)$,

LORD++: $\alpha_t = \gamma_t w_0 + (\alpha - w_0) \gamma_{t-\tau_1} + \alpha \sum_{j \geq 2} \gamma_{t-\tau_j}$,

SAFFRON: $\alpha_t = \min \{ \lambda, (1 - \lambda)[w_0 \gamma_{t-C_{0+}} + \dots] \}$,

ADDIS: $\alpha_t = \min \{ \lambda, (\tau - \lambda)[w_0 \gamma_{S^t-C_{0+}} + \dots] \}$,

online e-BH: reject H_i iff $E_i \geq 1/(k_i^* \alpha \gamma_i)$,

e-LORD (e-GAI): $\alpha_t = \omega_t (\alpha - \sum_{j < t} \frac{a_j}{R_{j-1}+1}) (R_{t-1} + 1)$.

SAFFRON (serial estimate of the alpha fraction that is futilely rationed on true null hypotheses) and ADDIS (adaptive discarding) are the adaptive descendants of LORD, and online e-BH is the e-value Benjamini–Hochberg procedure run at each step. SAFFRON's candidates are the $p \leq \lambda$; ADDIS additionally *selects* the $p \leq \tau$ and counts selected non-candidates. Here $k_i^* = \max\{k \leq t : \sum_{j \leq t} \mathbf{1}\{E_j \geq 1/(k \alpha \gamma_j)\} \geq k\}$ is the online-e-BH step-up fixed point, and ω_t is the e-GAI weight update [23], which we run **pre-committed** — the specialisation under which the e-LORD reparametrisation of Section III yields a fixed summable γ , and the only one our feasibility coverage claims. Throughout $\alpha = q = 0.05$, $w_0 = 0.025$, $k = 1$; ADDIS uses $\lambda = 0.25, \tau = 0.5$, SAFFRON $\lambda = 0.5$, e-SAFFRON $\lambda = 0.1$, and e-GAI $\phi = \psi = 0.5$, $\omega_1 = 1/T$. The two spending sequences are $\gamma_j \propto j^{-1.6}$ (normalised by $\zeta(1.6) = 2.2858$, no horizon needed) and horizon-uniform $\gamma_j = 1/T$ (max-min optimal, [HORIZON-AWARE]). We do not extend the feasibility bound to e-SAFFRON at $\lambda = 0.1$, a testing level the equivalence does not cover. mem-e-LORD is a different case and the tables mark it covered: its level has the family-I form, so Theorem 1 bounds its *feasibility* exactly as for e-LORD; what does not transfer is the error-rate reading, since it controls mem-FDR rather than FDR, and we never quote its rejections as FDR-controlled. The r-LOND variants inherit the classification verbatim, their rejection sets coinciding with e-LOND's on calibrated p -values. For e-TOAD we use the interpretation of the active set and step-up limit consistent with its stated limiting cases ($d_t = t \Rightarrow$ e-LOND, $d_t = \infty \Rightarrow$ online e-BH); the alternate literal reading is a sensitivity analysis in Table XIII. One equation of the compound-e construction admits two readings (a literal snapshot and a running union of the rejection history); we report both and default to the literal one.

Table VI gives the full procedure matrix at the primary and stress windows, including the first-flow rejection counts the taxonomy (Table VII) summarises.

C. Reporting conventions

Unless a sentence, caption or table says otherwise, every measured number in this paper is detector **seed 0** under the **canonical** within-bucket order at the stated window, in the **horizon-free** regime $\gamma_j \propto j^{-1.6}$; horizon-aware results are named as such or marked [HORIZON-AWARE]. Results that necessarily run under first-flow arrival — the alert-blur, smoothing, restart, asymmetric-weighting, host-detector and ADDIS contrasts, and the AIT host-conditioned arm — say so at the point of claim, and their appendix tables carry both seeds where a seed applies. Every per-alert padding cost r^* is an *oracle lower bound*, because exact sizing reads the realised group evidence and the live controller level; it is written r_t^* throughout, the subscript marking the step whose running level prices it, and a row priced against the static cold-start threshold instead says so in words and uses no such symbol (Tables XXIII and XXXVI); whole-detected-set padding totals and the front-load cost L^* are upper bounds, and the keyed-hash budgets $N_{50/90/99}$ are reliability figures over 400 draws (Table XX). Windows are named by chronological position: 0.55 primary, 0.62 secondary, 0.85 the known-invalid stress window, used for mechanism and cost measurement only.

D. Chronological split geometry

The five deployment positions are overlapping chronological views of one exercise. Each deployment block spans $[\text{pos}, \text{pos} + 0.15)N$ up to integer rounding, so adjacent positions share 46.7–53.3% of their deployment flows — the primary/secondary pair shares 53.3%, the largest in the set — and the deployment block at one position coincides, to within one flow, with the calibration block of the position 0.15 later. Deployment blocks two or more positions apart are disjoint, but no two positions are independent even then: the training pools $[0, i_1)$ are nested, so every earlier position trains on a prefix of every later one’s data and the fitted detectors are correlated across all five. Each position is internally chronological — training, then calibration, then deployment — so this couples the positions to one another without crossing any single position’s own split. This is why the paper reports the five windows separately and never pools them, and why 0.62 is a correlated robustness check rather than confirmatory replication of 0.55.

E. Datasets and preprocessing

LSPR23 [13] (Zenodo 8042347, CC-BY-4.0) is a single CSV of 16,353,511 flows; 89% of them fall in the final 25.6 hours, so window positions are chronological slices rather than random splits. The detector reads `Protocol` plus 32 per-flow timing and volume statistics (columns 9–40); the grouping key reads `SrcIP`, `DstIP` (columns 2–3) and, for the service and subnet families, the ports (columns 4–5); the alert audit of Appendix F-A reads the annotation columns. The dataset carries no flow-to-campaign labels: `Expoid_dst` is empty for

1,630,732 of the 1,644,599 malicious flows, and `Category`, `Severity` and `SigID` are effectively constant where present. Timestamps are in microseconds and 7,381,261 rows arrive out of order, so the pipeline sorts on load. With test-window start fraction `pos`, $i_2 = \lfloor \text{pos} \cdot N \rfloor$, $i_1 = i_2 - \lfloor 0.15N \rfloor$ and $i_3 = \min(N, i_2 + \lfloor 0.15N \rfloor)$ give training $[0, i_1)$, calibration $[i_1, i_2)$ and deployment $[i_2, i_3)$; benign subsampling is confined to $[0, i_1)$. A separate red-team task record (288 narratives, 83 machine-readable compromise reports, 295 timestamped step submissions) is used only as external ground truth in Appendices D and F, never as a feature.

AIT-LDSv2.0. The transfer tests use AIT-LDSv2.0 [14] (Zenodo 5789064, CC BY-NC-SA 4.0), eight organisation testbeds whose released packet captures we convert to per-organisation `tstat tcp_complete` netflows following the dataset’s own recipe (45,001 attack flows). Each flow carries the dataset’s own attack-label string; we treat as malicious exactly the attack-step labels it defines (`service_scan`, `host_discover_dmz/_local`, `online_cracking`, `wpscan`, `dirb_scan`, `upload_rce_shell...`) by exact string match, and as benign exactly the labels the dataset marks benign; flows whose label is unknown or empty are dropped rather than assumed benign. Numeric columns are kept per fold only if they are non-null on $> 90\%$ of rows and non-constant; all identity fields and all absolute-time fields are excluded so no fold can key on the capture epoch, and the six host-context features are the same strictly causal ones as on LSPR23. Splits are chronological within each organisation: calibration uses benign flows strictly preceding that organisation’s first attack flow, deployment is everything from that instant on.

Detectors and grouping. The detector is HistGradientBoosting; an unsupervised Isolation Forest runs alongside as a feasibility-versus-sufficiency probe, with the same margin yet tail reach 0.000 at all ten configurations. A group’s evidence uses every flow in its time bucket, so its hypothesis is emitted only when the bucket closes, using only observations available then. Groups closing in the same bucket are sequenced by the canonical splitmix64 hash of the group’s own (`SrcIP`, `DstIP`, `bucket`) key, collisions broken on the key; the first-flow arrival order is retained as a labelled upper bound. The headline grouping is not tuned on its reported window: under previous-window selection and five leave-one-out folds it is near-worst of the 30 groupings feasible at every window, the cap being the selection-stable choice while the grouping does not transfer (Table XVIII). Every closed form is verified numerically against an independent implementation, and each measurement script carries unit tests.

APPENDIX C FULL FEASIBILITY EVIDENCE

Table VIII gives the ten-configuration e-LOND detection matrix, Table IX the power of every procedure over the same grid, Table X the joint (q, γ) sweep, Table XI the smoothing arm, Table XII the restart arm, Table XIII the closure, donation

TABLE VI

PROCEDURES, ASSUMPTIONS, AND COVERAGE BY THEOREMS 1 AND 2. POWER IS REPORTED AT THE **PRIMARY ANALYSIS WINDOW** (0.55, $|C| = 2,448,993$, $T = 57,368$ EPISODES, 275 MALICIOUS) AND THE **STRESS-TEST WINDOW** (0.85, $|C| = 1,813,113$, $T = 31,568$, 255 MALICIOUS), TWO-HOUR GROUPING. HORIZON-UNIFORM γ REQUIRES THE STREAM LENGTH IN ADVANCE AND IS MARKED [HORIZON-AWARE] WHEREVER IT APPEARS. THE FDP COLUMN IS THE 0.85 STRESS-WINDOW VALUE, MEASURED AGAINST LABELS WHERE THE EVIDENCE IS NOT A VALID E-VALUE (APPENDIX F-A); AT 0.55 EVERY FDP HERE IS ≤ 0.010 . WHICH PROCEDURES ARE COVERED IS POSITION-INDEPENDENT; THE REJECTION COUNTS DIFFER BECAUSE THE TWO WINDOWS CARRY DIFFERENT TRAFFIC. [†]SAFFRON SITS OUTSIDE THE LETTER OF THEOREMS 1 AND 2 BECAUSE ITS INDEX COUNTS TESTED HYPOTHESES, BUT IT DOES NOT ESCAPE IN PRACTICE: IT IS SILENT FOR 99.3% OF THE STREAM AT 0.55 AND 75% AT 0.85. ONLY ADDIS AND ONLINE E-BH AVOID THE HORIZON’S PRACTICAL CONSEQUENCE, AND THEY DO SO DIFFERENTLY: ADDIS’S INDEX STOPS ADVANCING, A GENUINE ESCAPE FROM ARRIVAL-TIME ABSORPTION, WHEREAS ONLINE E-BH IS DECIDED BY A FIXED POINT OVER THE HISTORY RATHER THAN BY A LEVEL FIXED AT ARRIVAL, SO THE ABSORPTION PREDICATE IS NOT DEFINED FOR IT AT ALL. IT IS STILL UNABLE TO REJECT 90.1% OF HYPOTHESES *at their own arrival* HERE, EXACTLY AS OFTEN AS E-LOND, AND RECOVERS THEM ONLY BY REVISITING (SECTION III-C AND TABLE XIII). ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX); THE ORDER SHIFTS DETECTIONS AND THE LEVEL AN ALERT FIRES AT, AND SHIFTS NOTHING ELSE.

Procedure	Evidence	Spending index	Horizon needed	Covered by C1?	Rej. 0.55 (primary)	Rej. 0.85 (stress)	FDP (0.85)	Nominal guarantee / required assumptions
LOND [5]	p-value	elapsed time	no	yes (I)	18	72	0.000	FDR (super-unif. null p ; indep.)
e-LOND [7]	e-value	elapsed time	no	yes (I)	18	72	0.000	FDR under group e-validity
LORD++ [6]	p-value	lag over rejections	no	yes (II)	19	72	0.000	FDR (conditional / indep.)
e-LORD (=e-SAFFRON _{$\lambda=0$}) [23]	e-value	elapsed time (reparam.)	[HORIZON-AWARE]	yes (I)	101	150	0.020	FDR under group e-validity
mem-e-LORD [23]	e-value	elapsed time (reparam.)	[HORIZON-AWARE]	yes (I)	87	142	0.000	controls mem-FDR
SAFFRON [20]	p-value	non-candidates	no	no [†]	0	70	0.000	assumes independence
ADDIS [21]	p-value	selected non-cand.	no	no	110	152	0.033	null-cons. violated
online e-BH [7], [22]	e-value	fixed point over history	no	no	18	72	0.000	FDR under group e-validity

and decision-deadline procedures against the horizon, and Table XIV the low-prevalence sensitivity.

Question answered. Which procedures escape the finite horizon, at what price, and does a better detector change the boundary? *Takeaway.* Escaping takes a different spending mechanism, not a better one, and ranking quality moves the detection counts rather than the margin behind them.

A. Feasibility versus power, in full

Under the canonical order the primary window yields three e-LOND detections out of 275 malicious episodes and the secondary window eleven; at positions 0.70 and 0.77 no episode clears its own step anywhere in the stream, so the canonical count there is zero (Table XIX carries the full five-window order matrix). The separation appears directly in paired contrasts. At position 0.70 both detector seeds share $|C|$, T and the margin +0.536, yet under first-flow order one seed yields 30 detections and the other none. A host-conditioned detector raises AUROC from 0.916 to 0.954 at the primary window but reduces e-LOND detections from 18 to two (first-flow, both legs). An Isolation Forest has the same margin as the gradient-boosted detector and not one attack flow above the calibration maximum at any of the ten window-seed cells. Ranking quality can move power; it cannot move the margin in (10).

Two configurations outside the headline pair corroborate that the grouped counts are a property of the spending sequence rather than of the ceiling: e-LORD under the horizon-aware e-GAI weighting reaches 101 detections at the same window and seed with no false discoveries, and periodic restart at the same grouping and the same deployment-wide budget reaches 55.

Low prevalence compounds the cold-start problem, and live-fire prevalence is the optimistic case. Thinning malicious episodes to a target rate while replacing each with benign evidence from the same window holds T , the margin and every survivor’s index exactly fixed: at $\pi = 10^{-4}$ the probability

that e-LOND *ever* rejects falls to 0–40% across the evaluated cells, mostly below 10%, while the margin does not move (Table XIV). This is a sensitivity analysis, not a counterfactual deployment, but it shows why a short discovery prefix is most damaging in exactly the regime that motivates FDR control.

Prevalence as a swept parameter. LSPR23 is a live-fire exercise: 10.06% of flows and 0.48–0.81% of episodes are malicious, the optimistic case, so Table XIV treats prevalence as a swept parameter rather than asserting a figure for a typical SOC: the targets run from 0.7 to 2.7 orders of magnitude below the exercise’s episode rate, a sensitivity range, not a claim about any deployment. We thin malicious episodes — not flows, which would change group arity and evidence sums and so confound the feasibility question with the padding cost — to each target π , keeping a nested random subset so a lower target keeps a strict subset of a higher one. The primary mechanism replaces each thinned episode with benign evidence drawn from the same window, so the hypothesis stays at its own index and T , the margin and every survivor’s level are held exactly fixed; the deletion mechanism removes it instead, which shrinks T by up to 0.80% and raises the margin by up to +0.0129. The two agree to within 0.02 in bootstrap probability, so it is prevalence and not re-indexing that drives the collapse. Feasibility is untouched while detection collapses: the feasibility-is-not-detection separation, driven by base rate alone. The $\pi = 0$ arm produces no rejection on any stream, but that is 5 effective (overlapping) windows, and zero would be the likely outcome even at the 4.5% early-false rate quoted from simulation in Appendix C-B ($P = 0.79$): it is consistency, not a test of that number.

B. What escapes the horizon, and what it costs

Among the procedures we study the boundary is sharp: the horizon binds a procedure whose spending index advances on *every* hypothesis. One that advances it only on tested hypotheses can escape that bound; one that defers the decision falls outside

the predicate altogether (Table VII). SAFFRON sits outside the letter of the theorems, since its index counts non-candidates, but does not escape in practice (silent at 99.3% of steps at the primary window, first-flow procedure matrix). ADDIS advances its index only on hypotheses whose p-values are selected ($p \leq \tau$) but not candidates ($p \leq \lambda$); with two-point evidence at $k = 1$ no observed episode lands in $(\lambda, \tau]$, so the index stops advancing and the level stays high. That is a genuine escape from absorption, but on the real stream it coincides with a violation of ADDIS’s uniform-conservativeness premise, and Appendix G shows that conditioning the index on observed evidence also makes the state attacker-controllable. Online e-BH’s step-up threshold is a fixed point over the whole observed history [7], [25], [22], so it does not decide hypothesis t against a level fixed at t : (6), and with it the absorption predicate, is simply not defined for it. What it escapes is the *permanence* of an early infeasible prefix — a later observation raises the step-up count and can make a hypothesis already passed rejectable. It does not escape bounded evidence: index j is rejectable only if some attainable count R gives $R\alpha\gamma_j \geq 1/M$, so a horizon- T run still leaves an index no step-up count can reach. Decision deadlines interpolate between immediate e-LOND-style decisions and indefinite reconsideration [40], [27], and a bucket-close deadline is exactly the batched architecture a SOC receives, valid under arbitrary dependence where a batched BH is not. What deferral buys is latency, not feasibility: the fraction of hypotheses that cannot be rejected *at their own arrival* is 96.3% at the primary window whether decisions are immediate, held to bucket close, or never forced, because a step-up count of R needs R hypotheses to clear the bar at once and at most 0.48% of episodes carry any evidence. Under the canonical order it adds one detection in the stress window and none elsewhere (Table XIII).

Four constructions alter the boundary, but none removes the tradeoff for free, and Table VII prices each; the smoothing and restart contrasts are measured under first-flow order on both legs, so the order shifts the level and not the sign. Two remove a premise. *Randomised smoothing* dissolves the absorbing state by replacing the discrete p-value with a $\text{Unif}(0, 1)$ draw [3], [12], but its surplus is a lottery: the episodes detected with probability ≥ 0.9 are exactly the discrete ones, and the surplus does not survive arbitrary-dependence-valid merging (Table XI). *Periodic restart* renews the cold-start budget and recovers real power. Held at the headline grouping, the primary window, seed 0 and the *same* deployment-wide budget $\sum_i q_i \leq q = 0.05$, restarting every two hours takes e-LOND from 18 rejections to 55, recall 0.065 to 0.200, at a measured FDP of zero. Spending a fresh q per epoch instead buys more — recall 0.360 at a total budget of 0.25 — but that is a per-epoch FDR, a weaker promise than the deployment-wide one (Table XII, proof in Appendix A). Restart forces that choice rather than removing it, and the comparison has to hold grouping, seed and total budget fixed or it measures the wrong thing. Two work within the premises: the recent closure and compound-e donation constructions [27], [28], strict improvements over e-LOND under arbitrary dependence, are bounded on a rejection-free

prefix by a constant cold-start factor ($1/(1 - \delta)$, a 5% change against a $20\times$ shortfall) and by an additive horizon offset respectively (Propositions 1 and 2 and Corollary 3), and buy $+0$ true detections at every window and seed we run (Table XIII, seed 0 shown). Among the elapsed-time procedures bound by Theorems 1 and 2, the cold-start barrier is therefore a property of the evidence rather than of the controller’s sophistication.

The level- w_0 margin. The appendix tables report feasibility on one cross-controller scale rather than per procedure, so that a single column serves every row:

$$\text{margin} = M \frac{w_0}{T} - 1 = \frac{(|\mathcal{C}| + 1) w_0}{kT} - 1. \quad (10)$$

It is the body’s ρ of (7) rescaled, $\text{margin} = \rho_p c_{0,p}^{-1} w_0 - 1$, and for e-LOND, whose $c_0 = 2w_0$, $\rho_{\text{e-LOND}} = 2(\text{margin} + 1)$. A negative margin therefore does *not* imply e-LOND infeasibility: e-LOND can still fire down to margin $-\frac{1}{2}$, the rejection rule (4) being inclusive. This is exactly why the body plots ρ instead.

Feasibility inputs, window by window. Across the five positions $|\mathcal{C}|$ is 2,448,993, 2,449,031, 2,287,988, 2,116,418 and 1,813,113 benign calibration flows, against $T = 57,368$, 49,267, 37,231, 31,672 and 31,568 two-hour host-pair episodes, matching Table VIII. Within-bucket ordering changes the sequence, not T . The two headline windows are the two largest calibration sets, and 0.62 exceeds 0.55 rather than repeating it.

Full-exercise scaling. Under horizon-free $\gamma_j \propto j^{-1.6}$, testing all 16,353,511 flows in one uninterrupted run requires about 1.6×10^{13} calibration flows for LOND/e-LOND and 3.2×10^{13} for LORD++. These are the horizon-free counterparts to the horizon-uniform counts in Section III-B.

Cold-start arithmetic. The count ratio of Corollary 1 is dataset-independent, but its calendar form is not: for LORD++ the horizon-uniform requirement is 1.44×10^9 calibration flows for one hour of a 10k-flows/s link and 3.46×10^{10} for a day. Under $\gamma \propto j^{-1.6}$ the feasible prefix before any rejection is 902/902/865/824/748 steps at the five windows for e-LOND. At the $\pi = 10^{-4}$ sensitivity point that window almost always closes empty: the first true attack is not expected until index $\approx 10,000$, so a rejection arrives in time to bootstrap the controller with probability only $\approx 0.3\%$, and the alternative, an early false rejection, has probability at most a few percent (4.5% for e-LOND, 2.2% for LORD++, in simulation at a stated $|\mathcal{C}|$). The benign-only arm of Table XIV produces no rejection on any stream, which with five effective windows is consistent with that rate rather than a test of it ($\Pr(\text{zero} \mid 4.5\%) = 0.79$). Deleting thinned hypotheses instead of replacing them shortens T and raises the margin by up to $+0.013$, yet gives the same collapse to within 0.02 in bootstrap probability, so it is prevalence and not re-indexing that does the work.

Horizon misspecification (single cell). Online e-BH’s advantage over the covered procedures is robustness to a misspecified horizon. At the stress window and one seed, the only cell we sweep the horizon on, online e-BH keeps 144 of its 152 rejections under a $100\times$ over-estimate of T , where LOND keeps all 151 at $2\times$ but none at $100\times$, and LORD++ none

TABLE VII

PROCEDURES BY WHAT ADVANCES THEIR SPENDING SEQUENCE, AND HOW EACH RELATES TO THE FINITE HORIZON. “COVERED BY” IS COVERAGE OF THE PROCEDURE *together with* THE SPENDING SEQUENCES WE RUN, WHICH IS WHAT THEOREMS 1 AND 2 QUANTIFY OVER; A PROCEDURE NAME ALONE DOES NOT CARRY THE DECAY CONDITION (APPENDIX A-B). “ABSORBING” IS WHETHER A REJECTION-FREE PREFIX CAN ENTER THE ABSORBING INFEASIBLE STATE ON BOUNDED EVIDENCE. PRICES ARE MEASURED ON LSPR23 TWO-HOUR HOST PAIRS: DETECTION DELTAS UNDER THE CANONICAL ORDER (TABLE XIII), SAFFRON’S SILENCE FRACTION FROM THE FIRST-FLOW PROCEDURE MATRIX (TABLE VI). [†]DONATION’S BOUND NEEDS $\delta < 1$ AND HOLDS ON THE REJECTION-FREE PREFIX ONLY, SO IT IS NOT A GLOBAL MEMBER OF THEOREM 1’S FAMILY; THE CLOSURE’S ADDITIVE OFFSET NEEDS SPARSE EVIDENCE ($H + P_T \leq T$). BOTH ARE STATED EXACTLY IN APPENDIX A. [‡]THEOREM 1 BOUNDS MEM-E-LORD’S *feasibility*, NOT ITS ERROR RATE: IT CONTROLS MEM-FDR RATHER THAN FDR. [§]THE PREDICATE IS DEFINED ONLY FOR PROCEDURES THAT DECIDE AT ARRIVAL, SO IT DOES NOT APPLY TO ONLINE E-BH, NOR TO THE $d_t = \infty$ END OF THE DEADLINE FAMILY, WHICH IS ONLINE E-BH (APPENDIX C-B).

Procedure	Evidence	Spending index	Covered by	Absorbing?	Escape mechanism / measured price
LOND, e-LOND [5], [7]	p / e	elapsed time	Thm. 1	yes	—
e-LORD, mem-e-LORD [23]	e	elapsed time (reparam.)	Thm. 1 [‡]	yes	— (α -death fix = e-LORD before the first rejection)
LORD++ [6]	p	lag over rejections	Thm. 2	yes	—
Donation, closed e-LOND [27], [28]	e	elapsed time; zero-evidence subset	App. A	yes [†]	cold-start coefficient $\times 1/(1-\delta)$; horizon offset $\leq P_T$; +0 detections
SAFFRON [20]	p	non-candidates	no	in practice	silent at 99.3% of steps (primary window)
ADDIS [21]	p	selected non-candidates	no	no	index stops advancing; conservativeness premise violated; state attackable (Appendix G)
Decision deadline, e-TOAD [40], [27]	e	deferred to d_t	no	no	latency; +1 detection, stress window only
Online e-BH [7], [22]	e	fixed point over history	no	n/a [§]	revisiting; still needs $ C \geq T/(\alpha R) - 1$
Smoothing, restart	any	—	premise removed	no	unstable surplus; budget-split keeps deployment-wide q , fresh- q gives a per-epoch guarantee

already at $2\times$; read this as an illustration of the mechanism, not a measured rate.

Closure and donation, measured. Runs here do pass the $1/\delta$ rejection count at which Proposition 1 stops applying (30, 31 and 73 rejections at 0.70, 0.77 and 0.85, where the realised boost reaches 1.83 against the cold-start ceiling 1.05); what holds is the weaker fact that the denominator never vanishes on these streams. Both procedures buy +0 true detections under the canonical order across the ten cells and +1 and +0 under first-flow arrival (Table XIII).

APPENDIX D GRANULARITY AND ORDERING

Table XV gives the grouping families at the primary window, Table XVI the grouping-independent resolution cost against the fixed atomic reference (including the malicious-flow coverage the body omits), Table XVII the external red-team denominator, and Table XVIII the cross-window configuration transfer.

Grouping key and attack cost. Changing the primary window’s two-hour key from a host pair to a source host raises canonical horizon-free detections from 3 to 46 and the median exact per-alert suppression cost from 24 to 496 added flows. The grouping key therefore sets feasibility, alert semantics, and attack cost together. Source-host grouping is feasible at every evaluated bucket width, reducing each window to 763–24,043 episodes.

Restart as a granularity lever. Coarsening is not the only way to divide the hypothesis count a run must survive: hourly restart at one-hour grouping reaches recall 0.406 at ≈ 125 alerts, against 0.103 for six-hour coarsening, while preserving episode

resolution. A pre-committed budget split remains deployment-wide; assigning a fresh q to each epoch changes the guarantee to per-epoch (Section III-C).

The external denominator. LSPR23 ships the red team’s task record: 288 tasks and 295 timestamped step submissions, an attacker-defined decomposition of the same campaign whose declared network segments are the dataset’s own per-flow seg vocabulary (83 compromise reports agree, 0 disagree). Counting distinct red-team steps merged into each issued alert, the external count rises with the alerting unit as the proxy does (at 0.70, $4.3 \rightarrow 27.0$ across 5 min–24 h against the proxy’s $1.0 \rightarrow 28.3$; rank agreement +1.000 over six widths, +0.800 under the canonical order). We do not claim corroboration: a null that rotates the red-team timeline against the stream reproduces the observed counts at 18 of 19 cells; the record sees the alerts at only 3 of 10 window–order cells, because issued alerts sit in the feasible prefix, the first minutes of a window, which at 0.55 and 0.62 predates the exercise’s first submission; and every attributed alert sits on a labelled-malicious episode from the same exercise. The external unit agrees in direction and is too coarse to discriminate.

Attribution rule and nulls. The 5-minute atom is *our* proxy for one attack action, so Table XVII counts instead how many distinct red-team steps each issued alert merges, using LSPR23’s own task record (288 tasks, 295 timestamped submissions, 2023-03-09 07:03–15:08 UTC), an attacker-defined decomposition fixed before the dataset was published. A step is attributed to an alert when its submission falls in the alert’s bucket span *and* its task’s declared network segments meet the alert’s, or one of its compromise-report IPs is an alert endpoint. The link is the segment vocabulary

TABLE VIII

E-LOND ($\gamma \propto j^{-1.6}$) ACROSS FIVE WINDOW POSITIONS AND TWO DETECTOR SEEDS, TWO-HOUR HOST-PAIR GROUPING. THE MARGIN IS SEED-INDEPENDENT BY CONSTRUCTION; $|C|$ IS THE CALIBRATION SIZE $|C|$. “SILENT” IS THE FRACTION OF STEPS AT WHICH THE CONTROLLER IS *structurally infeasible* — THE OFFERED LEVEL α_t HAS FALLEN BELOW $1/M$, SO NO REJECTION IS POSSIBLE EVEN AT THE MAXIMAL ADMISSIBLE EVIDENCE M (IT IS *not* THE FRACTION OF STEPS WITHOUT A REJECTION). ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX); THE ORDER SHIFTS DETECTIONS AND THE LEVEL AN ALERT FIRES AT, AND SHIFTS NOTHING ELSE.

Position	Seed	AUROC	T	$ C $	margin	rejections	recall	silent
0.55	0	0.916	57,368	2,448,993	+0.067	18	0.065	90.1%
0.55	1	0.927	57,368	2,448,993	+0.067	18	0.065	90.1%
0.62	0	0.799	49,267	2,449,031	+0.243	13	0.033	90.5%
0.62	1	0.871	49,267	2,449,031	+0.243	12	0.030	90.9%
0.70	0	0.914	37,231	2,287,988	+0.536	30	0.105	80.1%
0.70	1	0.847	37,231	2,287,988	+0.536	0	0.000	97.7%
0.77	0	0.828	31,672	2,116,418	+0.671	31	0.126	77.3%
0.77	1	0.833	31,672	2,116,418	+0.671	31	0.126	77.3%
0.85	0	0.999	31,568	1,813,113	+0.436	72	0.282	65.4%
0.85	1	1.000	31,568	1,813,113	+0.436	71	0.278	65.7%

TABLE IX

PROCEDURE POWER OVER FIVE POSITIONS $\times$ TWO SEEDS (MIN / MEDIAN / MAX), TWO-HOUR GROUPING. ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX); THE ORDER SHIFTS DETECTIONS AND THE LEVEL AN ALERT FIRES AT, AND SHIFTS NOTHING ELSE. THE *uniform* ROWS ARE THE HORIZON-UNIFORM γ , AN [HORIZON-AWARE] THAT MUST KNOW T IN ADVANCE (TABLE V).

Procedure	γ	rejections			recall		
		min	med	max	min	med	max
LOND	$\propto j^{-1.6}$	0	24	72	0.000	0.085	0.282
LOND	uniform	0	100	151	0.000	0.273	0.576
e-LOND	$\propto j^{-1.6}$	0	24	72	0.000	0.085	0.282
e-LOND	uniform	0	100	151	0.000	0.273	0.576
LORD++	$\propto j^{-1.6}$	0	25	72	0.000	0.089	0.282
LORD++	uniform	0	100	151	0.000	0.268	0.576
SAFFRON	$\propto j^{-1.6}$	0	21.5	70	0.000	0.069	0.275
SAFFRON	uniform	0	0	0	0.000	0.000	0.000
ADDIS	$\propto j^{-1.6}$	0	106.5	152	0.000	0.274	0.576
ADDIS	uniform	0	0	0	0.000	0.000	0.000
online e-BH	$\propto j^{-1.6}$	0	31	72	0.000	0.094	0.282
online e-BH	uniform	0	105	152	0.000	0.273	0.576

TABLE X

ERROR TARGET q VS THE SPENDING SEQUENCE (MEDIAN EPISODE RECALL). FEASIBILITY SCALES LINEARLY IN q : THE MEASURED (MARGIN + 1)/BASE RATIO TRACKS THE PREDICTION $q/0.05$. ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX); THE ORDER SHIFTS DETECTIONS AND THE LEVEL AN ALERT FIRES AT, AND SHIFTS NOTHING ELSE. THE *uniform* COLUMN ARE THE HORIZON-UNIFORM γ , AN [HORIZON-AWARE] THAT MUST KNOW T IN ADVANCE (TABLE V).

q	median recall		(margin + 1) ratio	
	poly	uniform	measured	predicted
0.01	0.049	0.000	0.20	0.20
0.05	0.085	0.273	1.00	1.00
0.10	0.100	0.274	2.00	2.00
0.20	0.137	0.274	4.00	4.00

TABLE XI

RANDOMISED SMOOTHING AT THE PRIMARY WINDOW (LOND, $\gamma \propto j^{-1.6}$, POSITION 0.55, SEED 0): REMOVING THE FLOOR ADDS NO RELIABLE DETECTION THAT SURVIVES THE MERGE CHOICE. REJECTIONS ARE MEAN $\pm$ SD OVER 100 RANDOMISATION DRAWS WHERE APPLICABLE; “RELIABLE” IS THE NUMBER OF MALICIOUS EPISODES DETECTED WITH PROBABILITY ≥ 0.9 , “ANY” WITH PROBABILITY > 0 . ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX); THE ORDER SHIFTS DETECTIONS AND THE LEVEL AN ALERT FIRES AT, AND SHIFTS NOTHING ELSE.

merge	route	rejections	reliable	any
mean-e	discrete	18.0	18	18
simes	smoothed	24.1 ± 2.7	18	100
hommel	smoothed	12.5 ± 4.7	1	67
bonf-p	smoothed	23.2 ± 2.7	17	100

— the task record’s `bt_baf_int` is the dataset’s own per-flow `baf_int` — corroborated against the machine-readable compromise reports (36/39 IPv4 appear as a flow endpoint; 83 agree, 0 disagree). Two nulls are run. The **label** null permutes

which task each submission belongs to, holding the times fixed: it asks whether *which* task acted is informative, and goes degenerate at the daily bucket where every submission shares one bucket. The **shift** null rotates the whole red-team timeline against the stream, preserving task identity, ordering, attributes

TABLE XII

PERIODIC RESTART AT THE PRIMARY WINDOW (E-LOND, SEED PAIR 0/1 — DECISION-IDENTICAL TO LOND HERE, SINCE THE GROUP P-VALUE IS $\min(1, 1/E_v)$ SO $p \leq \alpha_t \Leftrightarrow E_v \geq 1/\alpha_t$): DETECTION RESTORED AT THE COST OF A PER-EPOCH GUARANTEE (TOTAL SPEND $n \cdot q$). BUDGET RESET (E2c) SPENDS $\alpha_{TOT} = n \cdot q$; INDEX RESET (E2b) KEEPS $\alpha_{TOT} = q$ VIA A PRE-COMMITTED ALLOCATION. REJECTIONS ARE SHOWN PER SEED (0/1); RECALL AND FDP ARE THE TWO-SEED MEAN. ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX); THE ORDER SHIFTS DETECTIONS AND THE LEVEL AN ALERT FIRES AT, AND SHIFTS NOTHING ELSE.

arm	epoch (h)	α_{tot}	rej. (s0/s1)	recall	FDP
uninterrupted (baseline)	—	0.050	18/18	0.065	0.000
restart 2 h, budget reset (E2c)	2	0.250	99/91	0.345	0.000
restart 1 h, budget reset (E2c)	1	0.450	130/121	0.406	0.000
restart 2 h, index reset, uniform (E2b)	2	0.050	55/52	0.195	0.000
restart 2 h, index reset, geometric (E2b)	2	0.050	28/26	0.098	0.000
[HORIZON-AWARE] uninterrupted	—	0.050	104/96	0.364	0.000

and inter-arrival structure: it asks the sharper question, whether the alignment *in time* is informative, and does not degenerate.

Reading. The result is agreement in direction without discrimination. The external count and the proxy rise together (panel (b) of Table XVII), so the resolution cost survives substituting an attacker-defined unit for ours; but the count clears the label null at only 6 of 19 cells and the shift null at only 1, so the rise is driven by a wider bucket capturing more submissions rather than by alert-level correspondence, and we report it as agreement rather than as confirmation. The record also sees the alerts at only 3 of 10 window-order cells: at 0.55 (canonical), 0.62 (canonical), 0.62 (first-flow) the issued alerts sit in the *feasible prefix* — the first minutes of a deployment window — which predates the exercise’s first submission, so at most 9% of them overlap it in time at all; at 0.85 (canonical), 0.85 (first-flow) they do overlap but only 2 of 17 tasks with a timed step declare a segment or file a compromise report. Four further limits. (i) At the 86,400 s bucket both nulls tie the observation exactly: every submission shares one daily bucket, so no permutation and no rotation can move a step between alerts (the spatial filter is *not* vacuous there — at 0.70/first-flow it admits 12 of 30 temporally overlapping alerts — it is the nulls that lose their power); only sub-daily widths inform. (ii) The whole positive result rests on six coarse segment labels: across every row the compromise-IP conjunct adds *no* alert the segment conjunct did not already admit. (iii) Every attributed alert sits on a labelled-malicious episode and the labels came from this same exercise, so nothing here is label-independent evidence that an alert is a true positive. (iv) A submission is when the red team *reported*: the lag, estimable from the compromise reports’ own timestamps, has median 170 s (IQR 115–279 s), small against every bucket width.

A. Within-bucket order

Ordering is robust to timestamp ties (at most 1.3% of episodes share a first timestamp; all 36 reported statistics are invariant across 50 randomised within-timestamp orders). The within-bucket order, by contrast, is load-bearing, and it is where Theorem 1 becomes visible: only a prefix of

902/902/865/824/748 steps can reject at all, so a detection needs a near-ceiling episode to land inside it, which under first-flow arrival is exactly what live-fire attack traffic does. Sequencing the same episodes by their own metadata takes the counts from 18/13/30/31/72 to 3/11/0/0/34 (Table XIX). Over 50 pre-committed orders the count has median 1 at the primary window and 23.5 at the stress window; the canonical order is not a uniformly typical draw (above the ensemble median at three windows, at its maximum at 0.62, below it at the two where nothing detects), while first-flow matches or exceeds the ensemble maximum at every window and seed.

Why that canonical order. First-flow arrival is an attacker timing lever, and the damage reaches beyond the attacker’s own hypothesis: one pad at the start of its bucket moves 373 other episodes at the primary window, shifting the γ_j and hence the $\mathcal{M}_j$ each is offered. A tie-break reading only the group’s own key makes that 0 by construction, so the Assumption 1-preservation argument of Appendix A describes the order the paper reports. Three qualifications are measured rather than assumed. It removes the timing lever, not the metadata one: appending flows moves no other group, but an adversary that *creates* keys moves ranks. A public hash is grindable over the attacker-selectable (SrcIP, DstIP): with the feasible prefix at 1.6–2.6%, an adversary trying 100 candidate keys lands one inside it with probability 0.80–0.93, so the canonical order is a canonicalisation, not a security mechanism. A keyed hash removes grinding at the price of a seed the adversary can neither learn nor infer: blind insertion still suppresses every one of the 84 targets in every draw at a budget of 10^6 keys, at 2.1–89 $\times$ (median 19 $\times$) the instantiated-key cost, with a per-window median N_{99} of up to 3.2×10^4 endpoint pairs (Table XX). That cost is a capability, not a computation: on a network where only a few hundred reachable pairs are available the seed is a hard stop, a regime these figures do not cover. The measurement assumes the target sits in the first bucket of its window, which holds at every cell here; where an earlier bucket exists the adversary inserts there and precedes the target with probability one. One keyed instance detects 0 at the primary window.

How the keyed-hash budget is measured. Under a public hash the adversary ranks candidate keys offline for free and instantiates only the G^* that land ahead of the target; under a keyed hash it cannot rank at all and must instantiate blind, and the cost scales as G^*/u with the target’s own hash quantile u , which is why the spread in Table XX is wide and heavy-tailed. Every one of the 84 targets falls in every draw at a budget of 1,000,000 keys, so the seed does not stop the attack; at the nearest matched reliability, N_{99} against the certain public G^* , it multiplies the instantiated-key cost by a median 19.3 $\times$ (range 2.1–89.1 $\times$). Suppression is verified by replay at every G^* , not asserted from a closed form, and the replayed success probability is compared with the binomial lower bound $P(\text{Bin}(N, u) \geq G^*_{\text{targeted}})$: it falls short of that bound by at most 0.030 in probability, the Monte Carlo tolerance of 400 draws. None of this touches Surface A padding (Theorem 3), which needs no key.

TABLE XIII

THE UAI 2026 PROCEDURES [27] AGAINST THE FEASIBILITY HORIZON ($\gamma \propto j^{-1.6}$, SEED 0, TWO-HOUR GROUPING; SECTION III-C). “DET.” IS TRUE DETECTIONS, “SILENT” THE FRACTION OF HYPOTHESES THAT CANNOT BE REJECTED *at their own arrival* EVEN AT THE EVIDENCE CEILING. FOR THE STEP-UP PROCEDURES THAT FRACTION IS COMPUTED *exactly*, BY RE-RUNNING THE STEP-UP ON THE COUNTERFACTUAL IN WHICH THE ARRIVING HYPOTHESIS CARRIES M ; THE CHEAPER TEST — DOES M CLEAR $1/(\alpha\gamma t r)$ AT THE LARGEST CONCEIVABLE r — IS ONLY *necessary*, SINCE REACHING r NEEDS r HYPOTHESES TO CLEAR THE BAR AT ONCE, AND IT REPORTS 0.0% AT EVERY WINDOW HERE. ALL COUNTS AND SILENCE FRACTIONS ARE UNDER THE CANONICAL WITHIN-BUCKET ORDER, AS TABLE I IS. **DEFERRING THE DECISION LEAVES AT-ARRIVAL FEASIBILITY ESSENTIALLY UNCHANGED** — IDENTICAL TO $d_t = t$ AT THREE OF THE FIVE WINDOWS — **AND BUYS POWER ONLY BY REVISITING**, WHICH IS WHY ITS PRICE IS ALERTING LATENCY. THE LAST ROW IS ONLINE E-BH, REACHED AS THE $d_t = \infty$ LIMIT AND VERIFIED REJECTION-IDENTICAL TO IT AT EVERY ROW. DONATION E-BH IS THE LITERAL READING OF ITS SOURCE’S EQ. (102); A SECOND READING OF THAT EQUATION GIVES ONE MORE DETECTION AT 0.85 AND IS REPORTED HERE AS A SENSITIVITY ANALYSIS, AND NEITHER IS QUOTED AS A POWER COMPARISON (SECTION III-C).

Procedure	0.55		0.62		0.70		0.77		0.85	
	det.	silent	det.	silent	det.	silent	det.	silent	det.	silent
e-LOND (<i>baseline</i>), $d_t = t$	3	96.3%	11	91.3%	0	97.7%	0	97.4%	34	78.1%
donation e-LOND	3	96.3%	11	91.3%	0	97.7%	0	97.4%	34	78.1%
closed e-LOND (e-LOND)	3	96.3%	11	91.3%	0	97.7%	0	97.4%	34	78.0%
donation e-BH	3	—	11	—	0	—	0	—	35	—
e-TOAD, bucket-close deadline	3	96.3%	11	91.3%	0	96.5%	0	97.4%	35	77.3%
e-TOAD, $d_t = \infty$ (= online e-BH)	3	96.3%	11	91.3%	0	96.5%	0	97.4%	35	77.3%

TABLE XIV

DETECTION AT LOWER BASE RATES (ITEM R5): MALICIOUS EPISODES THINNED TO A TARGET PREVALENCE π WITH BENIGN REPLACEMENT; E-LOND, SEED 0, BOTH WITHIN-BUCKET ORDERS (CANONICAL KEY-HASH AND **FIRST-FLOW**). UPPER PANEL: MEAN REJECTIONS R AND THE BOOTSTRAP PROBABILITY THAT THE CONTROLLER EVER REJECTS; LOWER PANEL: THAT PROBABILITY AT $\pi = 10^{-4}$ FOR THE ESCAPE PROCEDURES AND FOR THE DELETION MECHANISM. DESIGN AND READING: APPENDIX C-A.

pos	order	observed		$\pi = 10^{-3}$		$\pi = 10^{-4}$		$\pi = 10^{-5}$		$\pi = 0$	
		R	boot	R	boot	R	boot	R	boot	R	boot
0.55	key-hash	3.00	100	0.54	38	0.06	4	0.02	2	0.00	0
0.55	first-flow	18.00	100	1.40	56	0.08	6	0.00	0	0.00	0
0.62	key-hash	11.00	100	0.18	14	0.02	2	0.02	2	0.00	0
0.62	first-flow	13.00	100	1.36	62	0.20	18	0.04	4	0.00	0
0.70	key-hash	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0
0.70	first-flow	30.00	100	3.52	92	0.38	28	0.10	10	0.00	0
0.77	key-hash	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0
0.77	first-flow	31.00	100	2.22	70	0.06	6	0.04	4	0.00	0
0.85	key-hash	34.00	100	0.48	26	0.02	2	0.00	0	0.00	0
0.85	first-flow	72.00	100	8.28	100	0.54	40	0.28	28	0.00	0
bootstrap probability (%) at $\pi = 10^{-4}$: the escapes collapse too											
pos	order	e-LOND	online e-BH	e-TOAD	e-LOND (deletion)						
0.55	key-hash	4	4	5	4						
0.55	first-flow	6	6	5	6						
0.62	key-hash	2	2	0	2						
0.62	first-flow	18	18	25	18						
0.70	key-hash	0	0	0	0						
0.70	first-flow	28	28	40	30						
0.77	key-hash	0	0	0	0						
0.77	first-flow	6	6	5	8						
0.85	key-hash	2	2	5	2						
0.85	first-flow	40	40	40	40						

Notes. “pos”: the window’s chronological position; “observed”: the unthinned stream; “ R ”: mean rejections over 50 draws; “boot”: the percentage of draws in which the controller *ever* rejects; the observed and $\pi = 0$ arms are single deterministic streams. Lower panel: online e-BH and e-TOAD are the escape procedures of Appendix C-B; “e-LOND (deletion)” removes thinned episodes instead of replacing them. $\pi = 10^{-2}$ is unreachable: it is above the observed episode rate, and we cannot manufacture attacks. The 0.85 rows are the stress window, whose evidence is not a valid e-value.

APPENDIX E

ATTACK MEASUREMENTS AND MITIGATIONS

Table XXII gives the five padding pools under both orders, Table XXIII the per-window suppression cost (running-level

TABLE XV

GROUPING FAMILIES AT THE PRIMARY WINDOW (0.55, SEED 0); FLOW COVERAGE IS WINDOW-SPECIFIC. THE MARGIN COLUMN IS THE *level- w_0* FEASIBILITY MARGIN (EQ. (10)); THE RECALL AND COVERAGE COLUMNS ARE E-LOND UNDER THE HORIZON-UNIFORM [HORIZON-AWARE] γ , WHOSE COLD-START COEFFICIENT IS $\alpha = 2w_0$. [†] MARKS A *negative level- w_0 margin*; E-LOND CAN STILL FIRE THERE WHEN THAT MARGIN EXCEEDS $-\frac{1}{2}$ (ITS OWN $\alpha = 2w_0$ COLD-START BOUND), WHICH IS WHY SOME DAGGERED ROWS CARRY NON-ZERO RECALL/COVERAGE — A ROW IS 0 ONLY WHEN E-LOND IS ITSELF INFEASIBLE. FEASIBILITY AND DETECTION ARE THUS NOT THE SAME CONDITION (CF. TABLE I). ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX).

Family	bucket	margin	episode recall	flow coverage
src-dst	300 s (5 m)	$-0.637^{\dagger}$	0.000	0.000
src-dst	7200 s (2 h)	$+0.067$	0.378	0.276
src-dst	none	$+0.548$	0.253	0.032
src	300 s (5 m)	$+1.546$	0.501	0.408
src	7200 s (2 h)	$+20.505$	0.382	0.353
src	none	$+54.964$	0.286	0.350
dst	300 s (5 m)	$+0.085$	0.532	0.689
dst	7200 s (2 h)	$+1.736$	0.304	0.033
dst	none	$+3.646$	0.184	0.034
subnet24	300 s (5 m)	$-0.451^{\dagger}$	0.448	0.697
subnet24	7200 s (2 h)	$+0.501$	0.285	0.031
subnet24	none	$+1.251$	0.193	0.031
src-dport	300 s (5 m)	$-0.327^{\dagger}$	0.324	0.368
src-dport	7200 s (2 h)	$+3.025$	0.066	0.351
src-dport	none	$+6.189$	0.032	0.340

versus static) at both seeds and both orders, Table XXIV the real-flow padding-dilution replay on the shipped detector, Table XXVI the position-weighted schemes, Table XXVII the aggregation-cap sweep, Table XXVIII the insertion attack against padding, Table XXIX the host-conditioned detector, Tables XXX and XXXI the AIT organisation matrices, Table XXXII the AIT ordering arms, Table XXXVI both attacks in operational units, and Table XXXVII the joint zero-evidence controller-state rerun. *Question answered.* How much of the padding result survives changes of window, seed, order, pad

TABLE XVI

GROUPING-INDEPENDENT RESOLUTION COST (SRC-DST FAMILY, E-LOND $\gamma \propto j^{-1.6}$, SEED 0). THE *atomic evaluation reference* IS THE SET OF 5-MINUTE SRC-DST *malicious* EPISODES — A FIXED FINE-GRAINED DENOMINATOR WE CHOOSE, NOT EXTERNALLY VALIDATED INCIDENT TRUTH, SINCE LSPR23 CARRIES NO CAMPAIGN IDENTIFIER — 1899 AT THE PRIMARY WINDOW, 1973 AT THE STRESS-TEST WINDOW — AND ITS SIZE IS **FIXED** ACROSS EVERY ALERTING BUCKET. “COV (FIXED)” IS THE FRACTION OF THOSE ATOMIC UNITS COVERED BY ≥ 1 ISSUED ALERT; “BLUR” IS THE MEAN NUMBER OF DISTINCT ATOMIC MALICIOUS UNITS AN ISSUED ALERT LUMPS TOGETHER (THE RESOLUTION LOST); “RECALL (MOVING)” IS THE ORDINARY EPISODE RECALL, WHOSE DENOMINATOR CHANGES WITH THE BUCKET. AS THE UNIT COARSENS THE FIXED COVERAGE *rises* WHILE THE BLUR CLIMBS FROM 1 TO ~ 20 –40: THAT CLIMB, NOT THE MOVING RECALL, IS THE DENOMINATOR-INVARIANT RESOLUTION COST. $\dagger$ MARKS A *negative level- w_0 feasibility margin*; E-LOND CAN STILL FIRE THERE WHEN THAT MARGIN EXCEEDS $-\frac{1}{2}$ (ITS OWN $\alpha = 2w_0$ COLD-START COEFFICIENT), WHICH IS WHY SOME DAGGERED ROWS CARRY NON-ZERO COVERAGE/RECALL — A DAGGERED ROW READS 0 ONLY WHEN E-LOND IS ITSELF INFEASIBLE (CF. TABLES I AND XV). ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX).

bucket	cov (fixed)	blur	recall (moving)	flow cov.
<i>primary analysis window (0.55)</i>				
5 m $\dagger$	0.020	1.0	0.020	0.001
30 m $\dagger$	0.065	3.6	0.077	0.004
1 h $\dagger$	0.068	7.2	0.058	0.004
2 h	0.152	16.1	0.065	0.007
6 h	0.240	25.3	0.103	0.011
1 d	0.385	38.5	0.123	0.021
<i>stress-test window (0.85)</i>				
5 m $\dagger$	0.308	1.0	0.308	0.301
30 m $\dagger$	0.317	4.5	0.277	0.418
1 h	0.322	8.8	0.238	0.418
2 h	0.322	8.8	0.282	0.418
6 h	0.743	20.4	0.439	0.995
1 d	0.743	20.4	0.439	0.995

pool and attacker knowledge? *Takeaway*. The costs move; the suppressibility does not, with one labelled exception.

A. Windows, seeds and orders

Priced against the running level at each detected episode’s step, the canonical primary window’s three detections cost 23, 24 and 33 added flows and the eleven canonical secondary-window detections have a median of six. Over fifty pre-committed orders at the primary window the median is *one* detection, the range is 0–10, and 19 of the fifty produce none at all; the canonical order’s three alerts lie within this distribution, whereas first-flow arrival produces eighteen and exceeds all fifty audited metadata orders. At the secondary window the canonical order is the ensemble *maximum*, 11 against a median of 6. We use it because it reads only a group’s own metadata, so appending flows moves no other hypothesis: a validity-preserving canonicalisation, not a security mechanism, and not because it detects more. Conditional on an alert being issued, every evaluated alert is suppressible; the order changes *which* episodes are there to attack rather than whether they can be. At the stress window, whose evidence is not a valid e-value and which is therefore a cost measurement only, the canonical median is 116 over 34 detections. The first-flow order

TABLE XVII

AN EXTERNAL DENOMINATOR FOR THE RESOLUTION COST: DISTINCT RED-TEAM STEPS OF LSPR23’S TASK RECORD MERGED PER ISSUED ALERT, UNDER BOTH WITHIN-BUCKET ORDERS (**CANONICAL** KEY-HASH AND **FIRST-FLOW**). (A) STEPS PER ALERT AGAINST THE LABEL AND SHIFT NULLS; (B) WITHIN-WINDOW RANK AGREEMENT ACROSS BUCKET WIDTHS. DESIGN AND READING: APPENDIX D.

pos	order	bucket (s)	attrib.	steps/alert		label null		shift null		5-min
				bucket	flow	mean	>	mean	>	
0.55	first-flow	86400	12	18.33	18.33	18.33	–	18.33	–	38.47
0.62	key-hash	86400	1	35.00	35.00	35.00	–	35.00	–	35.00
0.62	first-flow	86400	6	41.00	41.00	41.00	–	41.00	–	32.00
0.70	key-hash	300	35	4.29	3.13	2.59	✓	2.02	–	1.00
0.70	key-hash	1800	7	12.57	12.57	11.06	–	5.83	–	3.00
0.70	key-hash	3600	2	18.50	12.00	14.81	✓	11.75	–	2.43
0.70	key-hash	86400	1	20.00	20.00	20.00	–	20.00	–	29.00
0.70	first-flow	300	32	4.31	3.18	2.63	✓	1.99	–	1.00
0.70	first-flow	1800	13	13.23	12.62	11.65	–	6.09	–	3.51
0.70	first-flow	3600	13	18.77	17.38	15.06	✓	11.95	✓	6.51
0.70	first-flow	7200	12	19.08	18.67	15.47	✓	11.45	–	13.00
0.70	first-flow	21600	12	19.08	18.67	15.28	✓	11.55	–	13.00
0.70	first-flow	86400	12	27.00	27.00	27.00	–	27.00	–	28.33
0.77	key-hash	3600	4	4.00	4.00	3.19	–	1.68	–	10.00
0.77	key-hash	86400	2	3.00	1.00	3.00	–	3.00	–	18.00
0.77	first-flow	3600	12	3.00	3.00	2.74	–	1.32	–	9.93
0.77	first-flow	7200	13	3.15	3.15	3.17	–	2.94	–	19.61
0.77	first-flow	21600	13	3.15	3.15	3.15	–	3.15	–	21.61
0.77	first-flow	86400	13	3.15	3.15	3.15	–	3.15	–	21.61

(b) within-window rank agreement across bucket widths

pos	order	widths	steps vs. atoms	steps vs. bucket	atoms vs. bucket
0.7	key-hash	4	+0.800	+1.000	+0.800
0.7	first-flow	6	+1.000	+0.986	+0.986
0.77	first-flow	4	+0.816	+0.775	+0.949

Notes. “pos”: the window’s chronological position; “attrib.”: alerts with at least one attributed step. “bucket”/“flow”: the same attribution rule priced against the alert’s bucket span or its actual first-to-last flow span. “mean”: the null’s mean steps per alert; ✓ marks an observed count exceeding that null’s 97.5th percentile. At the 86,400 s bucket both nulls tie the observation exactly, so only sub-daily widths inform. Panel (b): Spearman rank agreement across bucket widths within one window.

TABLE XVIII

CROSS-WINDOW CONFIGURATION TRANSFER: THE CAP TRANSFERS, THE GROUPING DOES NOT. EACH CELL IS THE PREVIOUS-WINDOW NORMALISED REGRET $\rho \in [0, 1]$ (0 = ORACLE, 1 = WORST FEASIBLE CONFIG) ON THE EVALUATION WINDOW, SEED 0; “–” IS A FLAT SELECTION WINDOW WHERE TRANSFER IS UNDEFINED. THE GROUPING AXIS IS NOT FLAT — ITS ORACLE FLOW COVERAGE VARIES BY 0.676 ACROSS WINDOWS (CAP: 0.253) — YET THE FROZEN GROUPING STILL FAILS TO TRANSFER. ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX).

Transfer fold (select $\rightarrow$ eval)	normalised regret ρ	
	cap axis	grouping axis
0.55 $\rightarrow$ 0.62	0.000	0.945
0.62 $\rightarrow$ 0.70	0.048	0.991
0.70 $\rightarrow$ 0.77	0.070	0.000
0.77 $\rightarrow$ 0.85	0.009	0.921
worst (previous-window sel.)	0.070	0.991
worst (leave-one-out)	0.048	0.945

detects more episodes and detects them deeper in the level sequence, so its costs are larger (118, 72 and 5,202 at the same three windows), but every target remains suppressible; that order-specific matrix is Table XXIII. Across 50 pre-committed orders and both detector seeds the per-order median stays at 16–178 flows, conditioned on the order having produced a detection at all — some orders produce none, so this is a

TABLE XIX

E-LOND TRUE DETECTIONS UNDER **PRE-COMMITTED**, **EVIDENCE-INDEPENDENT** WITHIN-BUCKET ORDERS, ALL FIVE WINDOWS, DETECTOR SEED 0. ALL ORDERS EMIT HYPOTHESES AT BUCKET CLOSE AND DIFFER ONLY IN HOW GROUPS CLOSING IN THE SAME BUCKET ARE SEQUENCED. THE COUNT IS **MATERIALLY ORDER-SENSITIVE**, AND FIRST-FLOW ARRIVAL IS THE *largest* OVER EVERY ORDER AND SEED WE AUDIT — WHICH IS WHY THE PAPER REPORTS THE **KEY-HASH** ROW, A DETERMINISTIC HASH OF THE GROUP’S *own* (SrcIP, DstIP, BUCKET) KEY (COLLISIONS BROKEN ON THE KEY), AND CARRIES FIRST-FLOW ONLY AS A LABELLED UPPER BOUND. THE MECHANISM IS THEOREM 1: BEFORE THE FIRST REJECTION ONLY A PREFIX OF 902 / 902 / 865 / 824 / 748 STEPS OUT OF 57,368 / 49,267 / 37,231 / 31,672 / 31,568 CAN REJECT AT ALL, SO A DETECTION NEEDS A NEAR-CEILING EPISODE TO LAND INSIDE IT. UNDER THE CANONICAL ORDER AT 0.70 AND 0.77 *no* EPISODE CLEARS ITS OWN STEP ANYWHERE IN THE STREAM, WHICH IS WHY THOSE ENTRIES ARE 0. THE KEY-HASH IS A TIMING-INDEPENDENT CANONICALIZATION, NOT A SECURITY MECHANISM: A PUBLIC HASH OF ATTACKER-SELECTABLE FIELDS CAN BE GROUND, AND THE KEYED ROW IS THE GRINDING-RESISTANT VARIANT, ITSELF AN UNPREDICTABLE DRAW FROM THE RANDOM ENSEMBLE BELOW. CONDITIONED ON THERE BEING DETECTED EPISODES, THE PER-ORDER MEDIAN PADDING COST IS 109.5 / 16 / 93 / 175 / 98 FLOWS — A SURVIVORSHIP-CONDITIONED ESTIMAND. BECAUSE THE KEY-HASH READS NO ARRIVAL TIME, IT IS ALSO WHAT MAKES THE PADDING ATTACK LEAVE ASSUMPTION 1 INTACT FOR EVERY OTHER HYPOTHESIS: INJECTING ONE PAD AT THE EARLIEST INSTANT OF ITS OWN BUCKET MOVES 373 / 348 / 4 / 7 / 3 OTHER HYPOTHESES UNDER FIRST-FLOW ARRIVAL, AGAINST 0 UNDER THE KEY-HASH AT EVERY WINDOW AND SEED. THE CANONICAL ORDER IS ALSO THE MORE *seed*-STABLE OF THE TWO: ACROSS DETECTOR SEEDS ITS COUNT MOVES BY AT MOST 3, AGAINST 30 FOR FIRST-FLOW ARRIVAL. FEASIBILITY (C1) IS ORDER-INVARIANT. SEE APPENDIX D-A.

within-bucket order	0.55	0.62	0.70	0.77	0.85
first-flow arrival (upper bound)	18	13	30	31	72
group-id (packed key)	0	3	0	0	24
last-flow arrival	0	0	0	0	0
fixed random permutation (one instance)	2	1	9	1	38
key-hash (canonical, reported)	3	11	0	0	34
keyed key-hash (secret seed)	0	6	3	3	24
<i>median over 50 orders</i>	1	6	2.5	1	23.5
<i>range over 50 orders</i>	0–10	0–11	0–12	0–6	0–44

survivorship-conditioned estimand.

B. Sizing without the oracle

For horizon-free spending, $c = 10$ is the smallest tested multiplier in $\{2, 3, 5, 10, 100\}$ that suppresses every canonical seed-0 alert in the primary and known-invalid stress windows, the two arms with stored per-episode arities. In the secondary window, the canonical alert with $m_t = 22$ and $r_t^* = 859$ requires $c = 41$. Across the LSPR23 analysis, all 674 detected episodes lie on host pairs containing no benign traffic, so episode arity is the attacker’s own generated flow count.

Multiplying the attacker’s own episode by c defeats an alert whenever $c \geq 1 + r_t^*/m_t$. That overshoot factor has median 1.57 at the canonical primary window and maximum 6.17 at the stress window, and it is what the attacker must cover — never the threshold itself. Tuning c per window is cheaper than the fixed $c = 10$ evaluated here ($c = 3$ suppresses all three primary alerts at a median of 80 added flows, $3.5\times$), but that choice reads the realised costs and is reported as an oracle-informed floor, not an attacker rule; so is the flat pad, where 33 flows suffice at the primary window and 17 for 90% of the secondary window’s eleven. The one exception is the first-flow

TABLE XX

WHAT A SECRET SEED FOR THE **CANONICAL** KEY-HASH ORDER COSTS THE INSERTION ATTACKER (E-LOND, SEED 0): THE PUBLIC-HASH COST G^* AGAINST THE KEYED-HASH BUDGETS $N_{50/90/99}$ OVER 400 DRAWS, AND THEIR RATIO (N_{99} AGAINST THE CERTAIN PUBLIC SUPPRESSION). DESIGN AND READING: APPENDIX D-A.

pos	order	targets	u	public	keyed: instantiated keys			blind
				G^*	N_{50}	N_{90}	N_{99}	N_{99}/G^*
0.55	public	3	0.037–0.072	217–217	5,663	6,639	8,962	41.3×
0.55	keyed	0	—	—	—	—	—	—
0.62	public	11	0.016–0.652	70–813	825	962	996	4.6×
0.62	keyed	6	0.060–0.413	288–506	4,994	5,843	6,053	12.3×
0.70	public	0	—	—	—	—	—	—
0.70	keyed	3	0.032–0.059	257–257	8,238	9,620	9,961	38.8×
0.77	public	0	—	—	—	—	—	—
0.77	keyed	3	0.016–0.037	416–416	26,101	30,432	31,502	75.7×
0.85	public	34	0.019–0.489	30–509	8,533	11,118	12,625	24.8×
0.85	keyed	24	0.016–0.385	78–518	8,262	9,640	9,981	19.3×

Notes. “pos”: the window’s chronological position; “targets”: detected episodes under that order; “ u ”: the target’s own hash quantile, unknown to a blind adversary; “ G^* ”: hypotheses an adversary who knows the ranking must instantiate ahead of the target on that same stream; “ N_p ”: distinct (SrcIP, DstIP) pairs a blind adversary must send traffic between to suppress the target with probability p , located on a six-points-per-decade grid and so exact to within $1.48\times$ and no further; the N_p cells are medians over the row’s targets; last column: N_{99} against the certain public G^* , the nearest matched-reliability comparison the grid offers (99% against 100%), a reliability budget rather than a lower bound. Rows with no detection carry no cost claim.

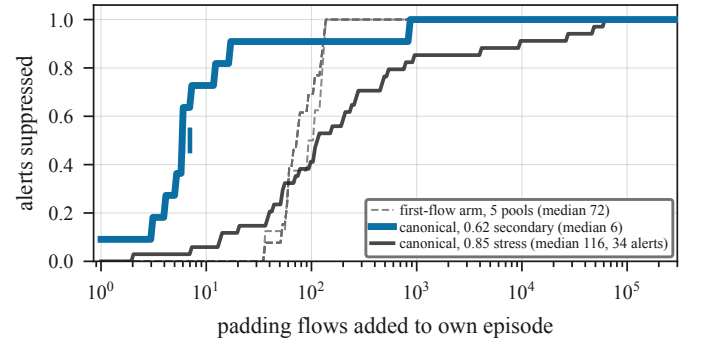


Fig. 5. The sensitivity arms the body’s Fig. 4 omits: the secondary window under the canonical order against the known-invalid stress window (canonical) and the five pad pools under first-flow arrival. Ticks are the five pools’ canonical medians at the secondary window, which span 6–7 flows — the pad-realism claim of this subsection. Costs move with window, order and pool; suppressibility does not.

stress arm, where overshoot reaches 4,302 and no multiplier we test suffices (Table XXV). The fixed $c = 10$ evaluated here costs a median $15.7\times$ the oracle at the primary window and $5.1\times$ at the stress window, so dropping to state-free sizing costs a factor of about ten, not the 10^4 – 10^6 span of $1/\alpha_t$ within a window. The resulting pad is large but not anomalous: the padded primary-window episode, arity 120, is exceeded by 0.9–9.9% of deployment episodes.

C. Pad realism and the pool construction

The realism of the pad barely matters. Generic benign, attacker-origin, protocol-matched and black-box pools have nearly identical median costs, because the running threshold dwarfs the largest pool mean e-value (216.8); service-matched padding is dearer — 1.2 – $1.5\times$ under the canonical order,

1.4–2.0 $\times$ under first-flow, and on a smaller episode subset — and does not change the conclusion (Table XXII). Under the headline key the adversary sends ordinary traffic between the source and destination already involved in its activity before the two-hour bucket closes. The flow-only detector reads no endpoint identity, but 17 of 33 features are backward or bidirectional, so whether a flow the victim answers transfers is empirical; the real-flow replay measures it rather than assuming it. Beyond the 3/3 and 11/11 canonical suppressions at the primary and secondary windows, the replay gives 34/34 canonical at the stress window, and 18/18, 13/13 and 72/72 under first-flow order; the measured cost equals the closed form on every episode of every arm (Table XXIV).

The black-box pool and the dilution replay. The pad flows of Table XXIV are real flows of the black-box pool, selected on network-observable frequency alone: the most common (proto,port) over the training prefix, which precedes both calibration and deployment — no labels, no detector output, no knowledge of the attacker’s own traffic. It picks proto 17/port 53 at both windows, the same service the label-using rule would, and the resulting pool is measured to be 0.0000% attack-labelled (an audit, not a selection criterion). The selection rule matters: taking the mode over *all* window traffic instead picks the attacker’s own flood at the stress window, a pool 90% attack-labelled that does not dilute. Zero pad fires out of the sampled flows at both windows is evidence for the structural premise (ordinary victim-service traffic scores like ordinary traffic), not proof that the firing probability is zero; the Clopper–Pearson bound in the table says how much that observation leaves open. Both within-bucket orders are attacked. The key-hash rows are the canonical order the paper reports, under which a pad provably moves no other hypothesis’s spending weight, so the Assumption 1 argument is clean; the first-flow rows re-run the identical chain — same detector, same pool, same pricing — on the detected set the initial first-flow order produces, which Table XIX shows is the largest over the orders we audit. Every detection is suppressible under both, and the canonical medians reproduce Table I. The canonical primary-window set is small, so read it as three individually-priced episodes (costs 23, 24, 33 flows) rather than as a rate; the canonical costs are lower than the first-flow ones because the two orders detect different episodes at different steps of the level sequence, not because the canonical order is weaker.

Residual risk at the horizon-aware arm. The pool’s zero firing count over 845,691 flows is a fact about the pool, and it leaves open a risk that replay cannot bound — so we simulate pads whose flows fire independently at the one-sided 95% Clopper–Pearson upper limit the observation admits, $p \leq 3.5 \times 10^{-6}$. Even there the worst alert suppresses on an estimated 90.5% of draws at the primary window and 82.5% at the secondary (Section VI-A). Both figures are Monte Carlo estimates over 200 draws and both treat pool flows as independent draws from a common firing rate, which the pool’s composition does not establish.

D. The AIT transfer arms

Host-conditioned detector on LSPR23. The detector of Table XXIX reads the 33 flow features plus 6 strictly causal ($ts < ts(\text{flow})$) host-context features — per-host prior counts, distinct-peer counts and failed-connection fractions — and no endpoint ground truth (label_src and label_dst both have attack rate 1.0 and are excluded). It has higher AUROC than the flow-only detector at both windows, yet lower tail reach and fewer e-LOND detections (2 against 18 at the primary window): higher ranking quality buying fewer, not more, operational alerts, the detector-quality-versus-feasibility gap of Section IV-A. To test the padding boundary we graft each attacked pair’s real causal context (strongest feature src-distinct-dst, $\rho = +0.34$; padding one victim cannot raise it) onto ordinary black-box pads — the most common (proto,port) on the training prefix, which precedes both calibration and deployment, so the pool is chosen on network-observable frequency alone with no labels and no detector output — and replay the append causally. The pads fire zero across the scored accumulation range at both windows — up to 100,000 appended flows per episode, 4,000,000 pad scorings at the primary window — and the suppression cost is the identical flow-level closed form ($0.55 r_t^* = 108$ host vs 118 flow; $0.85 r_t^* = 11,484$ host vs 5,202 flow), so within that range the transfer holds at both; costs above the cap are closed-form values, not replayed. A non-black-box pool rule does fire: selecting the modal service among detector-non-firing flows picks proto 6/port 80 at 0.85, a pool that is only 12% benign and fires 0.42. Grafting the attacker’s own flood onto attack context and observing the detector fire is not evidence of a defence, so that rule is reported only as a sensitivity. What LSPR23 cannot settle is the benign-inclusive question: its attacked pairs are 100% malicious, so no real ordinary-to-victim traffic exists there, which is what the AIT testbed supplies.

Resequencing the flow-only folds by first-flow arrival — the detector fit, scores, calibration and e-values are computed before ordering and are identical across the two arms — gives 84 of 85, and the canonical alerts are the *cheaper* to suppress: per-organisation median $r_{t,\text{emp}}^*$ falls from 212–13,142 flows under first-flow to 12–4,650 under the canonical order (Table XXXII). On shaw, ordinary victim traffic fires often in its *own* context (7%), which under a static-context replay made no episode suppressible; scored in the attacked pair’s evolving context the same pool fires at most 0.5%, and all three targets suppress (Tables XXX and XXXI). That contrast is why the host-conditioned arm recomputes context causally rather than scoring the pool in isolation.

Design of the AIT replay. Per organisation, Table XXXI trains a leave-one-organisation-out detector whose flow features are chosen per fold from the training organisations only, calibrates on in-organisation benign flows strictly preceding the first attack, forms (SrcIP, DstIP, 2h) episodes, runs e-LOND, and suppresses each detected malicious episode by replaying real benign flows to an attacked victim of the organisation (200 draws). A firing pad contributes the maximum e-value and

raises the cost, which the zero-pad closed form misses. The two arms replay differently because the two detectors read different things. Flow-only: flow features carry no context, so pads are drawn with replacement from the pool’s observed 0-or- M evidence, an empirical i.i.d. generator, so a modelled cost may exceed the finite observed pool. Host-conditioned: the six host features are causal, so appending pads changes the context every later pad is scored in, and holding them fixed would be a static-context diagnostic rather than a replay; we therefore recompute the victim’s context as pads accumulate (counts raised, distinct-peer counts pinned, failure fractions moved to the pool’s rate), score the whole pool under it, and report a success only over the accumulation range actually scored. The chain runs end-to-end at all eight organisations, in both arms: under first-flow order 84 of 85 flow-only detections are suppressible, per-draw success 0.79–1.00, and 57 of 63 host-conditioned detections are, per-draw success 0.61–1.00. Host conditioning therefore blunts the attack without stopping it. The causal replay is what shows both halves. It is why most episodes still suppress: *shaw*’s ordinary-to-victim flows fire 7% in their own contexts, which under a static-context replay made no episode suppressible, but a pad’s context is the attacked pair’s, and under that context the same pool fires at most 0.5% across accumulation, so all three episodes suppress. It is also why the remaining 6 do not (*harrison*, *wardbeck*, *wheeler*): under the accumulated context a pad can itself fire, and a firing pad adds the ceiling to the group’s evidence instead of diluting it, so on those episodes no draw brings the group mean below threshold in any of the 200 draws, within twice the zero-pad cost (at least 1,000 pads), which is the range the context curve is scored over. What decides the outcome is the pad count an episode needs against the rate at which pads fire along that accumulation path. These are censored simulations rather than proof that the episodes cannot be padded: a budget or a draw count large enough might still find a route. The pinning models a single already-active attacker–victim pair; padding from new hosts would move the distinct-peer counts and is not covered.

The joint attack on AIT. Table XXXIII re-runs e-LOND over the padded stream for each organisation, so the controller’s rejection count responds to the attack and, on the host-conditioned arm, every later flow on the attacked pair is re-scored as the pads arrive. Each figure is the median over 100 pad-sampling trajectories; the totals below sum those medians over the eight organisations and are not counts from a single run. Two template pools are separated throughout: the *deployment* pool is the benign-to-victim traffic the per-alert arms of Table XXXI use, and the *prior* pool is restricted to flows the victim had received before the attacker’s own last flow in the episode. An episode whose prior pool holds fewer than 20 flows is never padded.

With the deployment pool the campaign costs 244,250 and 91,713 flows against per-alert sums of 607,802 and 599,967, leaving 1 and 0 of the 79 and 80 own alerts. With the prior pool 24 and 27 remain and no organisation is cleared, at 430,626 and 242,206 flows: 30 of the 95 attacker-owned episodes have too

little prior traffic to imitate and 18 have none at all, and those episodes carry most of what survives. On the host-conditioned arm, *shaw* issues 3 and 3 own alerts and 1 and 6 non-attacker alerts; the deployment pool clears all of them for 28 and 7 flows, the victim’s prior traffic leaves 2 and 2; *wilson* issues 6 and 6 own alerts and 0 and 0 non-attacker alerts; the deployment pool clears all of them for 73,337 and 16,975 flows, the victim’s prior traffic leaves 1 and 1.

The state-free multiplier $c = \lfloor \rho \rfloor + 1$ of Corollary 2 leaves 65 and 60 alerts standing with the prior pool. Sizing it on the episode’s full arity rather than the attacker’s own flow count changes nothing there (65 and 60), and changes the outcome only with the deployment pool (57 and 1 against 59 and 6): what the attacker does not know about co-resident benign flows costs it nothing while cover traffic is the binding constraint.

Reliability over trajectories. A summed median can hide a bimodal outcome (a median of zero survivors is compatible with many trajectories having some), so Table XXXIV reports, per organisation, spending rule and attacker arm, the probability that every attacker-owned alert is silenced and the 95th percentile of the residual count and of the cost alongside the median. The residual count is essentially not a sampling quantity here: in 58 of the 60 cells its 5th and 95th percentiles coincide, and in the remaining 2 they differ by one alert. With the deployment pool the sequential oracle silences every own alert in every trajectory in 15 of the 16 flow-only cells; the exception is *russellmitchell* under horizon-free spending, where exactly 1 alert survives in every trajectory ($P = 0.00$). What pad sampling does move is the cost: the 95th percentile exceeds the median by up to a factor of 1.90 (median ratio 1.00 across cells). The variation that matters is therefore across organisations, not across pad draws, and the sums of medians in Table III should be read as eight nearly deterministic per-organisation outcomes added together, not as one typical campaign.

Testing the premise of Corollary 2 rather than a proxy. The corollary needs one thing of the rest of the stream: no non-attacker hypothesis carries evidence at or above the cold-start threshold T/α . A baseline non-attacker alert does not by itself show that this fails, because under either spending rule an alert can fire only after earlier rejections have raised the level. Table XXXV therefore measures the premise directly on the 10 streams the joint attack ran on, rebuilt with the same chain and asserted to reproduce every stored baseline. The cold-start premise fails on 3 of them (*russellmitchell* at 1.50 with 1 episode at or above it; *shaw* at 2.29 with 2 episodes at or above it; *shaw* (host) at 2.29 with 4 episodes at or above it); on the others the largest non-attacker evidence is below T/α , and in 5 streams no non-attacker episode carries any evidence at all. Of the 16 baseline non-attacker alerts over both spending rules, 9 clear the $R = 0$ level at their own position and 7 fire only at a raised level. An attacker episode attains the ceiling in every cell but *shaw*, *shaw* (host), where the largest attacker evidence gives $c_{\text{crit}} = 1.53 < \rho = 2.29$ in each, so the stream-specific integer $\lfloor c_{\text{crit}} \rfloor + 1$ can be smaller than c_{int} there. The other three conditions the corollary rests on are read

from the joint run itself: own episodes without a usable prior pool are never padded and account for most of what survives; admitted pads from the prior pool are not zero-evidence, and where they fire the multiplier’s dilution is partly undone; and the residual counts are the observed outcome. The corollary is stated for horizon-uniform spending, so its multiplier under horizon-free spending is a heuristic transfer, reported as such.

E. Aggregation alternatives and the capacity boundary

Caps large enough to preserve the intended validity domain yield zero detections at the median window–seed cell (first-flow, the detection-favourable order; Table XXVII). In the primary-window weight sweep every scheme that retains the mean’s recall is defeated by one to eight leading flows, against 111 appended flows for the mean on the same episodes, and buying a larger L^* costs about 58% of the recall (Table XXVI). For the group maximum, direct group calibration replaces 1.8–2.4M calibration flows by 37–57k groups while leaving T essentially unchanged; by Corollary 1 that is 20 groups per hypothesis the construction cannot supply, so the margin collapses to about -0.97 and the primary and secondary windows issue no alerts (Appendix F-B). Appending is not the only lever against it either: creating hypotheses ahead of a target pushes it past the cold-start window, and in that construction’s short prefix every group-maximum detection falls to 62–295 inserted hypotheses (Table XXVIII).

F. Capacity-constrained sequential padding

The resulting group sizes locate the membership-cap boundary. Under horizon-free spending, the median padded primary-window episode has arity 63, and 5% of deployment episodes are at least that large. Per-alert horizon-aware pads have median arity 2,989, reached by only 0.22% of episodes. Once controller state is propagated, however, the attacker has median padded arity 91 at the primary window and 55 at the secondary, sizes reached by 3.8% and 4.3% of episodes.

We therefore rerun the sequential attacker under a capacity constraint: it pads an episode only when $m_t + r_t \leq n$ and otherwise leaves the original alert in place. This experiment does not truncate or rescore the complete clean and attacked streams, so it is not an end-to-end evaluation of an admission-control policy. Under canonical order and $n = 100$, 96 of the 105 horizon-aware primary-window alerts still fire: 44 because their pad would not fit even at the cold-start level, and 52 because those first alerts raise R enough to price later pads above the cap; at the secondary window 79 of 107 fire (33 and 46) by the same mechanisms. At a cap of 300, we observe 19 and 60 alerts firing, whereas the per-alert reading where $m_t + r_t^* > n$ predicts 91 and 95, and at 1,000 the sequential attacker silences both windows. A cap of 100 would alter 3.4% and 2.9% of benign episodes as currently formed. This is the share an implemented cap would alter, not a measurement of its effect on evidence, clean detection, or benign service.

The same constraint does not stop the horizon-free attacker: all three primary-window alerts remain suppressible at $n \geq 100$. Reaching them requires $n = 30$, which would alter 7.4% of

benign episodes. Thus the experiment identifies a mitigation boundary under an enforced capacity model; it does not establish the performance of a working defence.

Static versus running pricing. Pricing against the static first-step threshold T/w_0 is the cheap extreme (median 34 flows at 0.85 under the canonical order); the body reports the larger running-level cost. Whole-detected-set totals are upper bounds, since suppressing an early rejection lowers R and can make the rest cheaper or moot, and are heavy-tail driven at their extreme.

Insertion against padding. Insertion beats padding for one alert at none of the twelve window–order cells, but silences the entire controller for 217–813 inserted hypotheses against $80\text{--}7.9 \times 10^8$ flows to pad every detection separately; per episode the targeted insertion cost is 434–5,674 hypotheses under the canonical and first-flow orders (down to 407 under a keyed hash). Under a public hash the ranking is an offline search we price at $1.0 \times 10^4\text{--}1.8 \times 10^5$ candidate keys, on three stated conditions (a public seed, enough endpoint pairs the adversary can send between, and a stable key encoding).

G. The joint controller-state rerun

The per-alert costs of Section VI-B price each alert at the level it received on the unperturbed trajectory. Table XXXVII instead propagates the zero-evidence pads that the real-flow replay of Section VI-A validates through the controller, which is re-run over the padded stream, so that its rejection count — and with it every later level $\alpha_t = \alpha\gamma_t(R_{t-1}+1)$ — responds to the attack. The black-box pool contains no flow that reaches the conformal tail at either window, so this is the zero-evidence limit of the real-flow replay, not a second replay. Three attackers are run. *Static*: the per-alert oracle pads of Table II, applied to every true detection at once. *Sequential*: a greedy oracle that pads each of its own episodes that would fire at the *live* level by the minimum; its total J_{seq} is one strategy’s cost on this stream, an upper bound on the minimum joint cost over all attacker-controlled episodes in the window. *State-free*: every own episode is multiplied by the integer $c_{\text{int}} = \lfloor \rho \rfloor + 1$ of Corollary 2, which is 3 at the primary and 3 at the secondary window, and silences the arm with no controller state. With $R = 0$ throughout, the stream-specific boundary is $c_{\text{crit}} = \max_j \text{Ev}_j \alpha\gamma_{t_j} \leq M\alpha/T = \rho$, with equality when a malicious episode attains the ceiling, as one does at each window ($c_{\text{crit}} = \rho = 2.134$ and 2.485); rejection is inclusive, so the boundary must be strictly exceeded, the real-valued checks just above and below c_{crit} are a continuous relaxation, and only the integer c_{int} is a realisable prescription. The lower panel re-runs the sequential attacker under a per-host-pair volume cap n : it pads only if $m_t + r \leq n$ and otherwise lets the alert fire and raise R , and each surviving alert is classified as structural or as a cascade victim by whether its pad would have fitted at $R = 0$. Every padded stream was regrouped from flows and asserted against the unperturbed stream on what an append must leave unchanged — the episode count, the (source, destination, bucket) key of every controller position and hence the permutation, each group’s first timestamp and

TABLE XXI

REPRESENTATIVE DESIGN DIRECTIONS AFTER THEOREM 3, EVALUATED ON LSPR23 TWO-HOUR HOST-PAIR EPISODES. AGGREGATION AND WEIGHTING USE FIRST-FLOW ORDER. THE MEMBERSHIP-CAP ROW USES A CAPACITY-CONSTRAINED ATTACKER UNDER CANONICAL ORDER, NOT A COMPLETE ADMISSION POLICY. SUPPORTING MEASUREMENTS APPEAR IN SECTION VII-B AND APPENDIX E-E.

Design direction	Gain	Cost / boundary
Capped normalisation $\sum_i e_i/n_0$	Invariant to members appended below the cap n_0	Leaves the symmetric e-merging class above n_0 ; validity-preserving caps detect nothing (Table XXVII)
Pre-committed asymmetric weights [34]	Valid invariance to <i>appended</i> flows	A few <i>leading</i> flows defeat every scheme that keeps the mean’s recall; a larger reach costs about 58% of it (Table XXVI)
Group-calibrated maximum	Padding-robust aggregate	The smaller calibration unit destroys feasibility: no alerts at the primary or secondary window (Appendix F-B)
Per-host-pair membership cap (<i>outside the merging class</i>)	Makes pads above the cap infeasible in the capacity model; an early unpadded alert can raise later costs	Not evaluated as an end-to-end policy: baseline traffic is not truncated or rescored; the constraint leaves residual suppression and does not stop the horizon-free attacker (Section VII-B)

TABLE XXII

FIVE PADDING POOLS PRICED AGAINST THE *running* E-LOND LEVEL $1/\alpha_t$, DETECTOR SEED 0, TWO-HOUR GROUPING, AT THE SECONDARY WINDOW (0.62) AND THE STRESS-TEST WINDOW (0.85), UNDER **BOTH** WITHIN-BUCKET ORDERS — THE CANONICAL METADATA HASH TABLE I REPORTS AND THE FIRST-FLOW ARRIVAL ORDER KEPT AS THE OPTIMISTIC UPPER BOUND. ENTRIES ARE THE MEDIAN SUPPRESSION COST r PER DETECTED EPISODE, OVER THE DETECTED SET THAT ORDER PRODUCES (COUNTS IN THE BLOCK HEADINGS); THE ORDER CHANGES *which* EPISODES MUST BE PAID FOR AND HOW DEEP INTO THE LEVEL SEQUENCE THEY SIT, WHICH IS WHY THE TWO BLOCKS DIFFER BY MORE THAN A CONSTANT. FOUR POOLS COINCIDE WITHIN EACH BLOCK; SERVICE-MATCHED IS DEARER, BOTH FROM A LARGER MEAN E-VALUE AND FROM BEING DEFINED ON A COSTLIER EPISODE SUBSET (SECTION V-C). μ IS THE POOL-LEVEL MEAN E-VALUE AT 0.85 WHERE DEFINED, A PROPERTY OF THE POOL AND NOT OF THE ORDER. PROTOCOL- AND SERVICE-MATCHED USE THE EPISODE’S MODAL PROTOCOL/PORT. COSTS ARE LOWER BOUNDS.

Padding pool	μ (0.85)	median r (0.62)	median r (0.85)
<i>canonical metadata-hash order (headline)</i> — 11 detected at 0.62, 34 at 0.85			
generic benign	50.9	6	116
attacker-origin	216.8	6	116
protocol-matched	—	6	116
service-matched	—	7	175
black-box	0.0	6	116
<i>first-flow arrival (optimistic upper bound)</i> — 13 detected at 0.62, 72 at 0.85			
generic benign	50.9	72	5,212
attacker-origin	216.8	72	5,247
protocol-matched	—	72	5,232
service-matched	—	99	10,299
black-box	0.0	72	5,202

its label — and asserted to carry arities $m_t + r_t$ and exactly the padded group means $S_t/(m_t + r_t)$ that the attacker was priced on; the padded means differ from the unperturbed ones by construction.

Capacity interpretation. The lower panel constrains the attacker’s additions; it does not apply an admission policy to the complete stream, and baseline groups are not truncated or rescored.

APPENDIX F VALIDITY AND DATA QUALITY

Table XXXVIII records what each result depends on; Appendix F-A audits the calibration tail against the labels, Appendix F-B calibrates the grouped unit directly, and Appendix F-C prices calibration-set contamination.

TABLE XXIII

PADDING SUPPRESSION COST PRICED AGAINST THE RUNNING CONTROLLER LEVEL $1/\alpha_t$ (REAL) VS THE STATIC T/w_0 THRESHOLD. PART (A) GIVES *both* WITHIN-BUCKET ORDERS AT BOTH DETECTOR SEEDS: THE **CANONICAL** METADATA-HASH ORDER THE PAPER REPORTS, AND THE FIRST-FLOW ARRIVAL ORDER RETAINED AS AN OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT (TABLE XIX). DETECTIONS AND THE LEVEL AN ALERT FIRES AT BOTH MOVE WITH THE ORDER; T , $|C|$, AUROC AND THE MARGIN DO NOT. PART (B)’S POOL COMPARISON IS ON THE FIRST-FLOW DETECTED SET, THE LARGER OF THE TWO; TABLE XXII GIVES THE SAME COMPARISON UNDER *both* ORDERS. FOUR POOLS COINCIDE; SERVICE-MATCHED IS DEARER. MEDIANS ARE ROUNDED TO THE NEAREST INTEGER; “—” MARKS A WINDOW/SEED/ORDER WITH NO DETECTIONS.

(a) per window $\times$ seed $\times$ order (e-LOND poly, two-hour grouping)							
pos	seed	canonical key-hash			first-flow (upper bd.)		
		det.	pad (real)	pad (static)	det.	pad (real)	pad (static)
0.55	0	3	24	3	18	118	4
0.55	1	2	27	2	18	101	3
0.62	0	11	6	4	13	72	4
0.62	1	8	5	2	12	69	0
0.70	0	0	—	—	30	1,851	62
0.70	1	0	—	—	0	—	—
0.77	0	0	—	—	31	1,165	85
0.77	1	0	—	—	31	1,289	90
0.85	0	34	116	34	72	5,202	33
0.85	1	32	109	34	71	6,802	33

(b) five padding pools, seed 0 (median suppression cost r)			
padding pool	real (0.62)	real (0.85)	static (0.85)
generic benign	72	5,212	33
attacker-origin	72	5,247	33
protocol-matched	72	5,232	33
service-matched	99	10,299	35
black-box	72	5,202	33

A. Calibration validity and the labels

The measured-to-nominal benign firing ratio stays near one across the ten position $\times$ seed configurations (median $1.94\times$; independently $0.86\text{--}2.03\times$ on the AIT wilson testbed), with a single exception, the stress window 0.85, where the benign firing rate runs $50.9\times$ nominal (exact interval $[37.3, 67.9]$), so $\mathbb{E}[e] \approx 51$ and the evidence is not a valid e-value there (Table XXXIX). The excess is sharply localised: three host pairs carry 44 of the 46 firing benign flows, and removing them restores the ratio to $2.22\times$. It is best explained by label error, which distribution shift, a new benign mode, duplicated records

TABLE XXIV

CONTROLLED PADDING-DILUTION REPLAY ON THE SHIPPED DETECTOR: REAL FLOWS OF THE BLACK-BOX POOL APPENDED TO EVERY DETECTED EPISODE UNDER BOTH WITHIN-BUCKET ORDERS (CANONICAL KEY-HASH AND FIRST-FLOW); SRC-DST TWO-HOUR UNIT, E-LOND $\gamma \propto j^{-1.6}$, SEED 0. DESIGN AND READING: APPENDIX E-C.

window	within-bucket order	pool flows	pad fires	95% upper	detected	measured = c.f.	median r_t^*
primary (0.55)	first-flow (<i>upper bd.</i>)	845,691	0/20,000	1.5×10^{-4}	18	18/18	118
	key-hash (<i>canonical</i>)	845,691	0/20,000	1.5×10^{-4}	3	3/3	24
stress-test (0.85)	first-flow (<i>upper bd.</i>)	584,850	0/20,000	1.5×10^{-4}	72	72/72	5,202
	key-hash (<i>canonical</i>)	584,850	0/20,000	1.5×10^{-4}	34	34/34	116

Notes. “pool flows”: the black-box pool’s size in the window; “pad fires”: how many of n sampled pad flows fire when run through the detector, with the exact one-sided 95% Clopper–Pearson upper bound on the firing probability in the next column; “measured = c.f.”: detected episodes whose padded group e-value $S_t/(m_t + r)$ crosses the firing threshold at exactly the predicted $r_t^* = \lfloor S_t \alpha_t \rfloor - m_t + 1$; “median r_t^* ”: over the detected episodes of the row, priced at the running level.

and Mondrian conformal each fail to account for, though we do not prove it. The other four windows are *not refuted* ($1.07\text{--}3.79\times$, every interval containing 1, and at 0.55 the interval $[0.03, 5.96]$ excludes essentially nothing); that is a failure to reject and not support.

The marginal rate is the weakest possible evidence for Assumption 1, so we interrogate it conditionally, stratifying the firing benign flows on the observable components of $\mathcal{M}_j$ (arity, service, bucket, destination host) at rank depths $k \in \{1, \dots, 1000\}$, resampling both the shared calibration tail and the test side (Table XL). At the depth the pipeline uses the data mostly cannot speak: 129 of 134 strata carry no test at $k = 1$ (428 of 536 cells across the four depths), and none that are estimable reject at the primary or secondary window. Deeper in the tail, 32 contrasts survive correction at all five windows (25 after removing the one confounder we can identify), the strongest an arity bin at 0.77. A departure at $k = 1000$ is compatible with a valid rank-1 rate, so the guarantee-bearing claims stay stated under Assumption 1, not supported by this diagnostic. The Bates calibration-conditional adjustment (Table XLIII) is stronger than e-LOND needs and is treated as optional.

Stratified design. Assumption 1 constrains benign flows of *true-null* groups conditionally on that group’s metadata, whereas the marginal diagnostic (Table XXXIX) reads the marginal rate at the shipped depth $k=1$, where the four non-stress windows fire 0–3 benign flows. Table XL stratifies those flows on the observable components of $\mathcal{M}_j$ — episode **arity** (the quantity padding manipulates), **service** (proto, port), the two-hour **bucket** that fixes a group’s position, and high-support destination **host**, all fixed by an outcome-blind support rule — and reads them at $k \in \{1, 10, 100, 1000\}$. Both sources of uncertainty are resampled jointly, and both are clustered: a window’s whole diagnostic depends on the calibration set only through its k -th largest score, which every stratum shares, so the calibration tail is resampled over calibration *host pairs*; the test side is a Poisson-multiplier bootstrap over deployment host pairs. Multiplicity is corrected with Benjamini–Yekutieli, valid under the arbitrary dependence this design has, over *every* cell: a cell firing fewer than ten times carries no test,

TABLE XXV

SURFACE A PRICED **WITHOUT ORACLE STATE**: A RE-ANALYSIS OF THE SAME PER-EPISODE (m, S, α_t) THE ORACLE COSTS COME FROM, WITH NO RE-SIMULATION — THE CLOSED FORM (9) IS RE-DERIVED FROM THE STORED FIELDS ON ALL 127 EPISODES AS A READ-BACK CHECK. *Multiplier*: APPEND $(c - 1)m$ FLOWS, THAT IS, MULTIPLY YOUR OWN EPISODE BY c . IT DEFEATS THE ALERT IFF $c \geq 1 + r_t^*/m_t$ AND NEEDS **NO** ESTIMATE OF S_t , α_t OR $|C|$ — ONLY m , WHICH ON LSPR23 IS THE ATTACKER’S OWN FLOW COUNT, SINCE EVERY HOST PAIR CARRYING ATTACK TRAFFIC IS 100% MALICIOUS. “PADDED ARITY” IS cm , AND THE COLUMN BESIDE IT ANSWERS THE DEPLOYMENT-SIDE QUESTION — WOULD A VOLUME MONITOR SEE THE PAD — AS THE FRACTION OF *deployment* EPISODES LARGER THAN THE PADDED ONE, BRACKETED BY THE TWO STORED GROUP-SIZE QUANTILES THAT STRADDLE IT. THOSE QUANTILES ARE *not* SHOWN AND ARE *not* BENIGN DEPLOYMENT VALUES: THEY ARE QUANTILES OF SOURCE–DESTINATION GROUP SIZE OVER THE *calibration* SLICE, UNFILTERED BY LABEL, WHICH IS WHY THE DEPLOYMENT EXCEEDANCE AND NOT THE QUANTILE IS WHAT THE CLAIM RESTS ON. ROWS ARE GIVEN UNDER **BOTH** WITHIN-BUCKET ORDERS — THE CANONICAL METADATA HASH THE PAPER REPORTS AND **FIRST-FLOW** ARRIVAL, RETAINED AS THE OPTIMISTIC UPPER BOUND. *Absolute*: APPEND A PRE-COMMITTED FLAT N FLOWS, WHICH NEEDS NOTHING AT ALL; N IS REPORTED AT THE FRACTION OF THAT CELL’S ALERTS IT SUPPRESSES. THE MULTIPLIER ARM COVERS THE TWO WINDOWS FOR WHICH PER-EPISODE ARITIES ARE STORED; THE SECONDARY WINDOW APPEARS IN THE ABSOLUTE ARM ONLY. BOTH STRATEGIES ARE UPPER BOUNDS ON THE ORACLE COST, AND BOTH ARE OPTIMISTIC ON A BENIGN-INCLUSIVE STREAM, WHERE m ALSO CONTAINS BENIGN FLOWS THE ATTACKER CANNOT SEE.

multiplier order	needs the attacker's own arity m and nothing else				c used	med. added	vs oracle	padded arity	% deployment larger
	pos	alerts	med. c	max c					
canonical	0.55	3	1.57	2.74	3	80	3.5 $\times$	120	0.9–9.9%
canonical	0.85	34	2.76	6.17	10	792	5.1 $\times$	880	0.1–1.1%
first-flow	0.55	18	3.54	7.44	10	414	3.6 $\times$	460	0.1–0.9%
first-flow	0.85	72	73.19	4302.15	none ≤ 100	—	—	—	—

absolute — a flat pad: executing it needs no state, but the N below is read off the realised r_t^* distribution									
order	pos	seed	alerts	N for 50%	N for 90%	N for all			
canonical	0.55	0	3	24	33	33			
canonical	0.55	1	2	30	30	30			
canonical	0.62	0	11	6	17	859			
canonical	0.62	1	8	6	659	659			
canonical	0.85	0	34	118	9,395	58,616			
canonical	0.85	1	32	111	9,426	58,620			
first-flow	0.55	0	18	125	645	837			
first-flow	0.55	1	18	108	645	829			
first-flow	0.62	0	13	72	127	134			
first-flow	0.62	1	12	70	93	98			
first-flow	0.77	0	31	1,165	3,736	7,886			
first-flow	0.77	1	31	1,289	4,258	7,886			
first-flow	0.70	0	30	1,877	4,145	5,445			
first-flow	0.85	0	72	6,714	69,195,773	145,525,527			
first-flow	0.85	1	71	6,802	69,086,348	145,096,310			

but still enters the family at $p = 1$, since dropping it would shrink the denominator using the same tail count that drives significance. A *contrast* is a stratum departing from its *own* window’s marginal on the same replicates — the quantity that speaks to the conditional statement, since the shared calibration draw largely cancels in a ratio of two rates read at one threshold.

Reading, in both directions. 428 of the 536 cells carry no test at all (419 of them fire fewer than ten times), which is itself the point: at the depth the pipeline uses, the data mostly cannot speak. Of the 40 cells that do reject, **none at $k=1$ rejects at the primary or secondary window** — all 5 $k=1$ rejections sit at 0.85, already an unambiguous violation, which is an *inconclusive non-rejection* and not evidence for Assumption 1: at $k=1$ most cells are simply uninformative. Deeper in the tail the conditional statement is another matter: 32 stratum contrasts survive correction, at windows 0.55, 0.62, 0.70, 0.77, 0.85. The largest lower bound is $m=21\text{--}100$ at 0.77, $k=1000$ ($9.3 \times [6.5, 13.7]$); the largest point estimate is proto 6/port 6444 at 0.62, $k=10$ ($14.4\times$). The *arity* family —

TABLE XXVI

PRE-COMMITTED WEIGHT SCHEMES AT THE PRIMARY WINDOW, LOND, $\gamma \propto j^{-1.6}$, MEAN OVER TWO DETECTOR SEEDS. $T = 57,368$ EPISODES, 275 MALICIOUS, $|C| = 2,448,993$. L^* IS QUOTED AT THE UNPERTURBED LEVEL AND IS THEREFORE AN UPPER BOUND. ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX); BOTH LEGS OF EVERY COMPARISON USE IT, SO THE ORDER MOVES THE LEVEL AND NOT THE RANKING.

Rule	pad-inv.	recall	L^*	appended	ratio
arithmetic mean (symmetric)	—	0.065	—	—	—
first-event-only	yes	0.062	1	111	112×
exp-decay $\rho = 0.5$	yes	0.062	2	111	56×
uniform over 10 slots	yes	0.062	8	111	14×
uniform over 100 slots	yes	0.027	81	492	6×
exp-decay $\rho = 0.99$	yes	0.027	164	492	3×
last-event-only	no	0.045	—	336	—
security-prior (<i>out of class</i>)	yes	0.000	—	—	—

TABLE XXVII

AGGREGATION-CAP SWEEP (MEDIAN OVER FIVE POSITIONS $\times$ TWO SEEDS): THE FROZEN $n_0 = p_{99}$ IS CHOSEN FOR VALIDITY AND ATTACK COST, NOT POWER. “% GROUPS $m > n_0$ ” IS THE SHARE OF GROUPS WHOSE SIZE EXCEEDS THE CAP; “FRONT-LOAD COST” IS THE NUMBER OF APPENDED FLOWS NEEDED TO DEFEAT THE FRONT-LOADING DEFENCE. ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX); THE ORDER SHIFTS DETECTIONS AND THE LEVEL AN ALERT FIRES AT, AND SHIFTS NOTHING ELSE.

cap	n_0	% groups $m > n_0$	det. raw	det. trunc.	front-load cost
mean	47	5.72	39.5	35	16
p50	2	37.58	99	71	1
p90	21	10.24	68.5	60	8
p99	408	1.08	0	0	116
p999	7,258	0.12	0	0	3,560
max	97,911	0.01	0	0	—

the component of $\mathcal{M}_j$ the padding attack moves — carries 3 of them, the strongest being $m=21-100$ at window 0.77, $k=1000$, firing $9.3 \times [6.5, 13.7]$ its own window’s rate. One dependence remains unresampled: host-pair clustering does not absorb a shock shared across many pairs at once, and a window’s calibration tail concentrates in its two to four buckets, so these counts may be optimistic — dropping the one stratum family aligned with that confounder leaves 25 contrasts at windows 0.55, 0.62, 0.70, 0.77, 0.85. A departure at $k=1000$ is nonetheless compatible with a valid rank-1 rate — the excess may sit entirely in ranks 2..1000 — so this tests a *necessary implication* where it is estimable, not the assumption itself. Every guarantee-bearing conclusion in this paper is therefore stated *under* Assumption 1, and this diagnostic is not offered as support for it.

At 0.85 every FDP is measured against labels. Auditing the ADDIS ($\gamma \propto j^{-1.6}$) set’s 152 alerts against the red-team task record, 146 agree (96.1%) and six disagree, so an under-estimate is not excluded; e-LOND’s realised FDP is 0.0066 (frozen) or $[0, 0.0265]$ (strict post-hoc). The audit is partly circular (99 of 152 verdicts rest on the label-generating identities), so it indicates the error’s direction, not a value

TABLE XXVIII

SUPPRESSION BY INSERTION, NOT BY PADDING. THEOREM 3 AND EVERY COST IN SECTION V-C CONCERN *appending* FLOWS TO A FIXED HYPOTHESIS. THIS ATTACK APPENDS NOTHING: it *creates* HYPOTHESES AHEAD OF THE TARGET, PUSHING IT OUT OF THE COLD-START WINDOW PAST WHICH NO REJECTION IS POSSIBLE. TWO MODELS ARE KEPT APART BECAUSE THEY ARE DIFFERENT ATTACKS. **TARGETED** INSERTS ONLY IMMEDIATELY AHEAD OF ONE EPISODE, WHICH LEAVES EVERY EARLIER HYPOTHESIS AND HENCE R UNTOUCHED, SO ITS COST IS CLOSED-FORM $G^* = \lfloor (\alpha(R+1)E/\zeta(1.6))^{1/1.6} \rfloor - p$ (REPLAY-VERIFIED). **WINDOW** INSERTS AHEAD OF EVERYTHING AND SILENCES THE CONTROLLER OUTRIGHT — THAT IS *not* a PER-EPISODE COST, SINCE IT ALSO DELETES THE EARLIER DETECTIONS THAT RAISED R . **THE SPLIT VERDICT IS THE RESULT:** INSERTION DOES *not* BEAT PADDING FOR ONE ALERT (0/12 CELLS), BUT SILENCES THE WHOLE WINDOW FOR A FEW HUNDRED HYPOTHESES (10/12 CELLS AT ONE FLOW PER INSERTED GROUP, 7/12 AT TWO). PADDING TOTALS ARE **UPPER BOUNDS** — THE JOINT COST IS LOWER, BECAUSE SUPPRESSING AN EARLY REJECTION LOWERS R — AND THE LARGEST RATIO IS HEAVY-TAIL DRIVEN. PANEL (B) IS THE CONSTRUCTION OF APPENDIX F-B, WHICH PADDING PROBABLY CANNOT SUPPRESS: CALIBRATING ON *groups* SHRINKS THE CEILING, SO ITS COLD-START WINDOW COLLAPSES TO 65–86 AND EVERY DETECTION FALLS TO A FEW DOZEN INSERTIONS. *Group validity by design plus padding robustness does not buy attack resistance.* GETTING AHEAD OF A TARGET IS FREE UNDER FIRST-FLOW ARRIVAL (SEND EARLIER IN THE BUCKET), COSTS AN OFFLINE KEYSPEC SEARCH UNDER A PUBLIC HASH, AND CANNOT BE RANKED OFFLINE AT ALL UNDER A KEYED ONE — WHICH REMOVES the *search*, NOT THE ATTACK. BOTH WITHIN-BUCKET ORDERS ARE REPORTED: THE CANONICAL METADATA KEY-HASH TABLE I USES, AND **FIRST-FLOW** ARRIVAL, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT (TABLE XIX).

(a) the shipped flow-level pipeline							
pos	order	det.	per episode		silence the window		cheaper
			insert	pad (flows)	insert	pad (upper)	
0.55	key-hash	3	434	24.0	217	80	—
0.55	first-flow	18	1,696	118.0	502	4,432	✓
0.62	key-hash	11	605	6.0	813	926	✓
0.62	keyed	6	452	14.5	506	126	—
0.62	first-flow	13	2,284	72.0	554	1,060	✓
0.70	keyed	3	407	97.0	257	310	✓
0.70	first-flow	30	4,088	1,850.5	706	62,250	✓
0.77	keyed	3	556	323.0	416	1,205	✓
0.77	first-flow	31	3,161	1,165.0	789	49,957	✓
0.85	key-hash	34	1,414	115.5	509	151,390	✓
0.85	keyed	24	998	160.5	518	162,496	✓
0.85	first-flow	72	5,674	5,201.5	743	786,257,264	✓

(b) the group-calibrated MAX pipeline — padding cannot suppress it at all						
pos	groups	firing	detections	cold start	targeted G^*	window G
0.55	57,368	110	0	81	—	—
0.62	49,267	109	0	83	—	—
0.70	37,231	69	1	86	81.0	81
0.77	31,672	60	2	78	76.5	68
0.85	31,568	152	42	65	207.0	62

(Table XLII).

B. Calibrating the group directly

The constructive alternative to Assumption 1 is to run split conformal on the grouped unit, same key and protocol, so that only calibration/test *group* exchangeability is required. Built and measured over two group statistics, five windows and two seeds (Table XLI), it discharges the premise but fails as a repair. Group exchangeability is still a deployment assumption, and our threat model is one under which test groups are partly adversary-constituted: insertion instantiates endpoint pairs that enter the stream as true nulls, so the group-formation process is

TABLE XXIX

HOST-CONDITIONED DETECTOR AND THE PADDING-TRANSFER BOUNDARY ON LSPR23 (SRC-DST TWO-HOUR UNIT, E-LOND $\gamma \propto j^{-1.6}$, SEED 0; **FIRST-FLOW** WITHIN-BUCKET ORDER ON BOTH LEGS, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL ORDER TABLE I REPORTS). “H/F”: HOST-CONDITIONED / FLOW-ONLY DETECTOR. DESIGN AND READING: APPENDIX E-D.

window	AUROC h/f	tail reach h/f	det. h/f	real benign fires	graft fire	median cost h/f	transfer
primary (0.55)	0.954/0.916	0.009/0.033	2/18	0/2,287,988	0.000	108/118	holds
stress-test (0.85)	1.000/0.999	0.295/0.610	37/72	3/1,638,722	0.000	11,484/5,202	holds

Notes. “AUROC”: area under the ROC curve on the window’s flows; “tail reach”: the fraction of attack flows scoring above every benign calibration score; “det.”: e-LOND detections; “real benign fires”: host-detector firings among the window’s benign flows; “graft fire”: the grafted-pad firing rate on detected episodes across accumulation; “median cost h/f”: the median suppression cost over detected episodes under the causally recomputed host context / the flow-level closed form r_t^* ; “transfer”: whether the flow-level closed form still prices the suppression. All counts here use the **first-flow** within-bucket order, the optimistic upper bound over the orders we audit, not the canonical metadata-hash order Table I reports (Table XIX); the order shifts detections and the level an alert fires at, and shifts nothing else.

something the adversary moves. The MAX statistic makes that fragility acute: its rank correlation with arity is $+0.23$ – $+0.41$ across the ten cells, and while marginal coverage is roughly right (0.55 – $2.32 \times$ nominal), arity-conditional coverage runs from $0.13 \times$ to $24.2 \times$ nominal under MAX against 0.00 – $5.6 \times$ under the mean. Stratifying calibration on arity restores a per-stratum guarantee and leaves each stratum a cold-start window of 2–52 hypotheses. The robustness and the fragility are one fact: appending cannot lower a maximum, which is why Theorem 3 does not run against it and why the arity dependence is there at all.

Grant the repair its premise and it still cannot alert. Group calibration replaces 1.8–2.4M flows by 37–57k groups, so M falls 40 – $49 \times$ at unchanged T : the margin moves from $+0.067$ to -0.977 at the primary window, no window is feasible, and detection at the primary and secondary windows falls from 18 and 13 (first-flow) to zero. At 0.55 the cold-start window is 81 hypotheses while the first of 110 firing groups arrives at position 401, and feasible-step counts match $\lfloor (\alpha M(R+1)/\zeta(1.6))^{1/1.6} \rfloor$ everywhere. Calibration groups containing any malicious flow are excluded, dropping up to 26.1% of calibration flows in groups whose mean arity runs to $45.4 \times$ the benign mean, a bias that favours firing, so the negative result survives its own bias.

C. Calibration contamination

At $k = 1$ the conformal rule is one order statistic, the calibration maximum, so contamination acts only through it. Injecting j attack flows above that maximum hands the threshold to the attacker once $j \geq k$, so tolerable adversarial mislabels are exactly $k - 1 = 0$. One mislabelled flow above the calibration maximum takes episode recall to 0.000, from 0.011 at 0.55 and 0.133 at 0.85 under the canonical order (0.065 and 0.282 under first-flow arrival). A random mislabel is milder but not much: median recall 0.004 at 0.55 and 0.035 at 0.85, with zero recall in 22.5% and 32.25% of draws, at a scale $\varepsilon^* = 1/(Nq_0)$ of 1.11–1.31 flows. A contaminating flow *below* the maximum only inflates the p-value denominator,

TABLE XXX

THE PADDING-TRANSFER BOUNDARY RE-MEASURED ON A *benign-inclusive* TESTBED (AIT-LDSv2.0, 8 ORGANISATIONS, **45,001** ATTACK FLOWS), WHERE VICTIMS SERVE HEAVY REAL BENIGN TRAFFIC WHILE ATTACKED, SO THE PADDING QUESTION IS ANSWERED ON *real* ORDINARY-TO-VICTIM FLOWS — **NO GRAFT**. LEAVE-ONE-ORGANISATION-OUT (TRAIN 7, TEST 1; A “DEPLOY ON A NEW NETWORK” TEST), IN-ORGANISATION BENIGN CONFORMAL CALIBRATION, FLOW FEATURES SELECTED PER FOLD FROM THE TRAINING ORGS WITH ALL IDENTITY AND *absolute-time* FIELDS EXCLUDED. “BENIGN→VICTIM” IS THE FIRE RATE OF REAL BENIGN FLOWS TO AN ATTACKED HOST (THE PAD ANALOGUE), HOST / FLOW-ONLY. HOST CONDITIONING ADDS LITTLE AND INCONSISTENTLY (MEDIAN HOST—FLOW DELTA -0.0000 ; HOST HIGHER ON $1/8$ ORGS, MAX 8.9%), WITH SHAW THE ONE ORGANISATION WHERE IT IS MATERIALLY HIGHER. **THAT PER-ORGANISATION EXCEPTION IS REAL BUT DOES NOT SURVIVE AS A DEFENCE**. A RAISED ORDINARY-TO-VICTIM FIRE RATE IS MEASURED HERE *in each flow’s own context*; THE PADS AN ATTACKER ACTUALLY SENDS CARRY THE *attacked* PAIR’S CONTEXT, AND REPLAYING THE WHOLE SURFACE A CHAIN UNDER THAT CONTEXT SUPPRESSES EVERY DETECTED SHAW EPISODE (TABLE XXXI). SO THE HONEST READING IS THAT HOST CONDITIONING IS USUALLY NEGLIGIBLE HERE AND CAN LOOK DECISIVE ON AN INDIVIDUAL ORGANISATION, BUT REPLAYING THE CHAIN END-TO-END AT ALL EIGHT ORGANISATIONS IT DOES NOT STOP THE ATTACK ON MOST EPISODES ONCE THE PAD’S OWN CONTEXT IS MODELLED, AND WHERE IT DOES THE BINDING CONSTRAINT IS THE NUMBER OF PADS REQUIRED (TABLE XXXI). THIS ALSO SETTLES THE LSPR23 SIDE: THE 0.43 GRAFTED FIRING AT THE STRESS WINDOW COMES FROM A NON-BLACK-BOX PAD POOL THAT IS MOSTLY THE ATTACKER’S OWN FLOOD (TABLE XXIX). AIT HAS NO HIGH-VOLUME FLOOD, WHICH REMAINS OPEN. THE FLOW-ONLY FOLDS ARE AUDITED UNDER **BOTH** WITHIN-BUCKET ORDERS IN TABLE XXXII: THE CANONICAL ORDER DETECTS FEWER EPISODES AND ITS ALERTS ARE CHEAPER TO SUPPRESS, AS ON LSPR23. ROWS SEQUENCED BY **FIRST-FLOW** ARRIVAL ARE CONDITIONAL ON THE DETECTED SET THAT ORDER PRODUCES ON THIS TESTBED.

organisation	AUROC h/f	atk. recall	benign→victim % (h/f)
fox	0.995/0.996	0.93	0.00/0.01
harrison	0.975/0.973	0.86	0.01/0.01
russellmit	0.988/1.000	0.47	0.00/0.02
santos	0.948/0.980	0.36	0.00/0.01
shaw	0.955/0.932	0.47	8.92/0.05
wardbeck	0.969/0.907	0.39	0.14/0.01
wheeler	0.999/0.998	0.97	0.01/0.01
wilson	0.974/0.984	0.85	0.07/0.00

degrading the guarantee to at most $q(1 + \varepsilon)$, so power dies at one flow while validity barely moves. The feasibility margin is not a safety indicator: it stays at $+0.436$ throughout the sweep that destroys detection.

APPENDIX G

ADAPTIVE CONTROLLER-STATE EXTENSION

Question answered. A controller can escape the finite horizon by conditioning its spending index on observed evidence (Appendix C-B). Does that escape create a new attack surface? *Takeaway.* It does: the attacker supplies the evidence that drives the state. The result is mechanism-specific and conspicuous in volume, which is why it is an extension rather than one of the paper’s main empirical claims.

The padding attack of Section V manipulates the composition of one hypothesis. A distinct surface appears when a controller escapes Theorem 1 by conditioning its spending index on observed evidence. ADDIS advances its index on selected non-candidates, whose p-values lie in $(\lambda, \tau]$, and never rejects

TABLE XXXI

FULL SUPPRESSION PIPELINE ON THE BENIGN-INCLUSIVE AIT TESTBED: PER ORGANISATION, A LEAVE-ONE-ORGANISATION-OUT DETECTOR, IN-ORGANISATION BENIGN CALIBRATION PRECEDING THE FIRST ATTACK, TWO-HOUR HOST-PAIR EPISODES, E-LOND, AND EACH DETECTED MALICIOUS EPISODE SUPPRESSED BY REPLAYING REAL ORDINARY-TO-VICTIM FLOWS AGAINST THE RUNNING CONTROLLER LEVEL (200 DRAWS). ROWS ARE SEQUENCED BY **FIRST-FLOW** ARRIVAL AND ARE CONDITIONAL ON THE DETECTED SET THAT ORDER PRODUCES; THE FLOW-ONLY FOLDS UNDER BOTH ORDERS ARE TABLE XXXII. DESIGN AND READING: APPENDIX E-D.

org	det.	pad fire	$r_{t,\text{emp}}^*$	r_t^* (closed)	suppr.	succ.
<i>flow-only detector</i>						
fox	13/14	1.3e-04	277	277	13/13	1.00
harrison	10/14	1.8e-04	1,412.5	1,412.5	10/10	1.00
russellmitchell	14/14	5.2e-04	3,551.5	3,365.5	14/14	0.97
santos	9/9	4.7e-04	212	212	8/9	0.79
shaw	3/6	0.0e+00	404	404	3/3	1.00
wardbeck	9/9	3.2e-04	13,142	8,294	9/9	0.90
wheeler	15/16	2.1e-04	1,609	1,609	15/15	1.00
wilson	12/13	4.3e-05	793	793	12/12	1.00
<i>host-conditioned detector, end-to-end</i>						
fox	9/14	0.0e+00	115	55	9/9	1.00
harrison	7/14	2.9e-04	792.5	692	6/7	0.86
russellmitchell	9/14	0.0e+00	1,140	1,140	9/9	1.00
santos	5/9	0.0e+00	43	43	5/5	1.00
shaw	3/6	7.0e-02	269	269	3/3	1.00
wardbeck	9/9	1.3e-03	8,471.5	8,263.5	8/9	0.74
wheeler	15/16	3.6e-04	1,609	1,486	11/15	0.61
wilson	6/13	0.0e+00	355	355	6/6	1.00

Notes. “det.”: e-LOND true detections over malicious episodes; “pad fire”: the real ordinary-to-victim firing rate as observed; “ $r_{t,\text{emp}}^*$ ”: the empirical median replay cost over suppressible episodes; “ r_t^* (closed)”: the zero-pad closed-form lower bound over the same episodes ($\leq$ empirical by construction); “suppr.”: detected episodes for which at least one of the 200 draws brought the group below threshold inside the scored range — a possibility count, not a reliability one; “succ.”: the mean per-draw success, which is the reliability the count does not carry. An episode outside “suppr.” had no successful draw within that budget, which does not establish that it cannot be padded.

them. Concretely it offers $\alpha_t = \min\{\lambda, (\tau - \lambda)[W\gamma_{S^t - C_{0+}} + \dots]\}$ at $\lambda = 0.25$, $\tau = 0.5$, where $S^t - C_{0+}$ counts exactly those selected non-candidates and γ is the 0-indexed ($j + 1$) $^{-1.6}/\zeta(1.6)$; the attack moves that index and nothing else (Appendix B). An attacker who prepends groups in the interval therefore advances the index without spending any discovery of its own, and can force the future level below the conformal floor.

Theorem 6 (ADDIS state budget). *Under ADDIS’s 0-indexed $\gamma_j = (j + 1)^{-1.6}/\zeta(1.6)$ with $0 < \lambda < \tau < 1$, rank- k conformal evidence, precursors injected at index 0, and a cap at or above the conformal floor; $\lambda \geq k/(|C| + 1)$, the number of precursor hypotheses that silences every later target is B^* , the smallest D with*

$$D + 1 > \left\lceil \frac{(\tau - \lambda) W (|C| + 1)}{k \zeta(1.6)} \right\rceil^{1/1.6},$$

where $W = w_0$ before the first rejection. After $R \geq 1$ rejections ADDIS’s reward terms sit at their own selected-non-candidate lags instead of collapsing to one index. Their coefficients still sum to exactly αR , but each lagged term carries $\gamma_{D-\text{lag}} \geq \gamma_D$ because γ decreases, so the driving level

TABLE XXXII

THE AIT TRANSFER UNDER **BOTH** WITHIN-BUCKET ORDERS, FLOW-ONLY FOLDS. THIS IS A *resequencing*: THE LEAVE-ONE-ORGANISATION-OUT FEATURE SELECTION, THE PER-FOLD DETECTOR HIT, THE SCORES, THE CHRONOLOGICAL IN-ORGANISATION CALIBRATION AND THE E-VALUES ARE ALL COMPUTED BEFORE EPISODES ARE ORDERED, SO THE TWO ARMS DIFFER BY THE PERMUTATION ALONE; THE CANONICAL ARM USES THE SAME KEY-HASH RULE AS THE LSPR23 RESULTS. THE STORED FIRST-FLOW ARM IS RE-DERIVED HERE AS A READ-BACK CHECK THAT THIS IS THE SAME CHAIN AND NOT MERELY A SIMILAR ONE (IT REPRODUCES 84/85). THE CANONICAL ORDER DETECTS FEWER EPISODES, AS ON LSPR23, AND

THE ALERTS IT DOES ISSUE ARE *cheaper* TO SUPPRESS — PER-ORGANISATION MEDIAN $r_{t,\text{emp}}^*$ FALLS FROM 212–13142 TO 12–4650 FLOWS. THE HOST-CONDITIONED ARM RUNS AT ALL EIGHT ORGANISATIONS AND IS LEFT ON FIRST-FLOW (TABLE XXX). “DET./MAL.” COUNTS *true* DETECTIONS AGAINST MALICIOUS EPISODES; THE EIGHT PER-ORGANISATION RUNS ISSUE 83 REJECTIONS UNDER THE CANONICAL ORDER AND 89 UNDER FIRST-FLOW, CARRYING 4 AND 4 FALSE DISCOVERIES RESPECTIVELY (POOLED REALISED FDP 0.048 AND 0.045); COSTS ARE EMPIRICAL REPLAY MEDIANS $r_{t,\text{emp}}^*$, AT OR ABOVE THE ZERO-PAD LOWER BOUND r_t^* (TABLE XXXI LISTS BOTH FOR THE FIRST-FLOW ARM).

organisation	det./mal.	median $r_{t,\text{emp}}^*$	suppressible	per-draw succ.
<i>canonical order — 78/79 suppressible</i>				
fox	11/14	12	11/11	1.00
harrison	9/14	50	9/9	1.00
russellmitchell	14/14	4,650	13/14	0.93
santos	8/9	35.5	8/8	1.00
shaw	3/6	26	3/3	1.00
wardbeck	8/9	65	8/8	1.00
wheeler	15/16	192	15/15	1.00
wilson	11/13	274	11/11	1.00
<i>first-flow order — 84/85 suppressible</i>				
fox	13/14	277	13/13	1.00
harrison	10/14	1412.5	10/10	1.00
russellmitchell	14/14	3551.5	14/14	0.97
santos	9/9	212	8/9	0.79
shaw	3/6	404	3/3	1.00
wardbeck	9/9	13,142	9/9	0.90
wheeler	15/16	1,609	15/15	1.00
wilson	12/13	793	12/12	1.00

is at least $(\tau - \lambda)\alpha R\gamma_D$ and feasibility survives longer. The displayed expression evaluated at $W = \alpha R$ is therefore a lower bound on B^* , not an upper one — with a lag of 100 at $R = 1$ it returns 313 where $B^* = 373$. The exact case is $R = 0$, which is the state the attack is mounted in, since the precursors precede every target. The cap hypothesis is what makes the driving term rather than the cap binding: if $\lambda < k/(|C| + 1)$ then $\alpha_t \leq \lambda$ is already below the floor and no target can be rejected at any D , so the budget is trivially zero and the displayed expression overstates it. Our configuration satisfies it by six orders of magnitude ($\lambda M = 4.5 \times 10^5$). Each precursor can be a group with one firing flow and enough zero-evidence flows to place its p -value in $(\lambda, \tau]$; the smallest such group has $m^* = \lfloor \lambda(|C| + 1)/k \rfloor + 1$ flows, which is a valid precursor whenever $m^* \leq \tau M$ — guaranteed by $(\tau - \lambda)M \geq 1$, and satisfied by six orders of magnitude here. The total injected volume is B^*m^* .

At the stress window (position 0.85, first-flow order) the closed form is verified against the implementation on the real stream: 202 precursors leave all 152 ADDIS rejections intact and 203 eliminate all of them, permanently (Fig. 6B, Table XLIV). Because that window’s evidence is not a valid

TABLE XXXIII

JOINT ATTACK ON THE BENIGN-INCLUSIVE AIT STREAM, FLOW-ONLY DETECTOR, CANONICAL ORDER, SEED 0, MEDIAN OVER 100 PAD-SAMPLING TRAJECTORIES. “OWN” AND “OTHER” ARE THE BASELINE ALERTS ON ATTACKER-OWNED AND OTHER EPISODES, HORIZON-FREE/HORIZON-AWARE. “NO POOL” COUNTS OWN EPISODES WHOSE PRIOR TRAFFIC TO THE VICTIM IS UNDER 20 FLOWS. FOR EACH SPENDING RULE THE ATTACKER IS RUN TWICE: FROM THE *deployment* POOL THE PER-ALERT ARMS USE, AND FROM THE VICTIM’S *prior* TRAFFIC ONLY; EACH PAIR OF COLUMNS GIVES THE ALERTS STILL STANDING AND THE FLOWS SPENT. READING AND THE HOST-CONDITIONED ARM: APPENDIX E-D.

org	own	other	no pool	horizon-free				horizon-aware			
				deployment		prior		deployment		prior	
				left	flows	left	flows	left	flows	left	flows
fox	11/11	0/0	4/14	0	972	2	601	0	5,610	3	61,704
harrison	9/9	0/0	5/14	0	4,144	3	11,624	0	21,741	3	69,940
russellmitchell	14/14	1/1	4/14	1	68,638	4	157,922	0	5,433	4	8,677
santos	8/9	0/1	4/9	0	397	4	2,718	0	2,003	6	4,267
shaw	3/3	3/3	4/6	0	536	2	906	0	353	2	659
wardbeck	8/8	0/0	3/9	0	225	3	550	0	4,728	4	15,198
wheeler	15/15	0/0	2/16	0	93,221	1	93,830	0	32,676	1	33,121
wilson	11/11	0/0	4/13	0	76,116	5	162,476	0	19,169	4	48,641

TABLE XXXIV

RELIABILITY OF THE JOINT AIT ATTACK OVER THE 100 PAD-SAMPLING TRAJECTORIES, CANONICAL ORDER, SEED 0. FOR EACH ORGANISATION AND SPENDING RULE, AND FOR EACH ATTACKER ARM OF TABLE III: $P(\text{all})$, THE FRACTION OF TRAJECTORIES IN WHICH EVERY ATTACKER-OWNED ALERT IS SILENCED; “LEFT”, THE MEDIAN AND 95TH PERCENTILE OF THE RESIDUAL OWN-ALERT COUNT; “FLOWS”, THE MEDIAN AND 95TH PERCENTILE OF THE FLOWS SPENT. THE 95TH PERCENTILE IS THE UPPER QUANTILE THE STAGE STORES (LINEARLY INTERPOLATED, SO A COUNT CAN BE FRACTIONAL); THE TRAJECTORIES THEMSELVES WERE NOT RETAINED, SO NO 90TH PERCENTILE IS AVAILABLE. FLOW-ONLY DETECTOR ON ALL EIGHT ORGANISATIONS; THE TWO HOST-CONDITIONED ROWS ARE THE JOINT HOST ARM.

org	spending	sequential oracle, deployment pool				sequential oracle, prior pool				state-free $c = \lfloor \rho \rfloor + 1$, prior pool			
		$P(\text{all})$	left med/p95	flows med/p95		$P(\text{all})$	left med/p95	flows med/p95		$P(\text{all})$	left med/p95	flows med/p95	
fox	horizon-free	1.00	0/0	972/1,366		0.00	2/2	601/601		0.00	5/5	8,896/8,896	
fox	horizon-aware	1.00	0/0	5,610/5,617		0.00	3/3	61,704/62,444		0.00	11/11	8,896/8,896	
harrison	horizon-free	1.00	0/0	4,144/4,538		0.00	3/3	11,624/11,671		0.00	8/8	29,217/29,217	
harrison	horizon-aware	1.00	0/0	21,741/21,755		0.00	3/3	69,940/70,865		0.00	8/8	29,217/29,217	
russellmitchell	horizon-free	0.00	1/1	68,638/130,582		0.00	4/5	157,922/168,760		0.00	14/14	1,242/1,242	
russellmitchell	horizon-aware	1.00	0/0	5,433/5,496		0.00	4/4	8,677/8,725		0.00	10/10	1,242/1,242	
santos	horizon-free	1.00	0/0	397/401		0.00	4/4	2,718/3,064		0.00	6/6	2,916/2,916	
santos	horizon-aware	1.00	0/0	2,003/2,039		0.00	6/6	4,267/6,307		0.00	6/6	2,916/2,916	
shaw	horizon-free	1.00	0/0	536/536		0.00	2/2	906/906		0.00	3/3	382/382	
shaw	horizon-aware	1.00	0/0	353/353		0.00	2/2	659/659		0.00	3/3	382/382	
wardbeck	horizon-free	1.00	0/0	225/225		0.00	3/3	550/594		0.00	4/4	5,400/5,400	
wardbeck	horizon-aware	1.00	0/0	4,728/4,789		0.00	4/4	15,198/16,887		0.00	8/8	5,400/5,400	
wheeler	horizon-free	1.00	0/0	93,221/93,382		0.00	1/1	93,830/94,031		0.00	14/14	40,524/40,524	
wheeler	horizon-aware	1.00	0/0	32,676/32,712		0.00	1/1	33,121/33,149		0.00	3/3	40,524/40,524	
wilson	horizon-free	1.00	0/0	76,116/76,144		0.00	5/5	162,476/162,476		0.00	11/11	20,334/20,334	
wilson	horizon-aware	1.00	0/0	19,169/19,175		0.00	4/4.05	48,641/48,641		0.00	11/11	20,334/20,334	
shaw (host)	horizon-free	1.00	0/0	28/28		0.00	2/2	258/258		0.00	2/2	382/382	
shaw (host)	horizon-aware	1.00	0/0	7/7		0.00	2/2	144/144		0.00	2/2	382/382	
wilson (host)	horizon-free	1.00	0/0	73,337/73,397		0.00	1/1	73,454/73,505		0.00	6/6	20,334/20,334	
wilson (host)	horizon-aware	1.00	0/0	16,975/17,010		0.00	1/1	17,187/17,261		0.00	3/3	20,334/20,334	

e-value, and the two-point structure that lets ADDIS escape is the same property that violates its conservativeness condition, this demonstrates the mechanism and prices it; it is not an FDR-valid real-data result. The attack survives a grey-box setting: the admissible group size is a τ/λ -wide interval, so a $\pm 33\%$ estimate of $|\mathcal{C}|$ suffices and unknown controller state costs about $36\times$ over-provisioning.

To isolate the mechanism from that validity failure we generate streams with genuinely conservative independent $\text{Unif}(0,1)$ null p-values and 120 strong targets. ADDIS’s empirical all-null FDR is 0.010 and its mixed-stream mean FDP is 0.047 at recall 1.00, both below $q = 0.05$. We then prepend *alternative*, non-null precursors in $(\lambda, \tau]$, so the

assumptions on true nulls are untouched and the guarantee holds on the attacked stream. The attack silences all 120 targets at a median $B^* = 140$ over 300 streams (range 140–255; Fig. 6C, Table XLV). The guarantee therefore remains valid while the security objective is defeated: controlling the fraction of false discoveries does not require making any discovery.

The real-stream cost is conspicuous, not cheap. At $|\mathcal{C}| = 1,813,113$ each precursor needs 453,279 flows and 203 precursors need 92,015,637, about 36.1 GB or 33.4 Mbit/s over the deployment span and $37.5\times$ the window’s own flow count; the total scales as $\lambda|\mathcal{C}|[(\tau-\lambda)w_0|\mathcal{C}|]^{0.625}/k^{1.625}$, so the calibration budget that buys feasibility also buys resistance. Suppressing one canonical e-LOND alert at the same window costs a median

TABLE XXXV

COROLLARY 2’S PREMISE TESTED ON THE JOINT-ATTACK AIT CELLS, CANONICAL ORDER, SEED 0. “MAX OTHER $\div T/\alpha$ ”: THE LARGEST NON-ATTACKER GROUP EVIDENCE DIVIDED BY THE HORIZON-UNIFORM COLD-START THRESHOLD (≥ 1 MEANS THE PREMISE FAILS ON THAT STREAM); “ $n \geq T/\alpha$ ”: NON-ATTACKER EPISODES AT OR ABOVE IT. “OTHER ALERTS (AT $R=0$)”: BASELINE NON-ATTACKER ALERTS UNDER HORIZON-FREE / HORIZON-AWARE SPENDING, WITH THE NUMBER THAT CLEAR THE $R = 0$ LEVEL AT THEIR OWN POSITION IN PARENTHESES; THE REST FIRE ONLY AT A LEVEL RAISED BY EARLIER REJECTIONS. “NO POOL”: OWN EPISODES WITH FEWER THAN 20 PRIOR TEMPLATES, NEVER PADDED. “PADS FIRED”: MEDIAN SHARE OF ADMITTED PADS THAT CARRY THE CEILING RATHER THAN ZERO, FOR THE STATE-FREE MULTIPLIER (REGIME-INDEPENDENT STREAM) AND THE SEQUENTIAL ORACLE (HORIZON-FREE / HORIZON-AWARE), BOTH FROM THE PRIOR POOL. “LEFT”: MEDIAN RESIDUAL OWN ALERTS, HORIZON-FREE / HORIZON-AWARE. UNDER HORIZON-FREE SPENDING THE MULTIPLIER IS A HEURISTIC TRANSFER OF A RESULT PROVED FOR HORIZON-UNIFORM SPENDING.

org	ρ	premise: other hypotheses			own no pool	pads fired (prior pool)		left (prior pool)	
		max other $\div T/\alpha$	$n \geq T/\alpha$	other alerts (at $R=0$)		c_{int}	sequential	c_{int}	sequential
fox	1.61	0	0	0 (0) / 0 (0)	4/14	3.76%	0.00% / 3.91%	5/11	2/3
harrison	3.48	0	0	0 (0) / 0 (0)	5/14	3.50%	2.51% / 3.87%	8/8	3/3
russellmitchell	2.99	1.497	1	1 (1) / 1 (1)	4/14	0.00%	0.01% / 0.05%	14/10	4/4
santos	4.03	0.336	0	0 (0) / 1 (0)	4/9	5.97%	11.35% / 8.57%	6/6	4/6
shaw	2.29	2.289	2	3 (1) / 3 (2)	4/6	0.00%	0.00% / 0.00%	3/3	2/2
wardbeck	5.48	0	0	0 (0) / 0 (0)	3/9	2.12%	7.73% / 4.57%	4/8	3/4
wheeler	3.31	0.019	0	0 (0) / 0 (0)	2/16	0.05%	0.05% / 0.05%	14/3	1/1
wilson	2.95	0	0	0 (0) / 0 (0)	4/13	0.13%	0.31% / 1.05%	11/11	5/4
shaw (host)	2.29	2.289	4	1 (0) / 6 (4)	4/6	0.00%	0.00% / 0.00%	2/2	2/2
wilson (host)	2.95	0	0	0 (0) / 0 (0)	4/13	0.00%	0.00% / 0.01%	6/3	1/1

TABLE XXXVI

BOTH ATTACKS IN OPERATIONAL UNITS (POSITION 0.85 — THE KNOWN-INVALID STRESS WINDOW; THESE ARE ATTACK *costs*, WHICH ARE MEASURED AGAINST THE CONTROLLER’S LEVEL AND DO NOT REST ON THE EVIDENCE BEING A VALID E-VALUE — BLACK-BOX PADDING POOL). WIRE BYTES ARE THE PER-FLOW MEDIAN; RATE IS PRICED OVER THE RELEVANT SPAN (7200 S WINDOW FOR PADDING, THE POSITION’S OWN DEPLOYMENT SPAN FOR THE STATE ATTACK). “ $\times$ WINDOW” AND “ $\times$ DATASET” ARE FLOW-COUNT RATIOS. **EACH PADDING ROW NAMES THE PROCEDURE WHOSE LEVEL IT IS PRICED AGAINST, WHETHER THAT LEVEL IS THE *running* $1/\alpha_t$ AT THE EPISODE’S OWN STEP OR THE *static* COLD-START τ , AND THE DETECTED SET IT IS SUMMED OVER, BECAUSE THESE DIFFER BY TWO ORDERS OF MAGNITUDE AND ARE EASILY CONFUSED, AND THE WITHIN-BUCKET ORDER THE DETECTED SET WAS PRODUCED UNDER.** ROW 1 IS THE HEADLINE COST: THE MEDIAN OVER E-LOND’S OWN 34 DETECTIONS AT 0.85 UNDER THE CANONICAL ORDER, PRICED AT THE RUNNING LEVEL — THE VALUE TABLE I QUOTES. ROW 2 IS THE SAME QUANTITY OVER THE 72 DETECTIONS OF THE **FIRST-FLOW** ARRIVAL ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT; IT IS $45\times$ DEARER BECAUSE THAT ORDER DETECTS MORE EPISODES AND DETECTS THEM DEEPER INTO THE LEVEL SEQUENCE, WHERE $1/\alpha_t$ IS LARGER. BOTH APPEAR IN TABLE XXIII. ROWS 3–4 ARE THE *cheap extreme*: r_{90} OVER THE 257 EPISODES OF POSITIONS 0.62 AND 0.85 (TWO SEEDS) AT WHICH ALL FIVE POOLS ARE DEFINED, PRICED AGAINST THE STATIC LEVEL- w_0 THRESHOLD T/w_0 , FIRST-FLOW. ROW 5 MULTIPLIES THAT STATIC MEDIAN BY ADDIS’S 147-EPISODE DETECTED SET (TABLE XLIV), SO IT MIXES AN E-LOND-SCALE PER-EPISODE COST WITH AN ADDIS-SCALE COUNT; IT IS THE CHEAPEST READING OF “SUPPRESS EVERYTHING” AND IS REPORTED ONLY FOR CONTRAST WITH THE LAST TWO ROWS. EVERY ROW BELOW THE FIRST IS UNDER FIRST-FLOW ARRIVAL.

attack budget	flows	wire bytes	rate	$\times$ window	$\times$ dataset
e-LOND, <i>running</i> $1/\alpha_t$: one alert (median), canonical	116	45.5 KB	50.5 bbit/s	4.7×10^{-5}	7.1×10^{-6}
same, <i>first-flow</i> (optimistic upper bd.)	5,202	2.0 MB	2265.8 bbit/s	2.1×10^{-3}	3.2×10^{-4}
level- w_0 <i>static</i> $\tau=T/w_0$: one alert (median)	34	13.3 KB	14.8 bbit/s	1.4×10^{-5}	2.1×10^{-6}
level- w_0 <i>static</i> $\tau=T/w_0$: one alert (p90)	227	89.0 KB	98.9 bbit/s	9.3×10^{-5}	1.4×10^{-5}
static median $\times$ ADDIS’s 147 detections (mixed)	4,998	2.0 MB	2176.9 bbit/s	2.0×10^{-3}	3.1×10^{-4}
ADDIS, <i>running</i> $1/\alpha_t$: one episode (median)	34,465,311	13.5 GB	15.0 Mbit/s	14.05 $\times$	2.11 $\times$
ADDIS spending-state attack (B^* precursors)	92,015,637	36.1 GB	33.4 Mbit/s	37.51 $\times$	5.63 $\times$

of 116 flows. The result is also mechanism-specific: online e-BH decides by a fixed point over the whole history rather than by a spending index, so this attack has no state variable to move. We do not claim that online e-BH is immune to every manipulation, only that Theorem 6 does not cover it. Where both surfaces apply to ADDIS they coincide: when its level cap λ binds, the minimal suppressing pad also lands

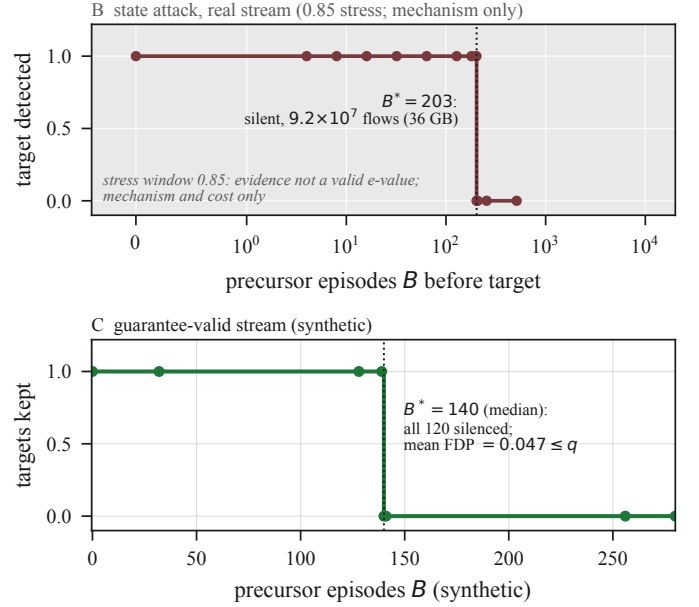


Fig. 6. Manipulating ADDIS’s spending state. **B**: on the real stress window, precursors silence every rejection permanently at exactly the Theorem 6 threshold — on evidence that is not a valid e-value, so this measures the mechanism and its cost, not an FDR-valid outcome. **C**: the same poisoning on synthetic streams where ADDIS’s guarantee genuinely holds still silences every target.

in $(\lambda, \tau]$ and advances the index (101 of ADDIS’s 147 true detections; Appendix A). Against ADDIS the state attack is also the cheaper one: its level is five orders of magnitude above e-LOND’s, so padding one episode costs a median 3.4×10^7 flows and state manipulation wins from the third episode on.

TABLE XXXVII

JOINT ZERO-EVIDENCE CONTROLLER-STATE RERUN: LSPR23, DETECTOR SEED 0, CANONICAL ORDER, TWO-HOUR HOST-PAIR EPISODES, E-LOND, BOTH WINDOWS AND BOTH SPENDING REGIMES. UPPER PANEL: THE STATIC, SEQUENTIAL AND STATE-FREE ATTACKERS OF SECTION VI-B; LOWER PANEL: THE PER-HOST-PAIR VOLUME CAP AGAINST THE SEQUENTIAL ATTACKER. DESIGN AND READING: APPENDIX E-G.

window	regime	true det. (FD)	median $r_t^* / \sum_{t \in \mathcal{D}} r_t^*$	static: remain	$J_{\text{seq}} (\div \sum_{t \in \mathcal{D}} r_t^*)$	med. pad (padded)	spared	padded arity (per-alert)	% episodes $\geq$	c_{int}
0.55	horizon-free	3 (0)	24 / 80	0	30 (0.375)	15	1	44.5 (63)	6.1%	2
0.55	horizon-aware	105 (0)	2,946 / 627,495	0	5,650 (0.009)	48.5	13	91 (2,989)	3.8%	3
0.62	horizon-free	11 (0)	6 / 926	0	859 (0.928)	859	10	881 (20)	0.5%	41
0.62	horizon-aware	107 (1)	3,908 / 977,567	0	9,992 (0.010)	33	4	55 (3,928)	4.3%	3

0.55 horizon-aware (105 true det.)				0.62 horizon-aware (107 true det.)			
cap n	fire (struct.+casc.)	per-alert fire	benign $> n$	fire (struct.+casc.)	per-alert fire	benign $> n$	
30	105 (87+18)	105	7.36%	107 (85+22)	107	7.30%	
100	96 (44+52)	100	3.40%	79 (33+46)	103	2.86%	
300	19 (1+18)	91	1.04%	60 (22+38)	95	1.59%	
1,000	0 (0+0)	80	0.50%	0 (0+0)	82	0.41%	
3,000	0 (0+0)	52	0.21%	0 (0+0)	59	0.19%	

Notes. “true det. (FD)”: true detections and false discoveries on the unperturbed run; “median $r_t^* / \sum_{t \in \mathcal{D}} r_t^*$ ”: the per-alert oracle cost and the per-alert sum of Table II. “remain”: true detections still rejected after the rerun (the one 0.62 horizon-aware false discovery also disappears). “padded” statistics are over the alerts the sequential attacker actually padded; “spared” alerts fire only at a level raised by earlier rejections and need no pad once those are gone. “padded arity (per-alert)”: the sequential attacker’s median padded arity, with the per-alert reading of Section VI-B in parentheses; “% episodes $\geq$ ”: the share of deployment episodes at least that large. $c_{\text{int}} = \lfloor \rho \rfloor + 1$ (Corollary 2). Lower panel: “fire (struct.+casc.)” splits the surviving alerts into structural (the pad would not fit even at $R = 0$) and cascade victims (it would have fitted at $R = 0$); “per-alert fire” is $m_t + r_t^* > n$ on the unperturbed pads; “benign $> n$ ”: the share of benign deployment episodes whose arity exceeds the cap, i.e. the episodes an implemented admission policy would alter; no baseline episode is truncated or rescored in this experiment. Caps are the sampled grid; nothing is claimed between grid points.

TABLE XXXVIII

WHAT EACH RESULT DEPENDS ON. “VALID e ” IS WHETHER THE CLAIM NEEDS THE EVIDENCE TO BE A VALID E-VALUE; “A1” IS ASSUMPTION 1; “LABELS / OWNERSHIP” IS WHETHER DATASET LABELS ENTER, INCLUDING AS THE ORACLE FOR WHICH EPISODES THE ATTACKER OWNS. WINDOWS ARE NAMED BY POSITION: 0.55 PRIMARY, 0.62 SECONDARY, 0.85 STRESS (EVIDENCE NOT A VALID E-VALUE; MECHANISM AND COST ONLY); AIT IS THE SECOND DATASET.

Result	valid e	A1	labels / ownership	windows
Finite horizon, exchange rate (§III)	no	no	no	all (algebraic)
Granularity buys feasibility (§IV)	no	no	no	all
Alert blur (§IV)	no	no	yes	0.55, 0.85
Padding impossibility (Thm. 3)	no	no	no	—
Suppression and r_t^* of a specified alert (§V)	no	no	no	0.55, 0.62, 0.85, AIT
Aggregates over $\mathcal{D}$, incl. $\sum_{t \in \mathcal{D}} r_t^*$ (§VI)	no	no	yes	0.55, 0.62
Joint rerun J_{seq} , whole-window multiplier (§VI)	no	no	yes	0.55, 0.62
Replay suppression counts, recall, FDP (§VI)	no	no	yes	0.55, 0.62, 0.85, AIT
Nominal e-LOND FDR guarantee	yes	yes	no	0.55, 0.62
ADDIS state attack, real stream (Appendix G)	no	no	yes	0.85 (mechanism)
ADDIS silenced, guarantee valid (Appendix G)	yes	—	no	synthetic

APPENDIX H

SECONDARY OPERATIONAL BASELINES AND NEGATIVE RESULTS

A. Matched operating points

Comparing a $q = 0.05$ procedure against an arbitrary classifier threshold measures nothing, so Tables XLVI and XLVII place every method on the achievable frontier of the episode-max-score threshold family at its own budget, an ex-post oracle whose thresholds use the test labels. Every method sits within 0.047 recall of the best achievable point at its budget, so the comparison that matters is choice of point: at the primary window e-LOND operates at 3 alerts, recall 0.011, FDP 0.000 under the canonical order (18 and 0.065 under first-flow arrival), while an ex-post oracle at the same zero error reaches recall 0.378 on 104 alerts. Sweeping q twenty-fold moves median recall by 0.088, while switching the spending sequence at

fixed q moves it by 0.188 (first-flow paired sweep, Table X): formal error control can reduce the operator’s freedom over the workload–recall tradeoff.

B. Analyst-feedback controllers

The three feedback baselines of Table XLVIII are threshold controllers, not FDR procedures. Each carries a single state $u_t \in [0, 1]$, a position in the calibration score distribution: it alerts on episode t when that episode’s maximum flow score exceeds the u_t -quantile of the calibration scores, and moves u_t only when an analyst’s disposition of an earlier alert arrives, after a wall-clock delay L . Writing $\widehat{\text{FDP}}_t$ for the false fraction over the last 200 dispositions received so far and $\varepsilon_t = \widehat{\text{FDP}}_t - q$,

$$\mathbf{P}: u_{t+1} = u_t + \eta_P \varepsilon_t,$$

$$\mathbf{PI}: u_{t+1} = u_t + \eta_P \varepsilon_t + \eta_I \sum_{r \leq t} \varepsilon_r,$$

$$\mathbf{AQT}: u_{t+1} = u_t + \eta[(1-q)\mathbf{1}\{\text{false}\} - q\mathbf{1}\{\text{true}\}],$$

TABLE XXXIX

BENIGN FIRING RATE VS NOMINAL BY WINDOW (SEED 0). “RATIO” IS THE MEASURED/NOMINAL BENIGN FIRING RATE AT $k = 1$, AND “95% CI” IS ITS EXACT CLOPPER–PEARSON BINOMIAL INTERVAL GIVEN THE OBSERVED FIRING COUNT OUT OF THE BENIGN FLOWS. THE INTERVAL SEPARATES THE THEORETICAL GUARANTEE FROM THE EMPIRICAL DIAGNOSTIC: AT THE FOUR NON-STRESS WINDOWS ONLY 0–3 BENIGN FLOWS FIRE (EXPECTATION ≈ 1), SO THE INTERVAL IS WIDE AND **INCLUDES 1** — THE DIAGNOSTIC CANNOT *prove* VALIDITY, ONLY FAIL TO REJECT IT; AT POSITION 0.85 THE INTERVAL **EXCLUDES 1** BY ORDERS OF MAGNITUDE, A CLEAR VIOLATION. THE LAST FOUR COLUMNS ARE THE RATIO AT RANK DEPTH $k \in \{1, 10, 100, 1000\}$. THIS DIAGNOSTIC IS **ORDER-INVARIANT**: IT SCORES BENIGN *flows* AGAINST THE CALIBRATION THRESHOLD, WITH NO CONTROLLER AND SO NO STREAM SEQUENCE TO PERMUTE.

position	AUROC	fired benign	ratio	95% CI	ratio at rank depth k			
					$k=1$	$k=10$	$k=100$	$k=1000$
0.55	0.916	1	1.07	[0.03, 5.96]	1.07	0.64	0.51	3.85
0.62	0.799	1	1.16	[0.03, 6.45]	1.16	7.17	0.91	5.09
0.70	0.914	3	3.79	[0.78, 11.06]	3.79	0.50	0.09	0.11
0.77	0.828	2	2.72	[0.33, 9.82]	2.72	0.95	0.98	2.41
0.85	0.999	46	50.90	[37.26, 67.89]	50.90	9.18	4.49	6.80

TABLE XL

METADATA-STRATIFIED INTERROGATION OF ASSUMPTION 1 (SEED 0; **ORDER-INVARIANT**: PER-FLOW FIRINGS AGAINST A FIXED THRESHOLD, NOT CONTROLLER DECISIONS). ROWS ARE WINDOWS; CELLS ARE (STRATUM, DEPTH) PAIRS READ AT $k \in \{1, 10, 100, 1000\}$; “SURVIVING BY” COUNTS BENJAMINI–YEKUTIELI REJECTIONS OVER ALL CELLS, AT $k=1$, AND OF STRATUM CONTRASTS. DESIGN AND READING: APPENDIX F-A.

position	strata	cells		surviving BY			marginal ratio [95% CI]	
		total	untested	ratio	at $k=1$	contrast	$k=1$	$k=1000$
0.55	28	112	90	6	0	5	1.07 [0.00, 4.56]	3.85 [1.69, 8.70]
0.62	28	112	82	13	0	9	1.16 [0.00, 94.53]	5.09 [1.74, 11.26]
0.70	26	104	98	0	0	3	3.79 [0.00, 9.02]	0.11 [0.04, 0.28]
0.77	26	104	86	5	0	9	2.72 [0.00, 12.90]	2.41 [1.45, 3.80]
0.85	26	104	72	16	5	6	50.90 [3.70, 204.68]	6.80 [3.38, 13.04]

Notes. “untested”: cells firing fewer than ten times, which carry no test but enter the Benjamini–Yekutieli family at $p = 1$. Ranks break ties conservatively, so each marginal ratio is a *lower* bound on the anti-conservatism; the interval is a 95% percentile interval from the joint resampling. “contrast”: strata departing from their *own* window’s marginal on the same replicates.

with the PI integral clamped against windup and u clipped to $[0, 1]$. Gains are chosen per window, delay and controller by grid search ($\eta_P \in \{0.005, 0.02, 0.05, 0.2\}$, window $\in \{50, 200\}$, $u_0 \in \{0.99, 0.999\}$, AQT $\eta \in \{10^{-4}, 10^{-3}, 5 \times 10^{-3}, 0.02\}$, $\eta_I = \eta_P/10$) minimising ($|\text{FDP} - q|, -\text{recall}$) on the realised test labels, an ex-post oracle, so the reported performance is an upper bound, the conservative direction for the negative conclusion. Under first-flow order they reach FDP 0.402 (P, PI) and 0.523 (AQT) at fifteen-minute disposition delay, agree with one another by four hours, and reach the no-feedback threshold (0.841) only by eight: a statement about delay on these controllers, not about feedback-aware testing [41], whose input our measurement prices.

C. Negative results

Inflating a benign group cannot create a discovery at a fixed level. The argument is pathwise, not distributional: for the arithmetic mean with evidence sum S over m members, appending r zero-evidence flows gives $S/(m+r) \leq S/m$, so a group that is not rejected at a fixed level with its existing evidence is not rejected after padding either. The

bound $\mathbb{E}[\text{Ev}(G) \mid \mathcal{M}_j] \leq 1$ of Assumption 1 does not by itself give this: an expectation bound permits large individual realisations (an e-value equal to 100 with probability 0.01 and zero otherwise has mean one and crosses a threshold of 20), which is exactly why a benign group can be rejected and why the FDR guarantee is a rate, not an exclusion. *Fragmentation leaves expected group evidence unchanged, and no more.* For a pure-attack group whose m flows fire independently with probability ϕ , $\mathbb{E}[\text{Ev}(G)] = \phi M$ does not depend on m ; the rejection probability does, $\Pr(\text{Ev}(G) \geq \theta) = \Pr(\text{Binomial}(m, \phi) \geq \lceil m\theta/M \rceil)$, and splitting a campaign also changes the number of hypotheses and the controller state, so we make no analytic claim about its effect on detection. *Boosting recovers nothing for two-point evidence:* the optimal e-to-e boosting factor for a $\{0, M\}$ -valued e-value is $b^* = 1$ by construction. *Raising k* removes the operating point rather than trading power for validity; *randomised smoothing* removes the floor but not the reliably-detected set (Appendix C-B); *asymmetric weighting* is not a free mitigation (Section VII); *stratified (Mondrian) conformal* does not repair the position-0.85 anomaly; and the published α -death fix (mem-e-LORD) is identical to e-LORD before the first rejection.

TABLE XLI

WHAT CALIBRATING THE *grouped unit* COSTS (SEED 0; APPENDIX F-B). LEFT: THE SHIPPED CONSTRUCTION, CALIBRATED ON BENIGN *flows*. RIGHT: SPLIT CONFORMAL ON THE GROUP, SAME KEY AND PROTOCOL, WHICH WOULD NEED ONLY CALIBRATION/TEST *group* EXCHANGEABILITY. THE EVIDENCE CEILING FALLS BY THE FLOWS-PER-GROUP RATIO ($40\text{--}49\times$) WITH T UNCHANGED, SO THE MARGIN COLLAPSES AND DETECTION AT THE PRIMARY AND SECONDARY WINDOWS GOES TO ZERO. THAT IS THE FEASIBILITY BOUNDARY, NOT A WEAK STATISTIC: THE COLD-START WINDOW IS TENS OF HYPOTHESES AND THE FIRST *firing* GROUP ARRIVES HUNDREDS OF POSITIONS LATER (AT 0.55, 110 GROUPS FIRE AND NONE LANDS INSIDE THE WINDOW); FEASIBLE-STEP COUNTS MATCH $\lfloor (\alpha M(R+1)/\zeta(1.6))^{1/1.6} \rfloor$ AT EVERY CONFIGURATION. THE LAST TWO COLUMNS QUALIFY THE REPAIR WITHOUT REFUTING IT: THE *max* STATISTIC — THE PADDING-ROBUST ONE — IS STRONGLY ARITY-DEPENDENT, AND ARITY IS WHAT THE ADVERSARY CONTROLS, SO GROUP EXCHANGEABILITY MAY BE FRAGILE UNDER GROUP-FORMATION SHIFT, THOUGH THE CORRELATION ALONE NEITHER REFUTES IT NOR SHOWS THAT IT FAILS; THE *mean* IS ARITY-INVARIANT TO WITHIN NOISE AND IS EXACTLY THE RULE THEOREM 3 DEFEATS. POSITION 0.85 IS THE STRESS WINDOW AND ITS DETECTIONS ARE NOT GUARANTEE-BEARING (APPENDIX F-A). THE FLOW-CALIBRATION COLUMNS USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER (THE OPTIMISTIC UPPER BOUND), NOT THE CANONICAL ORDER OF TABLE I; THE GROUP ARM MATCHES ITS TIE-BREAK TO THE SAME ORDER, SO THE TWO REMAIN COMPARABLE (TABLE XIX).

Position	flow calibration			group calibration						arity corr.	
	$ C $	margin	det.	$ C $	margin	det. max	det. mean	cold start	1st fire	max	mean
0.55	2,448,993	+0.067	18	52,279	−0.977	0	0	81	401	+0.35	+0.04
0.62	2,449,031	+0.243	13	53,897	−0.973	0	0	83	349	+0.36	+0.01
0.70	2,287,988	+0.536	30	57,093	−0.962	1	0	86	6	+0.23	−0.12
0.77	2,116,418	+0.671	31	48,869	−0.961	2	1	78	11	+0.34	−0.01
0.85	1,813,113	+0.436	72	36,944	−0.971	42	42	65	4	+0.30	−0.07

TABLE XLII

ADJUDICATED ALERT AUDIT AGAINST THE RED-TEAM TASK RECORD (POSITION 0.85, THE **KNOWN-INVALID** STRESS WINDOW). TABLE XXXIX IS WHAT ESTABLISHES THAT INVALIDITY, FROM THE BENIGN FIRING RATE;

THIS IS A *follow-up label-quality audit* AT THE SAME WINDOW, NOT A GUARANTEE-BEARING RESULT AND NOT THE DIAGNOSTIC THAT MADE THE CALL. PANEL (A): ADJUDICATED VERDICT COUNTS WITH THEIR LABEL SPLIT. PANEL (B): FALSE-DISCOVERY PROPORTION BY METHOD UNDER THE LABEL, THE PRE-REGISTERED AUDIT LOWER BOUND, AND THE STRICT LOWER/UPPER BOUNDS. ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX); THE ORDER SHIFTS DETECTIONS AND THE LEVEL AN ALERT FIRES AT, AND SHIFTS NOTHING ELSE.

(a) adjudicated verdicts				
verdict	n	labelled mal.	labelled ben.	
clearly malicious	49	46	3	
probably malicious	102	100	2	
ambiguous	0	0	0	
probably benign	1	1	0	
clearly benign	0	0	0	

(b) false-discovery proportion					
method	R	FDP _{label}	FDP _{audit} ^{lo}	FDP _{strict} ^{lo}	FDP _{strict} ^{hi}
e-LOND/uniform	151	0.026	0.007	0.000	0.026
ADDIS/poly	152	0.033	0.007	0.000	0.026
union	152	0.033	0.007	0.000	0.026

TABLE XLIII

BATES CALIBRATION-CONDITIONAL ADJUSTMENT ($n_{\text{CAL}} = 1000$): IT FIXES CONDITIONAL VALIDITY AT A POWER COST SET BY THE SAME QUANTITY.

THE FIRST THREE COLUMNS ARE THE CONDITIONAL EXCEEDANCE PROBABILITY — ANALYTIC PREDICTION, MEASURED NOMINAL, AND β -ADJUSTED; THE LAST TWO ARE THE e-LOND DETECTION COST (MEDIAN REJECTIONS, NOMINAL VS β -ADJUSTED). ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX); THE ORDER SHIFTS DETECTIONS AND THE LEVEL AN ALERT FIRES AT, AND SHIFTS NOTHING ELSE. THE β -BOUND REQUIRES 0.22–0.62 OF $|C|$ ACROSS ERROR TARGETS $\delta \in [0.01, 0.2]$.

k	conditional exceedance			e-LOND rej. (med)	
	predicted	nominal	β -adj.	nominal	β -adj.
1	0.368	0.370	0.092	100	62.5
10	0.459	0.439	0.084	0	0
100	0.489	0.483	0.092	0	0

TABLE XLIV

ADDIS CONTROLLER-STATE ATTACK: $B^* = 203$ PRECURSOR EPISODES PERMANENTLY SILENCE THE CONTROLLER AT POSITION 0.85 — THE **KNOWN-INVALID** STRESS WINDOW, WHICH IS WHERE THE REAL-STREAM DEMONSTRATION MUST SIT, SINCE ADDIS’S HORIZON ESCAPE AND ITS GUARANTEE-INVALIDITY ARE THE SAME PROPERTY ON TWO-POINT CONFORMAL EVIDENCE (TABLE XLV REPRODUCES THE MECHANISM WHERE THE GUARANTEE DOES HOLD). THE BREAK-EVEN IS THE NUMBER OF DETECTED EPISODES WHOSE PER-EPISODE PADDING COST EQUALS THE WHOLE ONE-OFF STATE ATTACK. ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX); THE ORDER SHIFTS DETECTIONS AND THE LEVEL AN ALERT FIRES AT, AND SHIFTS NOTHING ELSE.

quantity	value
B^* (precursor episodes, $R = 0$)	203
per-precursor cost $\lfloor \lambda(C + 1)/k \rfloor + 1$	453,279 flows
total attack flows	92,015,637 ($\approx 9.2 \times 10^7$)
level-saturated coincidence	101 of 147
minimal pad lands in window	101 of 147
median pad per episode (alternative)	34,465,311 flows
break-even vs padding	2.67 episodes
grey-box landing band $(\hat{C} / C)$	[0.67, 1.33]

TABLE XLV

THE ADDIS STATE ATTACK ON A **SYNTHETIC STREAM WHERE ADDIS’S GUARANTEE GENUINELY HOLDS**, ISOLATING THE MECHANISM FROM THE REAL-WINDOW VALIDITY FAILURE. THE TWO-POINT CONFORMAL EVIDENCE USED ELSEWHERE CANNOT SUPPLY THIS: ITS ADDIS HORIZON-ESCAPE (THE SPENDING INDEX NEVER ADVANCES) IS THE *same* PROPERTY THAT MAKES ITS NULL P-VALUES NON-CONSERVATIVE, SO NO TWO-POINT STREAM IS BOTH ESCAPE-FEASIBLE AND GUARANTEE-VALID.

WE THEREFORE USE GENUINELY UNIFORMLY-CONSERVATIVE NULL P-VALUES ($\text{Unif}(0, 1)$), INDEPENDENT, THE ASSUMPTION ADDIS NEEDS). THE STREAM IS SYNTHETIC, SO LSPR23’S GROUPING PLAYS NO PART AND THE RESULT IS **ORDER-INVARIANT** BY CONSTRUCTION. *Top*: ADDIS CONTROLS FDR — AN EXPECTATION, NOT A PER-REALISATION BOUND (THE MIXED ARM’S WORST SINGLE REALISATION IS $\text{FDP} = 0.104$, ABOVE q , WHICH THE GUARANTEE PERMITS) — ON THIS STREAM: THE HONEST *all-null* TEST (EVERY REJECTION FALSE) GIVES MEAN

$\text{FDP} = 0.010 \leq q=0.05$, AND WITH 120 STRONG TARGETS PRESENT MEAN $\text{FDP} = 0.047$ AT RECALL 1.00 (200 SEEDS EACH). *Bottom*: THE

ATTACK IS ADVERSARIAL STATE *poisoning* — PREPENDING B ATTACKER-GENERATED *alternative* (NON-NULL) PRECURSOR EPISODES WITH $p \in (\lambda, \tau]$. ADDIS CONSTRAINS ONLY *true-null* P-VALUES, SO ALTERNATIVES MAY TAKE ANY VALUE: THE LEGITIMATE NULLS STAY $\text{Unif}(0, 1)$ AND ADDIS’S GUARANTEE HOLDS ON THE *attacked* STREAM, NOT MERELY THE CLEAN ONE. THESE SELECTED NON-CANDIDATES ADVANCE THE SPENDING INDEX AND SILENCE EVERY REJECTION AT A MEDIAN $B^* = 140$ OVER 300 GUARANTEE-VALID STREAMS (RANGE 140–255; THE TAIL FROM A RARE NULL FALSE POSITIVE THAT TRANSIENTLY RESTORES WEALTH). SO EVEN WHERE ADDIS’S GUARANTEE HOLDS *after* THE ATTACK, AN ADVERSARY WHO ONLY ADDS TRAFFIC DRIVES THE CONTROLLER TO SILENCE — A PROPERTY OF THE MECHANISM, NOT AN ARTEFACT OF THE DATASET ALREADY VIOLATING THE ASSUMPTION.

	mean FDP	$\leq q=0.05?$
all-null (every rejection false)	0.010	yes
mixed (120 targets, recall 1.00)	0.047	yes
<i>state poisoning: median $B^*=140$ (range 140–255) silences all</i>		

TABLE XLVI

MATCHED OPERATING POINTS AT THE **PRIMARY ANALYSIS WINDOW** (POSITION 0.55, TWO-HOUR GROUPING, $T = 57,368$ EPISODES OF WHICH 275 ARE MALICIOUS; EVERY FDR STATEMENT CONDITIONAL ON ASSUMPTION 1 AND LEMMA 1). ROWS ARE UNDER THE **CANONICAL** WITHIN-BUCKET ORDER, AS TABLE I IS, WITH THE FIRST-FLOW ARM — THE OPTIMISTIC UPPER BOUND — BENEATH FOR THE FOUR ROWS THE ORDER MOVES. THE STRESS-TEST WINDOW 0.85 IS TABLE XLVII; THE NEGATIVE RESULT, EVERY METHOD WITHIN 0.047 RECALL OF THE FRONTIER, HOLDS AT BOTH AND UNDER EVERY ORDER AUDITED. THE FRONTIER IS AN EX-POST ORACLE (THRESHOLDS USE THE TEST LABELS) AND IS *measured* ORDER-INVARIANT HERE, SO IT IS QUOTED ONCE. THE SLOT RULE IS AN EXPECTATION OVER SLOTS AND THE NO-FEEDBACK THRESHOLD NEVER UPDATES, SO BOTH ARE ORDER-FREE BY CONSTRUCTION AND VERIFIED UNCHANGED.

Method	alerts	FDP	recall	gap
<i>canonical metadata-hash order</i>				
online FDR (mean rule)	3	0.000	0.011	+0.000
online FDR (slot rule)	91.0	0.000	0.331	+0.000
feedback, $L=0$	117	0.068	0.396	+0.000
feedback, $L=20$	137	0.204	0.396	+0.015
feedback, $L=50$	163	0.331	0.396	+0.044
fixed threshold	1,047	0.836	0.625	+0.000
<i>first-flow arrival (optimistic upper bound; unmoved rows omitted)</i>				
online FDR (mean rule)	18	0.000	0.065	+0.000
feedback, $L=0$	119	0.076	0.400	−0.004
feedback, $L=20$	135	0.185	0.400	+0.004
feedback, $L=50$	150	0.267	0.400	+0.040
<i>frontier</i> , $\text{FDP}=0.000$	104	0.000	0.378	—
<i>frontier</i> , $\text{FDP}=0.050$	110	0.009	0.396	—

TABLE XLVII

MATCHED OPERATING POINTS AT THE **STRESS-TEST WINDOW** (POSITION 0.85, TWO-HOUR GROUPING, $T = 31,568$ EPISODES OF WHICH 255 ARE MALICIOUS). FDP IS MEASURED AGAINST LABELS AT A WINDOW WHERE THE EVIDENCE IS *not* A VALID E-VALUE (APPENDIX F-A). ROWS ARE UNDER THE **CANONICAL** WITHIN-BUCKET ORDER, WITH THE FIRST-FLOW ARM BENEATH FOR THE 4 ROWS THE ORDER MOVES; THE FRONTIER IS MEASURED ORDER-INVARIANT AND QUOTED ONCE. THE COMPANION PRIMARY-WINDOW TABLE IS TABLE XLVI.

Method	alerts	FDP	recall	gap
<i>canonical metadata-hash order</i>				
online FDR (mean rule)	34	0.000	0.133	+0.000
online FDR (slot rule)	126.6	0.001	0.496	−0.003
feedback, $L=0$	160	0.069	0.584	+0.004
feedback, $L=20$	171	0.117	0.592	+0.024
feedback, $L=50$	190	0.184	0.608	+0.047
fixed threshold	875	0.763	0.812	+0.000
<i>first-flow arrival (optimistic upper bound; unmoved rows omitted)</i>				
online FDR (mean rule)	72	0.000	0.282	+0.000
feedback, $L=0$	161	0.068	0.588	+0.000
feedback, $L=20$	160	0.081	0.576	+0.012
feedback, $L=50$	171	0.140	0.576	+0.039
<i>frontier</i> , $\text{FDP}=0.000$	117	0.000	0.459	—
<i>frontier</i> , $\text{FDP}=0.050$	156	0.045	0.584	—

TABLE XLVIII

ANALYST-FEEDBACK CONTROLLERS UNDER WALL-CLOCK DISPOSITION DELAY ($q = 0.05$) AT THE **PRIMARY ANALYSIS WINDOW** (0.55); FDP IS THE TWO-SEED MEAN. “OPEN-LOOP” AND “AT LIMIT” ARE FOR THE **P** CONTROLLER (OPEN-LOOP IS IDENTICAL FOR **PI**). THE FINAL ROW IS THE LONG-SPAN WINDOW AT A 24 h DELAY, WHERE FEEDBACK COLLAPSES TO THE NO-FEEDBACK OPEN-LOOP BASELINE. ALL COUNTS HERE USE THE **FIRST-FLOW** WITHIN-BUCKET ORDER, THE OPTIMISTIC UPPER BOUND OVER THE ORDERS WE AUDIT, NOT THE CANONICAL METADATA-HASH ORDER TABLE I REPORTS (TABLE XIX); THE ORDER SHIFTS DETECTIONS AND THE LEVEL AN ALERT FIRES AT, AND SHIFTS NOTHING ELSE.

delay (h)	FDP (P)	FDP (PI)	FDP (AQT)	% open-loop	% at limit
0	0.053	0.049	0.172	0.9%	95.0%
0.25	0.402	0.402	0.523	51.4%	87.3%
1	0.514	0.514	0.592	60.5%	73.4%
4	0.671	0.671	0.671	77.8%	44.7%
8	0.834	0.834	0.834	99.8%	12.0%
long-span, 24 h	0.877	0.877	0.876	no-feedback FDP = 0.887	